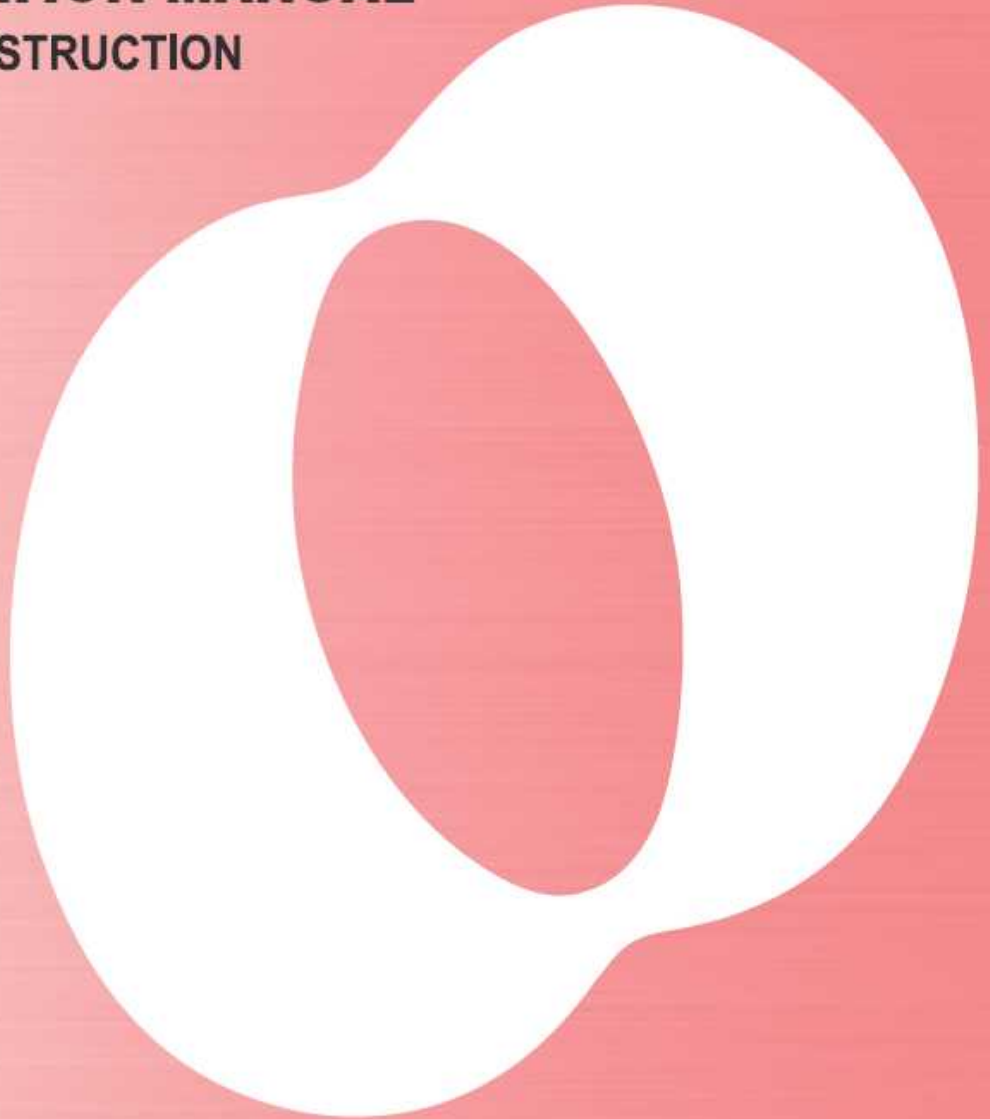


JH-SERIES

JH5

Professional U

INSTALLATION MANUAL ORIGINAL INSTRUCTION



JH-SERIES Professional U

INSTALLATION MANUAL

MAKINO

1. Do not install this machine without carefully reading and understanding this manual.
2. Store this manual in a clearly marked location for easy reference.
3. All rights reserved. No part of this document may be reproduced, copied, or modified in any form or any means without direct permission of MAKINO ASIA PTE LTD.

	English	This manual should only be used by skilled maintenance personnel.
	Japanese	このマニュアルは訓練された保守要員がご使用ください。
	Albanian	Ky manual duhet të përdoret vetëm nga personeli i trajnuar i mirëmbajtjes.
	Bosnian	Ovo uputstvo bi trebalo koristiti samo osoblje obučeno za održavanje.
	Brazilian Portuguese	Este manual apenas deve ser utilizado por técnicos de manutenção qualificados.
	Bulgarian	За използване само от специално обучен персонал за поддръжка.
	Chinese	本手册仅供已经培训的维护人员使用。
	Croatian	Ovaj priručnik namijenjen je samo obučenom osoblju za održavanje.
	Czech	Tato příručka může sloužit k použití pouze pro školený personál údržby.
	Danish	Denne manual må kun anvendes af specielt uddannet servicepersonale.
	Dutch	Dit handboek mag enkel door speciaal opgeleid onderhoudspersoneel gebruikt worden.
	Estonian	Manuaal on mõeldud kasutamiseks ainult vastavate oskustega hoolduspersonalile.
	Finnish	Vain koulutetut huoltohenkilöt saavat käyttää tätä ohjekirjaa.
	French	Ce manuel ne doit être utilisé que par du personnel de maintenance compétent.
	German	Dieses Handbuch darf nur durch speziell ausgebildetes Wartungspersonal verwendet werden.
	Greek	Το παρόν εγχειρίδιο πρέπει να χρησιμοποιείται από ειδικά εκπαιδευμένο προσωπικό συντήρησης.
	Hungarian	Ezt a kézikönyvet csak szakképzett karbantartó személyzet használhatja.
	Indonesian	Manual ini harus digunakan hanya oleh petugas pemeliharaan yang berketerampilan.
	Italian	Il presente manuale deve essere utilizzato solo da parte del personale di manutenzione appositamente addestrato.
	Korean	이 작업지도서는 기술이 능숙한 정비인원이 사용해 주십시오

	Latvian	Šo rokasgrāmatu drīkst izmantot tikai kvalificēts tehniskās apkopes personāls.
	Lithuanian	Šį vadovą turėtų naudoti tik kompetentingi techninės priežiūros darbuotojai.
	Macedonian	Овој прирачник треба да го користи само квалификуван персонал за одржување.
	Malaysian	Manual ini hendaklah digunakan oleh kakitangan penyenggaraan yang mahir sahaja.
	Norwegian	Denne håndboken skal kun brukes av autoriserte reparatører.
	Polish	Ten podręcznik jest przeznaczony do użytku wyłącznie przez odpowiednio przeszkolony personel konserwacyjny.
	Portuguese	Este manual apenas pode ser utilizado por técnicos de manutenção qualificados.
	Romanian	Acest manual trebuie utilizat numai de personal de întreținere calificat.
	Russian	Это руководство предназначено только для квалифицированного обслуживающего персонала.
	Serbian (cyr)	Ово упутство би требало да користи само особље обучено за одржавање.
	Serbian (lat)	Ovo uputstvo bi trebalo da koristi samo osoblje obučeno za održavanje.
	Slovak	Príručku môže používať len špeciálne vyškolený personál údržby.
	Slovenian	Ta priročnik naj bi uporabljalo le osebje, usposobljeno za vzdrževalna dela.
	Spanish	Este manual sólo puede utilizarlo personal de mantenimiento especialmente cualificado.
	Swedish	Denna handbok får endast användas av speciellt utbildad underhållspersonal.
	Thai	คู่มือนี้ควรใช้โดยเจ้าหน้าที่ซ่อมบำรุงที่เชี่ยวชาญเท่านั้น
	Turkish	Bu kullanım kılavuzu sadece özel eğitim almış uzman bakım personeli tarafından kullanılmalıdır.
	Ukrainian	Ця настанова призначена лише для кваліфікованого обслуговуючого персоналу.
	Vietnamese	Tài liệu hướng dẫn này chỉ nên sử dụng bởi nhân viên bảo trì lành nghề.

Introduction

This machine is an NC machine tool (machining center) that incorporates an automatic tool change function and performs a wide range of machining operations such as milling, drilling, and tapping without changing how the workpiece is mounted.

Operation of the machine in optimum condition requires meeting of various needs for the machine itself, workpiece, utilities, operating environment conditions, and operator skills.

This manual is provided to ensure that the customer understands the correct operating procedures and uses this machine safely and efficiently.



- **Be sure to fully understand the information in the Safety Chapter before starting any work, and perform work according to the instructions provided.**
- **Be sure to thoroughly read and understand this manual before attempting to operate, maintain, or inspect the machine.**

Warranty and Disclaimer

The information contained in the purchase agreement exchanged with the customer serves as the warranty with Makino.

For details about the warranty, see the purchase agreement.

Important Information

General

- Do not attempt to modify the machine.
- Operation, maintenance, and inspection of this machine must be performed by staffs who have received technical training for the machine, training in machine hazards and their prevention, and safety training.
- Observe the laws, regulations, and other rules of the relevant national and local administrative agencies.
- This machine, including technical data and software, may be subjected to local and international laws governing trade and import/export.
Prior to any resale, transfer or re-export of controlled items, contact Makino to obtain any required authorization or approval.
- The specifications and design are subject to change without prior notice.

This manual

- This manual is provided to ensure that the customer understands the correct operating procedures and uses this machine safely and efficiently.
- This manual was prepared for usage by experienced operators. For this reason, it does not include safety precautions for operators who do not have mechanical or technical knowledge of machine operation, programming, and maintenance.
- If the machine is operated by persons who are not native speakers of the language in this manual, the customer must ensure that the operators receive complete safety training. Also, warning labels must be affixed in a language that the operators can understand.
- The copyright for the entire content of this manual belongs to MAKINO ASIA PTE LTD. The copying, reproduction, or transfer of this manual, in whole or in part, without the express written permission of MAKINO ASIA PTE LTD, is strictly prohibited.
- Illustrations and other details may differ from the actual machine due to the selected options, modified specifications, or other reasons.
- Store the manuals needed for operation, maintenance, and inspection of this machine in a location where they can be easily accessed by the operator.
- Be sure to perform periodic inspection and maintenance of the machine according to the periodic maintenance manual or the legend plate to prevent breakdown of the machine.

Important points for work safety

- Familiarize yourself with the safety precautions and functions before attempting to operate, maintain, or inspect the machine.
- The points that the operator must observe when performing machine operation and maintenance vary depending on the situation. All possible points cannot be covered in the content of this manual. Be sure to fully understand the machine, and remain constantly aware of safety and the potential hazards while doing work.
- If the safety devices or protective devices do not operate properly, stop operation of the machine and notify the supervisor or manager. The supervisor or manager must immediately notify your authorized Makino dealer or Makino service representative.
- When the machine is stopped due to an unknown cause, immediately contact the supervisor or manager, and wait for permission before restarting operation.

Maintenance of machining accuracy

- After installing the machine, to maintain machining accuracy, conduct periodic inspection such as performing level adjustments.
- If the stability level of the machine changes, high-accuracy machining cannot be performed. In addition, normal machining cannot be performed if the machine vibrates.
- For approximately six months after installation, it is possible for the stability level of the machine to change significantly, while the foundation stabilizes due to the machine weight.
- Depending on the condition of the foundation or the machine usage frequency, conduct inspection and adjustment approximately every six months or every year.



- **Be sure to fully understand the information in the Safety Chapter before starting any work, and perform work according to the instructions provided.**
- **Be sure to thoroughly read and understand this manual before attempting to operate, maintain, or inspect the machine.**

Manuals and How to Use Them

Manuals belonging to this machine.

Name	Description
Instruction Manual	This manual includes the basic information (overview, specifications) needed for operation, practical operating procedures (operation), and troubleshooting procedures.
Periodic Maintenance Manual	This manual explains the intervals for periodic maintenance and work that is required for maintaining optimum performance of this machine.
Peripheral Device Manual	This manual describes the operating procedures for the peripheral devices connected to the machine body.
FANUC Set of NC Manuals	These manuals describe the operating procedures for FANUC equipment.
Maintenance Manual	This manual describes the mechanisms of the machine and how to perform the maintenance and adjustment work.
Installation Manual	This manual describes the preparation, transportation, and installation procedures for setup of the machine.
Electrical Schematic Manual	This manual is the circuit diagram of the Machine, for reference during machine maintenance and service.
Parts Manual	This manual provides the list of parts used in the machine for reference during maintenance and service. Order of spare/replacement parts can be made to your Makino Service Representative using the part numbers indicated in this Manual.
Other manuals for options	These manuals describe the operating procedures for the optional devices.

-NOTE-

Illustrations and other details may differ from the actual machine because the machine configuration is different depending on the specifications.

Notation used in this manual

 DANGER	Indicates an imminent hazard which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a potential hazard which, if not avoided, will result in death or serious injury.
 CAUTION	Indicates a potential hazard which, if not avoided, may result in minor or moderate injury.
NOTICE	Indicates a potential hazard which, if not avoided, may result in physical damage to the machine or adversely affect the work environment.
-NOTE-	Indicates supplemental information for appropriate and efficient work or better understanding of the operation.

General Contents

Chap. 1 Safety

This chapter describes the safety devices and warning labels, work and operating precautions, and other information for ensuring safety operation of the machine. Be sure to read this chapter before using the machine.

Chap. 2 Preparation

This chapter describes the activities and check points required to be completed before actual installation of the machine.

Chap. 3 Installation

This chapter describes the procedures for hoisting and installing the machine and peripheral equipment.

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1 Safety Precautions

- Disregarding the specific instructions or precautions included in this manual may result in serious injury or death to the operators or surrounding workers, or damage to the machine.
- Only qualified personnel who have adequate mechanical and technical knowledge are allowed to operate and maintain the machine.
- Never disable or remove any safety device. Operating the machine while the safety devices are disabled may result in serious injury, death, or damage to the machine.
- Observe the safety precautions provided in this manual at all times and fully implement safety measures.
- Inspect and maintain the machine regularly to keep it in optimum operating condition. Do not run the machine if it shows any signs of abnormal operation.
- The keys (release key for door switch, machine controller panel key, etc.) which are not necessary for regular operation and maintenance must be removed from the machine and managed by supervising personnel.
- The lubricating oil, cutting fluid, and other chemical substances used with the machine must be arranged by supervising personnel.
- Workpiece materials such as magnesium and titanium may cause a fire if mishandled, so be particularly careful when machining workpieces and handling cutting chips from these materials.

1.1 Operators Checks

- Only qualified personnel who have adequate mechanical and technical knowledge are allowed to operate and maintain the machine.
- Only qualified electrical engineers may perform electrical work.
- Only qualified personnel may use a crane or forklift.
- Wear suitable work clothes whenever operating or maintaining the machine. Do not operate the machine while wearing loose-fitting clothes, a necktie, jewelry, or any other clothing or objects which may become entangled with the moving parts of the machine.
- Tie up long hair, wear a hat.
- Wear safety glasses, safety shoes, safety helmet, and safety gloves as needed.



Safety goggles



Safety shoes



Safety helmet



Safety gloves

- Protective gear should be worn to protect hearing when excessive noise may be generated during operation or maintenance.
- Never operate any machinery while under the influence of alcohol or drugs.
- The operator should be in proper physical condition. If the operator suffers from a condition that impairs judgement, it may result in serious injury or death.

1.2 Work Environment Checks

- Make sure the machine and surrounding area are fully lighted.
- Make sure the machine and surrounding area are tidy and clean at all times.
- Clean up any oil, cutting fluid, or chips scattered around the machine.
- When performing work at high locations, use a stable footstool or stepladder.
- Keep all flammable substances away from the work area.
- Maintain adequate working space.

1.3 Precautions for Potential Fire Hazards

Use the machine by following the precautions below to protect the machine equipment, plant, and surrounding environment from the danger of fire and to ensure the safety of operators.

- 1 When using cutting fluid, be sure to use a water-soluble cutting fluid (type A1).

There is no fire hazard when using water-soluble cutting fluids (except when using them with special materials).

- 2 Oil-based cutting fluids present a potential fire hazard.

If an oil-based cutting fluid must be used due to unavoidable circumstances, be sure to observe the precautions below.

- Do not run the machine in an unmanned operation mode.
- Install the proper fire-extinguishing equipment near the machine.
- Provide alarm devices to detect a fire, automatic fire-extinguishing devices, and other equipment to the greatest extent possible.
- Do not create situations which may potentially start a fire.
 - Machine under the proper cutting conditions.
 - Perform proper tool management to prevent the occurrence of abnormal frictional heat and sparks.
 - Do not allow chips to accumulate in the machining chamber.
 - Check that a constant and full supply of cutting fluid is provided.
 - Always clean up and organize the area around the machine, and do not place flammable objects in the area.

- 3 Precautions for machining of flammable solids, resins, wood, and other flammable materials.

When machining flammable solids or other special materials, be sure to fully implement safety measures after gaining a thorough understanding of the material properties. Be sure to also pay careful attention to safety when machining resins, wood, and other materials.

When machining materials that generate dust and powder, be sure to provide equipment that takes into account the danger of a dust explosion for certain material types.

- 4 Precautions for machining while blowing air

Because air blowing has weak cooling performance, the chips that spray and fly out in the surrounding area are extremely hot. Do not place flammable objects in the machining chamber or in the area surrounding the machine.

1.4 Confirmation of Machine Status

- Machine inspections and maintenance must be performed regularly to maintain optimum machining accuracy and long-term performance, and increase machine operating efficiency.
- Confirm that all safety devices are functioning normally.
- Make sure the operator knows the location of the "Emergency Stop" switches to enable easy access in the event of an abnormal or dangerous situation (Refer to [3.1 Emergency Stop Switch](#)  20).
- Check for any loose, damaged, or worn parts on the machine. Operating the machine in a condition in which any of the parts has an abnormality may cause abnormal noises or damage to the machine during machine operation.
- Check for any loose or damaged piping or wiring. Operating the machine with the piping or wiring left damaged may cause oil leakage, electrical shock, or fire during machining operation.
- Use the most appropriate cutting tool, tool holder, retention knob, and workpiece, and make sure that they are all secured firmly in place. Otherwise, the workpiece may fall or the tool may fly out, and this may result in damage to the machine, serious injury, or death.
- Check that the tool numbers are registered correctly. Otherwise, the spindle may rotate at a speed outside the allowable range, the tool may fly out, and this may result in damage to the machine, serious injury, or death.

1.5 Pre-operation Checks

- Be sure that you fully understand the work procedures and precautions before operating and maintaining the machine. Never operate any machinery if you are unsure about any points.
- Check that the clothes you are wearing are suitable for operation.
- Perform periodic maintenance.
- Confirm that all safety devices are functioning properly before operating and maintaining the machine.
- Periodically back up the parameters when the machine was shipped and the program and offset data that have been prepared by the customer. Makino is not liable for any program or offset data that is corrupted or lost.
- Makino does not accept responsibility for any trouble caused by apparatus or programs prepared by the customers, such as damage to workpieces or the machine.
- For details on replacement parts, contact your Makino service representative. Use of improper parts may result in reduced machine performance or safety, damage to the machine, or operator injuries.
- Perform the lock-out and tag-out procedures.
- Make sure the operator knows the location of the "Emergency Stop" switches for each device so that they can be easily operated in the event of an abnormal or dangerous situation.
- Be sure to observe the information on the warning labels. Contact your Makino service representative if a warning label comes off or becomes illegible.
- When handling a hazardous or toxic material (oils, cutting fluids, and other chemical substances), obtain the safety data sheet (SDS), and follow the instructions. The safety data sheet (SDS) contains information about the safe handling of hazardous and toxic materials, and emergency measures.
- For models with graphite specifications, do not suck in anything other than dust (graphite) into the dust collector. Failure to observe this precaution may result in a dust explosion. It may also result in filter damage.

- For models with graphite specifications, to prevent health hazards due to dust (graphite), observe the following:
 - After machining a workpiece (product), wait until floating dust is collected into the dust collector to open the operator door.
If you open the operator door immediately after machining, floating dust scatters outside the machining chamber.
 - To perform any of the operations below, be sure to wear a high efficiency dust proof mask. (Recommended: 99.9% or higher particle collection efficiency)
Sucking in dust may cause pneumoconiosis.
 - When opening or closing the operator door
 - When working with the operator door being open
 - When replacing a tool in the tool magazine
 - When handling dust
 - While the dust collector is running, do not remove the dust car or the chip bucket. Otherwise, dust scatters around.

1.6 Implementing Lock-out and Tag-out

Lock-out consists of shutting down the power source to the machine or devices and locking it.

Example: Set the main power switch to the "OFF" position and secure it using a padlock or a lockout device such as a cover.

Tag-out consists of placing a warning tag to prevent anyone from turning ON the power supply.

Example: Place a "Do Not Operate" or "Under Maintenance" sign with the operator's name and department and indicating that machine operation is prohibited on the main power switch and main operation panel.

Lock-out/tag-out should be performed to prevent inadvertent operation and ensure operator safety.

- Performing lock-out/tag-out alone does not completely ensure operator safety. The operator must read and thoroughly understand the work procedures and safety precautions, and always be aware of potential hazards.

Each operator should perform lock-out or tag-out by himself or herself. Perform this procedure before starting the work, and release the lock and remove warning tag yourself after finishing work. Never release a lock and remove a sign without confirming with the operator or without the presence of the operator himself or herself.

- Be sure to clearly define and implement the lock-out and tag-out procedures of your company.

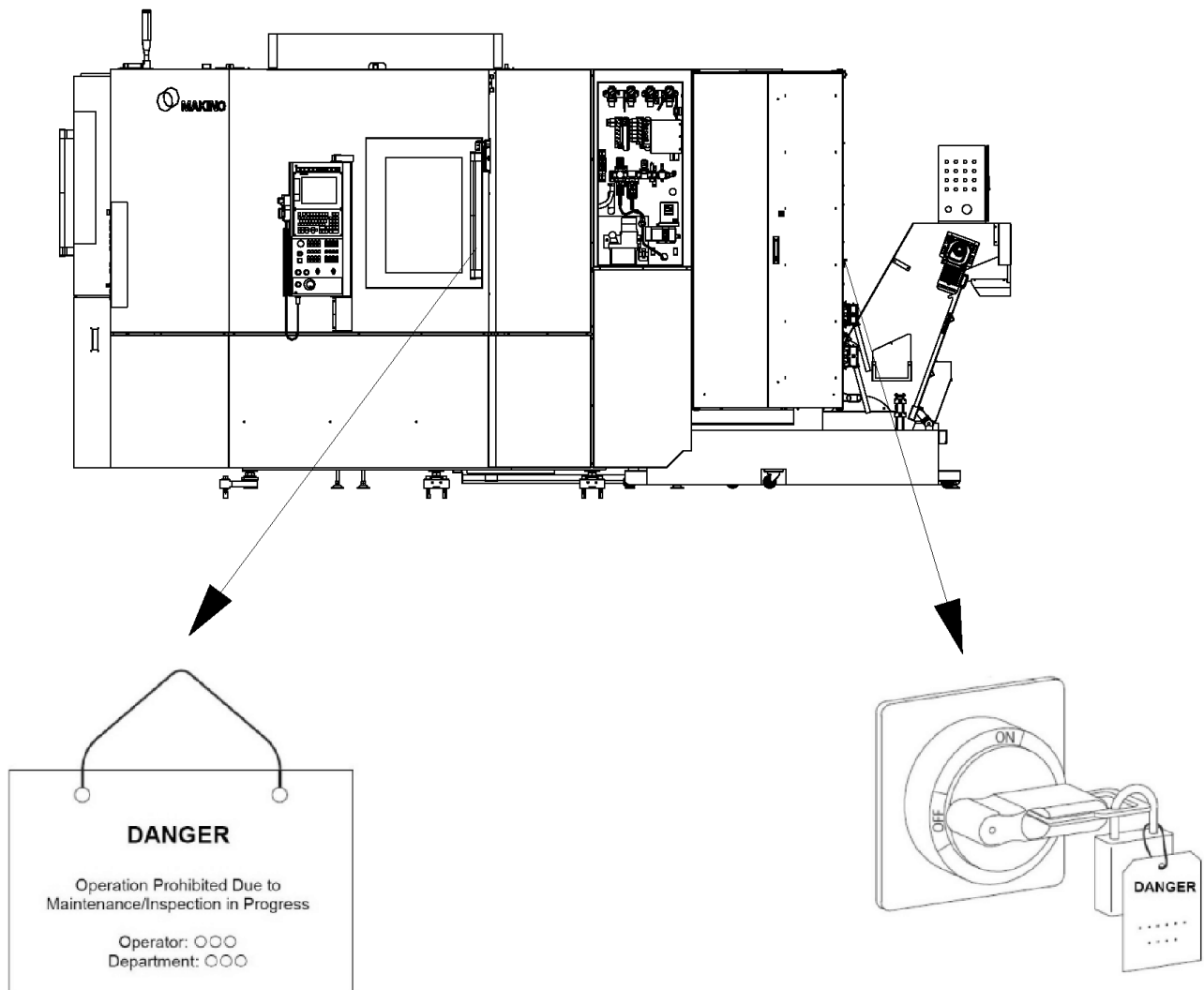


Figure 1.1 Implementing Lock-out & Tag-out

1.7 During Work

- Turn OFF the power before performing maintenance of the machine. When work must be performed with the machine power ON, confirm that all machine operations are completely stopped. Check that any residual energy in the machine is completely discharged.
- Never go near the moving parts of the machine. If you must approach moving parts to perform maintenance, be sure to take adequate safety precautions. Inadequate attention to safety may result in death or another serious accident.
- Always keep the doors and covers closed during operation. If you must work with a door or cover open when performing maintenance, be sure to take adequate safety precautions. Inadequate attention to safety may result in death or another serious accident.
- If an operator gets trapped inside the machine, press the "Emergency Stop" switch regardless of whether the machine power is ON or OFF.
- If the machine is stopped by a power failure or power supply fault, turn OFF the machine power. If the power is not turned OFF, the machine may start operating unexpectedly when the power is restored, and this may result in serious injury, death, or damage to the machine.
- If the machine is stopped by a power outage or fault in the power supply, check that the parameter, program, and offset data have not been corrupted. The machine may be damaged if it is operated using corrupted data.
- Be aware of the movement range of the machine and auxiliary components (each axis stroke, rotation range, etc.), and keep all body parts clear of moving components.
- When two or more people are required for maintenance work, be sure to maintain clear communication at all times to ensure operator safety. When performing work, be ready to press the "Emergency Stop" switch at any time.
- Be sure to always pay attention to the safety precautions listed on the warning labels affixed to the machine (Refer to [2 Warning Labels](#) .
- Do not move the switches or change the circuits except for adjustment purposes. In particular, operating the machine with the interlock(s) or other safety devices or functions disabled is extremely dangerous and may result in death or damage to the machine.
- If a circuit or other component needs to be changed for adjustment purposes, be sure to return it to the original setting after adjustment is completed.
- The optimum values for the NC parameters and machine parameters are set when the machine is shipped. Do not change any parameter setting unless it is described in the manual. Also, be sure that you fully understand the function of a parameter before attempting to change the parameter setting, and return the parameter to its original setting after the work is completed. If you try to operate the machine without the proper settings, the machine may operate unexpectedly, and this may result in serious injury, death, or damage to the machine.

- If the memory clear operation needs to be performed, be sure to contact your Makino service representative beforehand.
- If an alarm is triggered, eliminate the cause of the alarm using the appropriate procedure. If the remedy procedure is unclear, contact your Makino service representative.
- Never climb onto the covers. This may deform the covers or result in injury.
- When using a stepladder or stool, it should be sturdy, safe, and have anti-slip surfaces.
- If any oils or cutting fluids get into your eyes, body, or on your skin surface, they may cause severe health problems. Wear safety gloves, mask, safety glasses, and other safety equipment.
- Wear safety gloves whenever handling chips, tools, and workpieces.
- Protective gear should be worn to protect hearing when excessive noise may be generated during operation or maintenance.
- If lubricating oil, grease, cutting fluid, or other substances are spilled on the floor, it may result in slippage, causing injury. Wipe up any spilled fluids as soon as possible.
- Never touch a switch, button, or key while your hands are wet. Failure to observe this precaution may result in electric shock.
- Some devices (motors, lighting equipment, valves, etc.) may become very hot while the machine is operating and remain hot soon after the power is turned OFF, so be careful to avoid burns.
- Do not subject the machine to sudden impact or jolts. This may cause the machine to perform an unexpected motion or damage it.
- Do not use the machine for operation outside the specifications or exceeding the performance range. This may cause the machine to perform an unexpected motion or result in serious injury, death, or damage to the machine.
- Use the most appropriate cutting tool, tool holder, retention knob, and workpiece, and make sure that they are all secured firmly in place. Otherwise, the workpiece may fall or the tool may fly out, and this may result in damage to the machine, serious injury, or death.
- Be careful that you do not leave objects such as tools or jigs inside the machine.
- Do not place the tools, workpiece, or other parts on an unstable location.
- When a lifting sling or attachment is necessary, verify that it is strong enough to support the weight of the parts. Confirm that no one is close to the machine and the parts are well balanced, and be careful not to hit to the machine.
- Never go under a load that is being lifted. While transferring the machine, constantly pay careful attention to the hoisted load during the operation.
- Check that the tool numbers are registered correctly. Otherwise, the spindle may rotate at a speed outside the allowable range, the tool may fly out, and this may result in damage to the machine, serious injury, or death.

- Never insert hands or feet into the lift-up chip conveyor or internal chip conveyor. They may be pulled in, and this may result in death or another serious accident.
- When machining material that generates dust particles, to prevent health hazards due to inhaling the dust particles, observe the following:
 - After machining a workpiece (product), wait until floating dust has settled or cleared before opening the operator door.
If you open the operator door immediately after machining, floating dust scatters outside the machining chamber.
 - To perform any of the operations below, be sure to wear a high efficiency dust proof mask.
(Recommended: 99.9% or higher particle collection efficiency)
Sucking in dust may cause pneumoconiosis.
 - When opening or closing the operator door
 - When working with the operator door being open
 - When replacing a tool in the tool magazine
 - When handling dust

1.8 Handling of Hazardous and Toxic Materials

Handlers of hazardous and toxic materials (such as oils and cutting fluid) must receive information, education, and training in accordance with your local laws and regulations.

Particular attention must be paid to the following points.

- Be sure that there is adequate ventilation in areas where hazardous and toxic materials are used.
- Hazardous and toxic materials must be handled and stored based on the handling procedures recommended by the manufacturer.
- Identify hazardous and toxic materials by affixing labels to their containers.
- Assign a person in charge to handle the hazardous and toxic materials, and provide education and training in emergency response procedures and handling procedures.
- Before handling any hazardous or toxic material, be sure to check the safety data sheet (SDS). The safety data sheet (SDS) contains detailed information on health and safety hazards, safe handling procedures, and responses to emergency situations.

2 Warning Labels

Warning labels are affixed to machine parts that are potentially hazardous to warn operators about the hazard and its level of danger and ensure the safety of operators.

The warning labels include symbols to indicate the source of the danger, signal words to indicate the level of danger, and warning text to describe how to prevent the danger. When working at a location where a warning label is affixed, make sure that you fully understand the warning label information and definitions and follow the warning text that is provided. Failure to observe the information in the warning labels may result in death or another serious accident or damage to the machine.

2.1 Signal Word Definitions

Signal words are divided into four classes based on the degree of expected risk.

Signal Word	Description
 DANGER	Indicates an imminent hazard which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a potential hazard which, if not avoided, will result in death or serious injury.
 CAUTION	Indicates a possible hazard which, if not avoided, may result in minor or moderate injury.
NOTICE	Indicates a potential situation which, if not avoided, may result in physical damage to the machine or adversely affect the work environment.

2.2 Using Warning Labels

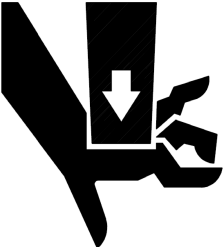
- Do not cover up or peel off the warning labels.
- Confirm that the operators and maintenance personnel are familiar with the language on the labels. If labels in other languages are required, contact your Makino service representative.
- Check that all the information in the warning label is legible. If any portion of the warning text or symbol is not visible, clean by wiping with a soft cloth dipped in water or household cleanser. Do not use organic solvents or gasoline. These may damage the surface of the warning label.
- Replace the warning label if the information in the warning label is no longer visible. To obtain new warning labels, contact your Makino service representative.
- If a part is replaced where a warning label was affixed, obtain a new warning label and affix it at the same position as before on the new part. To obtain new warning labels, contact your Makino service representative.

2.3 Information Contained in Warning Labels

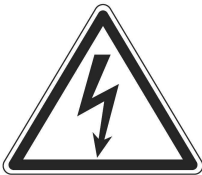
The two types of warning labels are shown below.

- Labels with warning text and a warning mark
- Labels with a warning mark only

2.3.1 Warning Labels with Warning Text

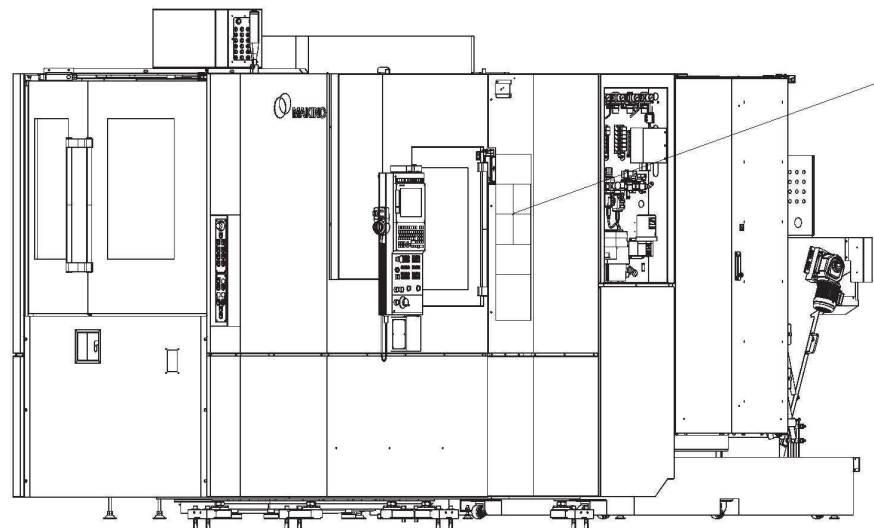
Signal Word	Symbol	Description
 DANGER		Entering enclosed spaces during operation can cause serious injury or death. Turn OFF the main power before opening door.
 WARNING		Improper insertion of tool can cause the tool to be thrown during ATC cycle, leading to serious injury. Close door fully before ATC cycle.
 WARNING		Carelessness when removing workpiece can cause serious injury. Machined edges are sharp and can cut. Exercise extreme care when removing workpiece.

2.3.2 Warning Labels with Warning Marks Only

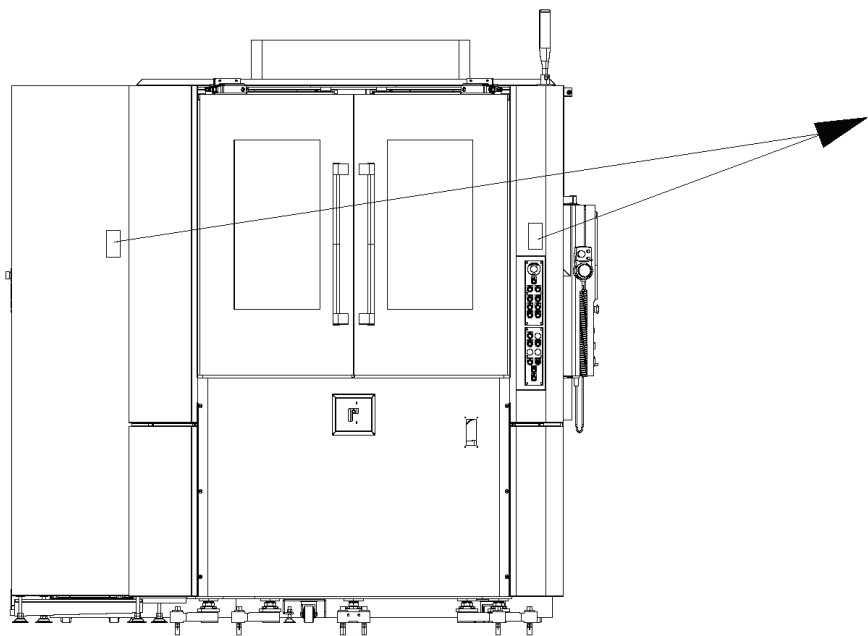
Symbol	Description
	This warning label is affixed to parts where touching the internal high-voltage components may result in electrical shock. Workers who are not qualified electrical engineers must not access the parts where this warning label is affixed.
	This warning label is affixed to parts that may become extremely hot. Do not touch parts where this warning label is affixed. Be particularly careful immediately after operation because these parts are extremely hot at this time.

2.4 Warning Labels Locations

2.4.1 Warning Labels Locations on Standard Machine

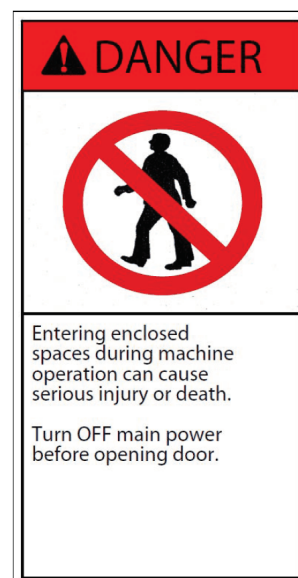
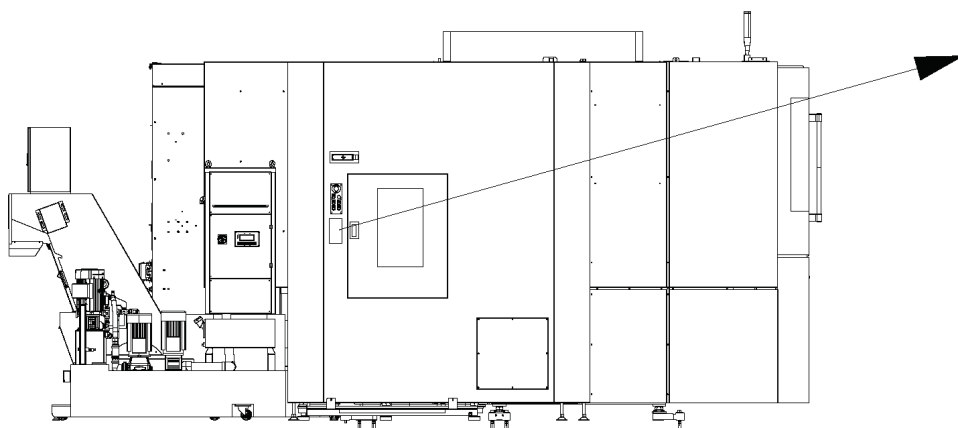
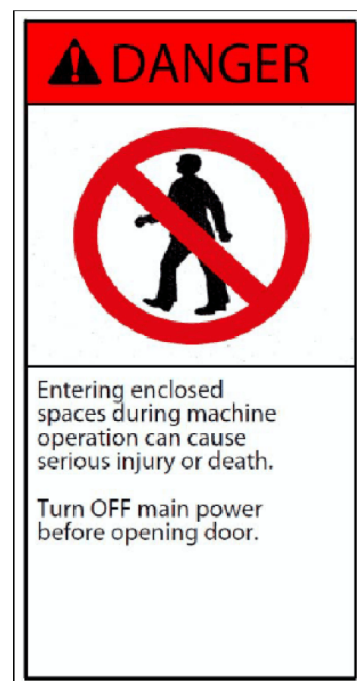
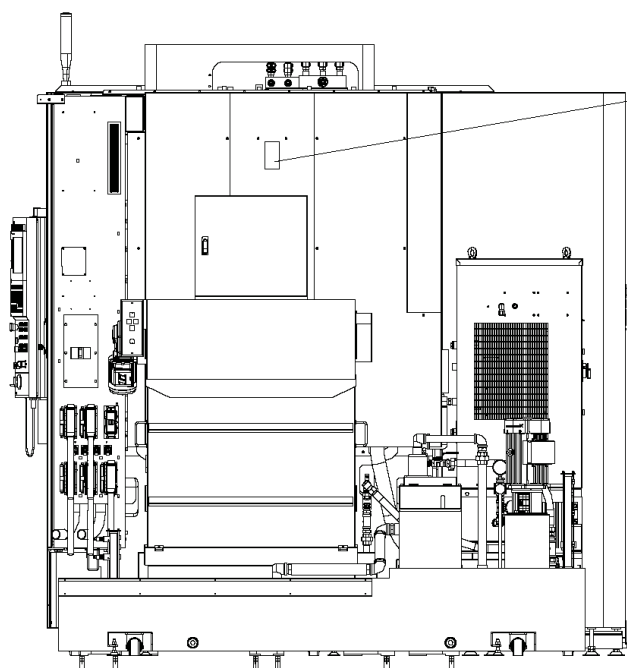


⚠ CAUTION SETTING OF ATC SPEED Select medium or low ATC speed mode if tool weight exceeds 6kg. High speed ATC mode with heavy tool (>6 kg) will damage the ATC.	⚠ DANGER Entering enclosed spaces during machine operation can cause serious injury or death. Turn OFF main power before opening door.
⚠ WARNING Improper insertion of tool can cause the tool to be thrown during the ATC cycle, leading to serious injury. Close door fully before ATC cycle.	⚠ WARNING Carelessness when removing workpiece can cause serious injury. Machined edges are sharp and can cut. Exercise extreme care when removing workpiece.

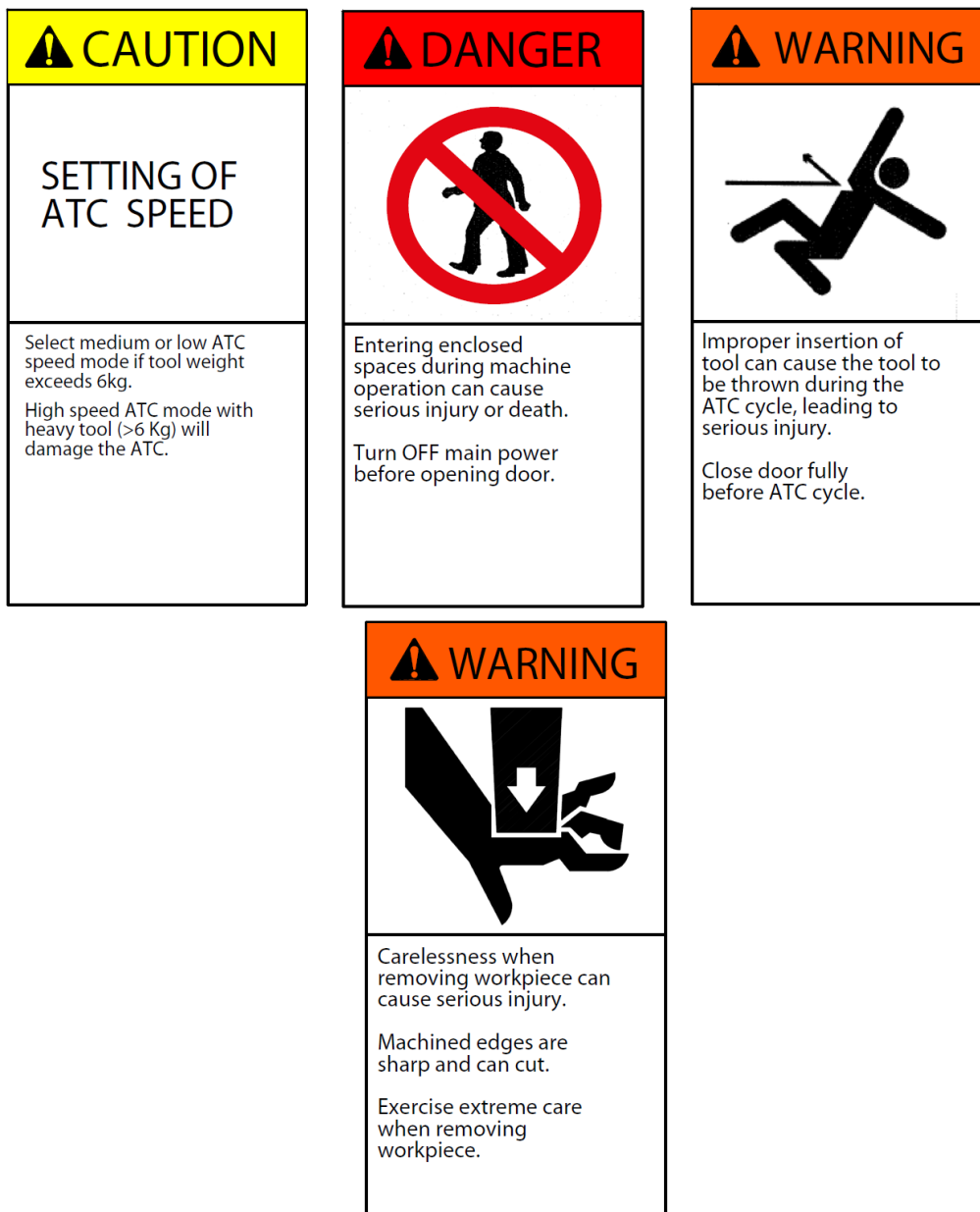


⚠ DANGER

Entering enclosed spaces during machine operation can cause serious injury or death.
Turn OFF main power before opening door.



2.4.2 Labels



-NOTES-

- 1 Locations and types of labels differ depending on the machine type or specifications (the above diagram is an example) and subject to revision and change.
- 2 Confirm the Warning labels on the actual machine.

3 Safety Devices



- **Never disable or remove any safety device.**

Safety devices are installed on the machine to protect operators and maintenance personnel. The safety devices also include ones that function on the condition that the operator observes the safety procedures.

3.1 Emergency Stop Switch



- **Make sure operators know the locations of all the "Emergency Stop" switches prior to performing machine operation or maintenance to enable use in an abnormal or dangerous situation.**

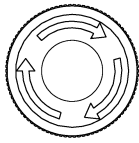
The machine goes to the following status when the "Emergency Stop" switch is pressed:

- Feeding of the axes is stopped immediately.
- Spindle rotation stops if it is rotating.
- The spindle is clamped if it has been unclamped.
- When orientation of the spindle has been performed, orientation is reset.
- When a tool is being changed or the tool magazine is operating, operation stops immediately (even during motion).
- The lift-up chip conveyor and internal chip conveyor come to an immediate stop during operation.
- The hydraulic unit is stopped.
- The cutting fluid supply is stopped.
- The air supply of air blow inside the machining chamber is stopped.
- Energizing of all solenoid valves is reset.
- The NC is reset.

Once the "Emergency Stop" switch is pressed, it is locked in the pressed position. The lock can be released by turning the switch in the direction indicated by the arrow or pulling out the switch. Then, pressing the [CONTROL POWER ON] switch cancels the emergency stop state.

"Emergency Stop" switches are installed at the following locations. All buttons function identically.

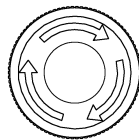
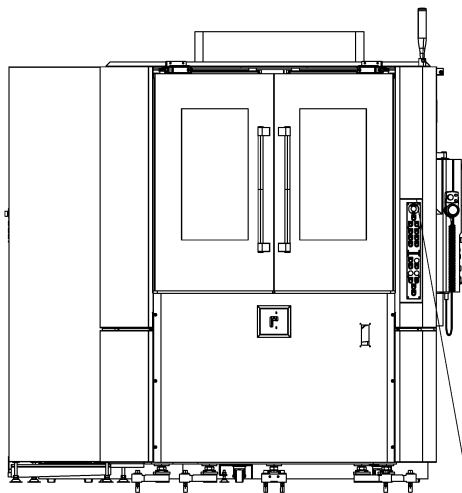
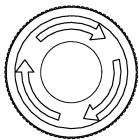
"Emergency Stop" switch
(Main operation panel)



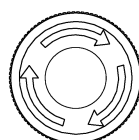
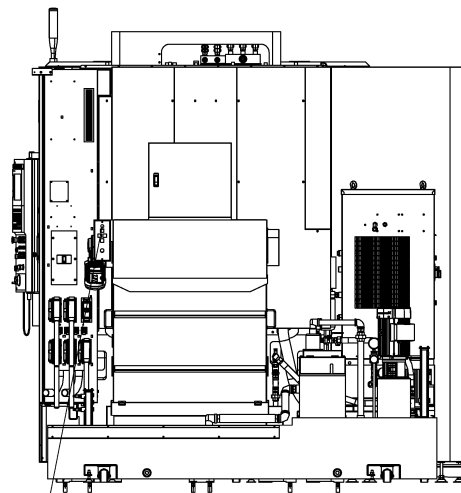
"Emergency Stop" switch
(Manual pulse generator)



"Emergency Stop" switch (ATC
operation panel)



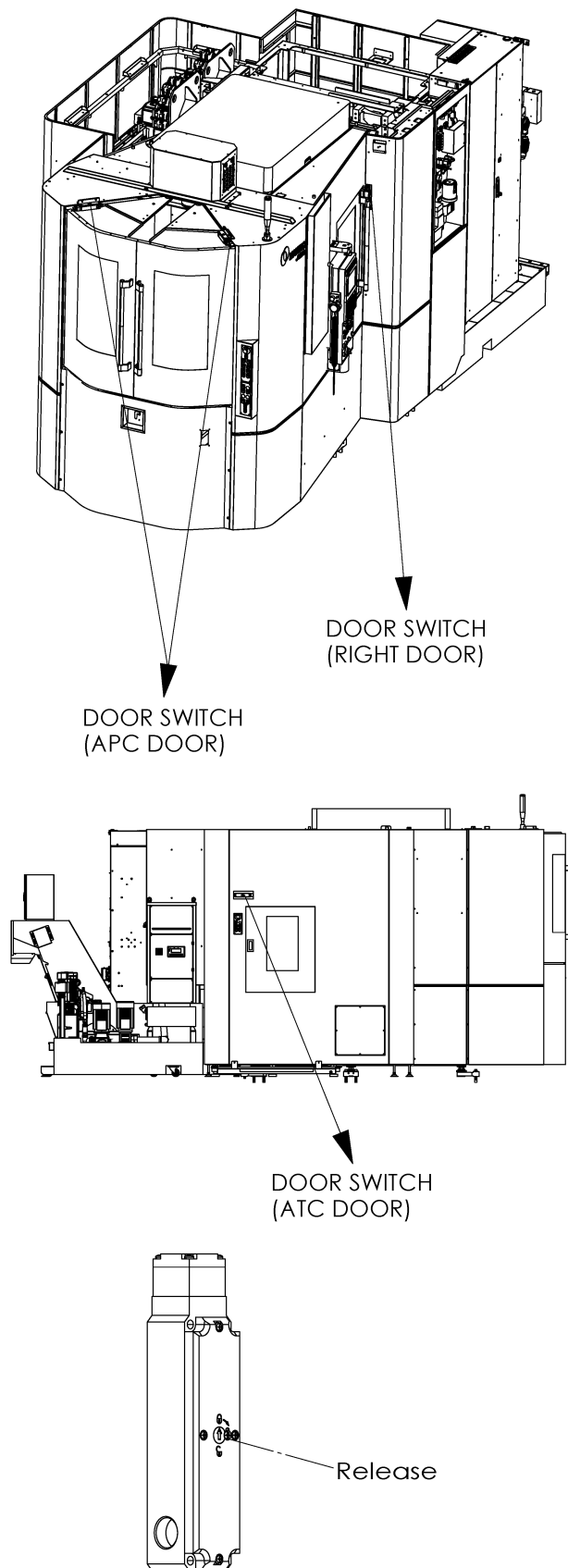
"Emergency Stop" switch
(APC operation panel)



"Emergency Stop" switch
(Chip conveyor operation panel)

3.2 Door Switch

The door switch is a device for detecting and controlling the status of doors and covers.



4 Work Hazards

The following tables list examples of hazardous actions and situations during machine operation and maintenance and examples of the resulting accidents and incidents that may occur.

The points that the operator must observe when performing machine operation and maintenance vary depending on the situation. All possible points cannot be covered in the content of this manual. Operators and maintenance personnel must have a thorough understanding of the machine and must constantly be aware of safety and potentially dangerous situations when performing operation.

Improper operation may result in death or damage to the machine.

4.1 Area Surrounding Machine

Action/Situation	Result
Tripping over cables, piping, or change in height	Falling
Operator working on oily floor	Fall, bone fracture, injury
Bringing an open flame into the area	Fire, burn
Improper handling of chemical substances	Skin lesions, eye injury, or respiratory problems

4.2 Electrical System

Action/Situation	Result
Performing inspection or maintenance operation without turning OFF the machine power (Refer to NOTE)	Electric shock, breakdown of the machine, abnormal operation, fire
Touching a device with residual voltage (Refer to NOTE)	Electrical shock
Miswiring	Breakdown of the machine, abnormal operation, fire
Loose screw in terminal block or other location	Breakdown of the machine, abnormal operation, fire
Machine controller door or junction box cover is left open	Electric leakage, breakdown of the machine, abnormal operation, fire
Touching switches with wet hands	Electrical shock
Damage to wiring on floor surrounding machine	Electric leakage, breakdown of the machine, abnormal operation, fire
Operation or maintenance by non-qualified personnel	Breakdown of the machine

-NOTES-

- 1 Prior to performing maintenance of the servo amplifiers, spindle amplifiers, and inverters, turn OFF the main power switch and confirm that the LED which indicates charging (red) for each amplifier and inverter is off.
- 2 High voltage current flows through components inside the junction box.
- 3 High voltage current continues to flow on the primary side of the main power switch even after the main power supply is turned OFF.
- 4 Electrical current continues to flow to lamps and outlets in the machine controller even after the main power supply is turned OFF.

4.3 Parameters

Action/Situation	Result
Changing NC/machine parameters not listed in manual	Injury or death, workpiece damage, breakdown of the machine
Forgetting to return the parameter to its original value after adjustment	Injury or death, workpiece damage, breakdown of the machine

4.4 Inside the Machining Chamber

Action/Situation	Result
Working without performing lock-out and tag-out	Injury or death, damage to the machine
Operator entering machining chamber with safety devices or functions disabled	Injury or death
Getting near the operating range of an axis when the power is turned ON or during axis feeding	Injury, bone fracture, bruises
Replacing the gravity axis motor without using the fixing jig	Injury or death, damage to the machine
Rotating spindle prior to cleaning of tapered section while operator door is open	Injury or death, damage to the machine
Rotating unbalanced tool at high speed while operator door is open	Injury or death, damage to the machine
Hand or other body part becomes wedged in operator door when it is opened or closed	Injury, bone fracture, bruises
Operator touching rotating spindle	Hands or fingers cut off, body part becomes entangled, or other serious injury or death
Touching a feed axis motor	Burns
Workpiece has not been clamped	Injury, bone fracture, bruises
Operator being struck by ATC Mechanism	Injury or death
Touching operating Machine shutters	Injury
Getting entangled in the internal chip conveyor belt	Body part pulled in, resulting in injury or death
Climbing on top of covers	Fall, bone fracture, injury
Climbing on top of the movable covers, inserting your hand into the gaps	Injury, bruises
Operator working on oily floor	Fall, bone fracture, injury
Touching the wipers	Cuts to hands
Operator touching bladed tools	Cuts, injuries to hands
Holding a heavy tool	Strained back, hands become wedged between tool and object
Operator being struck by chips and cutting fluid/dust scattered during machining	Damage to eyes, cuts or burns to skin
Operator being splashed by cutting fluid dripping from ceiling in machining chamber	Damage to eyes, skin irritation
Operating machine while there is abnormal vibration or abnormal noise	Damage to the machine
Operating machine with a tool incorrectly clamped	Injury or death, damage to the machine

Chapter 1 Safety

Action/Situation	Result
Opening the operator door and entering the machining chamber while mist is still present	Respiratory problems
Cleaning without wearing protective gloves	Cuts to hands
Fastening tools or implementing other work during spindle orientation	Injury, bone fracture, bruises

When machining materials that generate dust particles:

Action/Situation	Result
Opening the operator door and entering the machining chamber while dust is still air borne in chamber	Respiratory problems
Going inside the machining chamber without a dust proof mask	Respiratory problems
Breathing in dust	Respiratory problems

4.5 Tool Magazine

Action/Situation	Result
Working without performing lock-out and tag-out	Injury or death, damage to the machine
Operator entering the tool magazine without turning OFF the power	Injury or death
Operator entering the tool magazine with the safety devices and functions disabled	Injury or death
Manually changing tools by entering inside the tool magazine	Injury or death
Performing T command or tool change while tool blade or tool is mounted incorrectly	Injury or death, damage to the machine
Tool number is registered incorrectly	Damage to the machine
Operator inserting hands into tool magazine during tool magazine operation	Cuts, bone fracture of hands
Touching the tool magazine motor	Burns
Operator working on oily floor	Fall, bone fracture, injury
Operator touching bladed tools	Cuts, injuries to hands
Holding a heavy tool	Strained back, hands become wedged between tool and object
Operator working at elevated locations	Fall, bone fracture
Hand or other body part becomes stuck in tool magazine door when it is opened or closed	Injury, bruises
Leaving a safety guard or maintenance cover opened or removed	Injury
Performing cleaning or maintenance while a tool is stored in the tool magazine	Cutting or stabbing of hand or body
Cleaning without wearing protective gloves	Cutting or stabbing of hand

4.6 Cutting Fluid Supply Unit/Chip Disposal Device/Cutting Fluid Temperature Controller/Dust Collector (Graphite Specifications)

Action/Situation	Result
Working without performing lock-out and tag-out	Injury or death, damage to the machine
Inserting hands or feet into conveyor or tank without turning OFF the power	Serious injury due to body part becoming entangled, or cuts to hands and feet
Touching part or component immediately after operation	Burns
Operator touching moving parts when chip discharge outlet cover is removed	Serious injury due to body part becoming entangled
Stepping on a tank that is not fixed	Fall, bone fracture, injury
Operating with low levels of cutting fluid	Fire, damage to the machine
Touching cutting fluid or chemical additives	Skin irritation
Touching chips	Cuts, injuries, burns to hands
Replacing the filter without prior cleaning	Cuts, injuries to hands
Inhaling large quantities of cutting fluid mist	Respiratory problems
Mixing different brands of oils	Breakdown of the machine
Failing to properly clean or collect flammable cutting chips or sludge	Fire
Mixing cutting chips made from different materials	Fire
Performing operation without wearing protective gloves	Cuts, injuries to hands
Breathing in dust particles from machining	Respiratory problems

4.7 Spindle Lubricant Controller/Feed Axis Temperature Controller/Hydraulic Unit

Action/Situation	Result
Touching part or component immediately after operation	Burns
Oil temperature exceeds flash point Flash point of Mobil Velocite No. 3: >76°C, Flash point of standard hydraulic oil: Approximately 200°C	Fire
Changing filters without reducing the internal pressure	Injury or death, damage to the machine
Performing operations without safety gloves, mask, and other protective gear	Damage to eyes, skin irritation, accidental ingestion, or respiratory system damage
Mixing different brands of oils	Breakdown of the machine

4.8 Pneumatic Unit

Action/Situation	Result
Removing the pneumatic unit source piping on the machine side or disassembling the primary-side line filter without shutting of the air supply from the factory facilities	Injury due to flying off of parts or impact
Removing the pneumatic unit source piping on the machine side or disassembling the primary-side line filter without releasing the residual pressure in the air piping (inside the air circuit)	Injury due to flying off of parts or impact

4.9 Splash Guard

Action/Situation	Result
Working without performing lock-out and tag-out	Injury or death, damage to the machine
Operator entering splash guard with the safety devices and functions disabled	Injury or death
Operating the machine with the covers removed	Injury or death
Operator working on oily floor	Fall, bone fracture, injury
Operator working at elevated locations	Fall, bone fracture

4.10 Other Peripheral Equipment

Action/Situation	Result
Performing operation without turning OFF the machine power	Injury or death, electric shock

5 Occupational Health and Safety Management

5.1 Safety Device Inspection

Perform periodic maintenance and inspection of the safety devices used on this machine in order to ensure that they continue to function normally. Make sure that all maintenance staff fully understand the types of safety devices, functions, and locations described in [3 Safety Devices](#) before performing any maintenance work.

Before starting operation or maintenance work, inspect the safety devices, and if any are not functioning properly, abort machine operation.

5.2 Noise

The noise level of the machine is shown below.

- Noise level: 80 dB maximum (A-weighted sound level)

These values are measured values at full operation (maximum rotation of the spindle without machining a workpiece) with the safety guards and covers correctly mounted and all doors closed.



- **When a workpiece is machined, noise at 80 dB or higher may occur under certain customer cutting conditions. When performing work near the machine during machining, be sure to wear ear protection and other equipment to ensure that your health is not harmed.**
Performing work without wearing ear protection may lead to hearing loss.

-NOTE-

The above noise level is measured at a location that is a distance of 0.5 m from the machine and a height of 1.2 m above the floor.

5.3 Personal Protective Equipment

When conducting installation, operation, and maintenance work on the machine, be sure to protect your body from the following potential hazards.

- Mechanical hazards
- Hazardous and toxic materials
- Heat
- Noise

Be sure to always use protective equipment to protect your body from these hazards. Some examples of protective equipment are shown below.

Protected Body Part	Hazard Type	Protective Equipment	Work Requiring Protective Equipment
Head	Bruises and cuts	Helmet	• Work where parts or other objects may fall on the head • Work where the head may get hit
Eyes, nose, mouth, face	Flying out of dust, chips, chippings, and other substances Splashing of mist, lubricants, grease, and other oils	Protective glasses, dust-proof masks	• Work where objects may spray or fly out
Ears	Noise	Earplugs	• Work where the environment constantly generates noise above the stipulated level
Hands	Cuts, burns, and adhesion of chemical substances	Protective gloves, heat-resistant gloves, chemical-resistant gloves	• Work where the hands must be protected
Feet	Bruises	Safety shoes	• Handling and transporting of heavy objects
Torso	Falling from high locations	Safety belt	• Work at locations 2 meter or higher

5.4 Disposal of Waste Products

5.4.1 Disposal of Waste Oil, Waste Fluids and Waste Materials

Dispose of liquids such as cutting fluid, grease, hydraulic oil and lubricant, and chips or graphite dust, all of which are discharged during operation of this machine, in accordance with the laws, regulations, and ordinances established in your country.

Also, separate and dispose of recyclable materials properly.

5.4.2 Disposal of This Machine

When disposing of this machine or its parts, disassemble and dispose of them in accordance with the laws, regulations, and ordinances established in your country.

Also, separate and dispose of recyclable materials properly.

1 Preparation for Installation



- Before performing the installation work, be sure to carefully read Chapter 1: Safety, and perform work while paying attention to safety.
- Some installation processes can only be performed while operating the machine, and so careful attention is necessary.
- Be sure to place a warning tag on each operation panel, main power switch, and other locations when performing work to ensure that other personnel do not turn ON the power inadvertently which could result in the machine being operated.

This section specifically describes safety information for installation as well as what needs to be checked when performing installation. Be sure to carefully read this together with Chapter 1: Safety, and perform while paying careful attention to safety.

Installation Location

Assign an installation location based on the elevated view drawing.

Power Supply and Compressed Air Source

Refer to the elevated view drawing for the required capacities for the power supply and compressed air source and for the locations of the connection ports.

Installation Location Requirements

When installing the machine, select a location that meets the requirements below.

- Location not subjected to vibrations or physical impact. Vibrations and impact can cause malfunctions and breakdowns of the machine.
- Location not exposed to dust. Installation in locations exposed to dust can adversely affect machine accuracy and result in breakdowns.
- Well-lit location. Install in a well-lit location for enabling work to be performed safely.
- Location not exposed to direct sunlight. Exposure to direct sunlight can adversely affect machine accuracy and result in breakdowns.
- Location not exposed to extreme changes in temperature. Install in a location having the temperature and humidity environment below.
 - 1) Ambient temperature: 10°C to 40°C (Optimal temperature: 20±1°C)
 - 2) Relative humidity: 35% to 70% (No condensation)
 - 3) Changes in temperature: 1°C/30 minutes or less
- Location providing ample space for a maintenance area. Refer to the elevated view drawing.
- Location providing ample space for easily loading and unloading tools and workpieces and where maintenance work can easily be performed.
- Location with ample space for routes for carrying in and out the machine.
- Location that can fully support the machine weight. The weight of the heavyweight machine model is about 7300 kg.
- Location with anti-slip floor surfaces in the area around the machine.
- Well-ventilated location.

1.1 Preparation of Installation Space

Ensure sufficient clearance along the entire transportation route prior to transportation and installation.

The Machine body is the largest item during transportation, requirements for all other peripheral devices and loose items will comply, unless specifically indicated in the table below.

If transporting by crane, ensure sufficient clearance from the crane to the item height, including hoisting attachment indicated below.

If transporting by forklift, ensure sufficient space to maneuver the forklift for positioning the machine, as well as to extract the forks from the machine.

Table 1.1 Machine Body Dimensions During Transportation

Unit	Height (mm)	Width (mm)	Length (mm)	Weight (kg)
Standard Machine (SPD-15k)	3000	3000	5000	12,000
Coolant Tank with Lift-Up Conveyor	2050	2470	2310	1030
Lift-Up Conveyor	2027	1042	1995	570

Refer to the following figures of machine general view and layout to ensure sufficient installation space is available at the installation location.

The maintenance areas shown in the Machine Layout is the maintenance space required after installation.

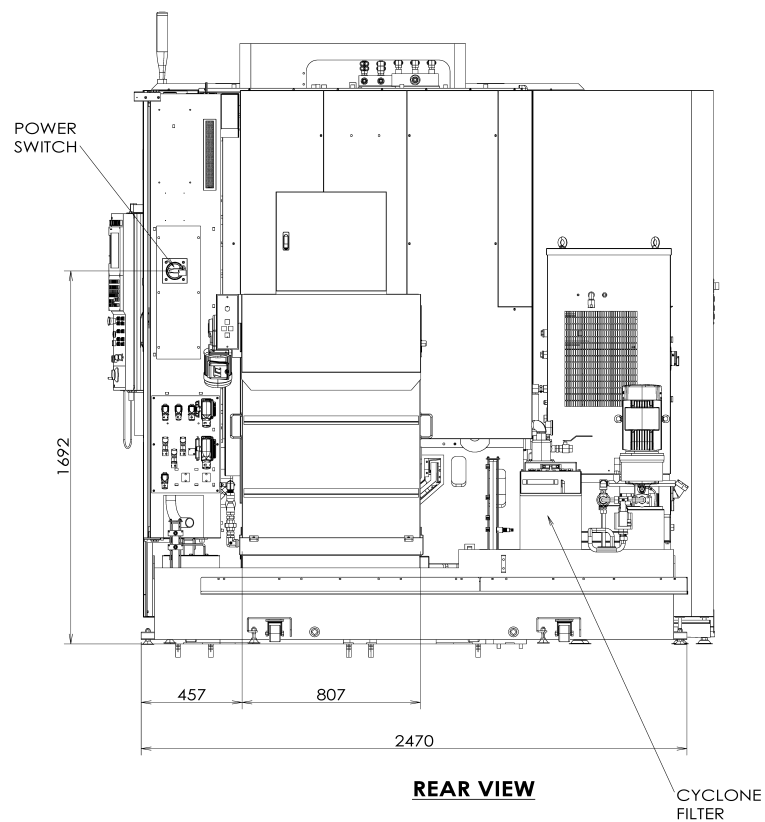
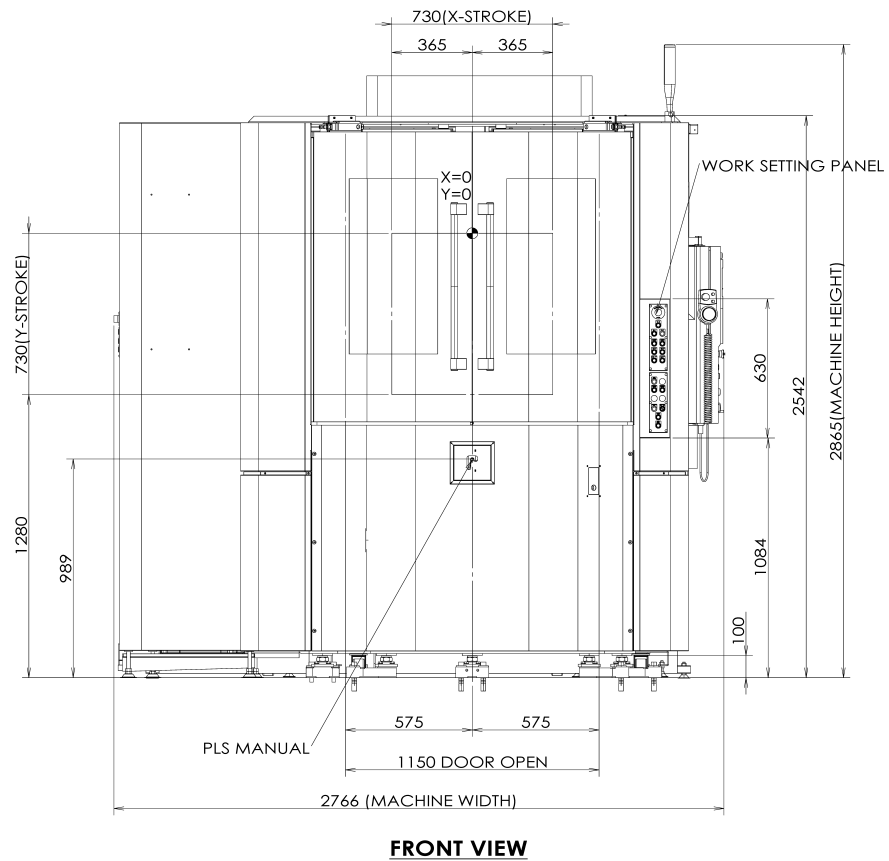


Figure 1.1 Standard Machine General Views & Dimensions 1

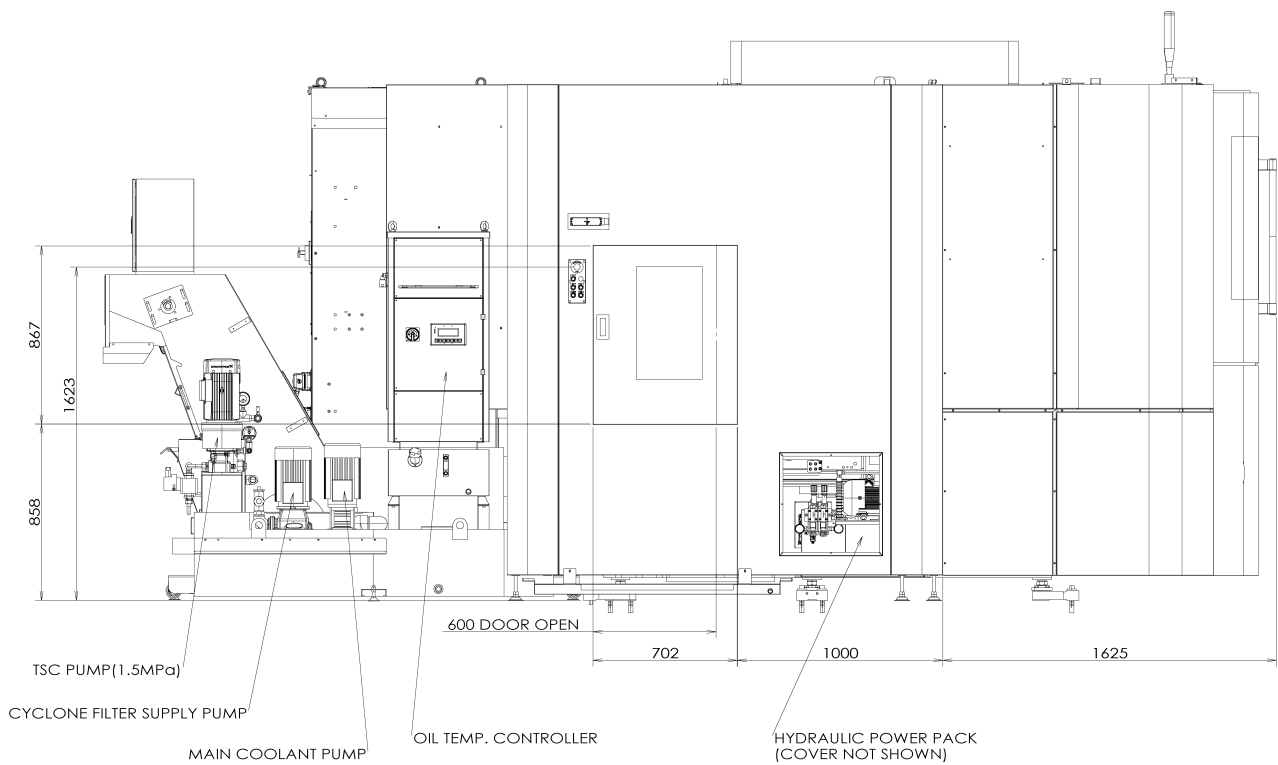
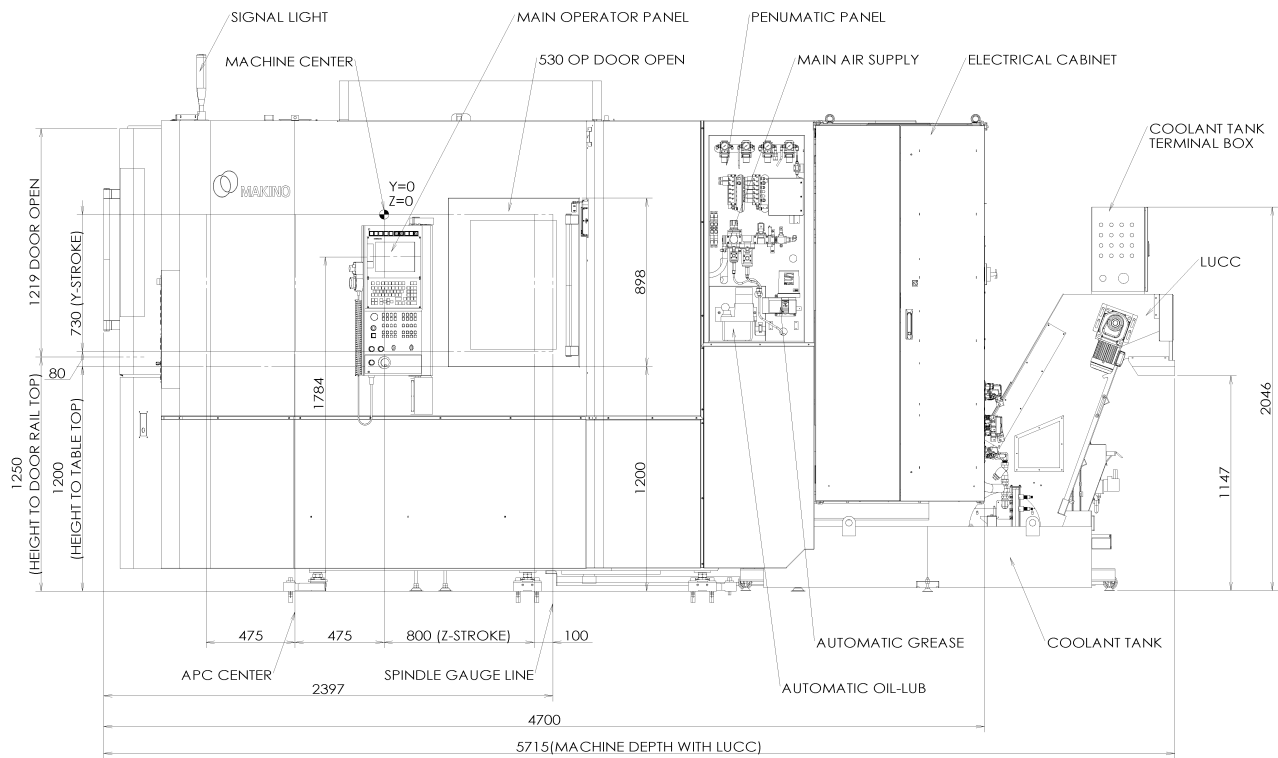


Figure 1.2 Standard Machine General Views & Dimensions 2

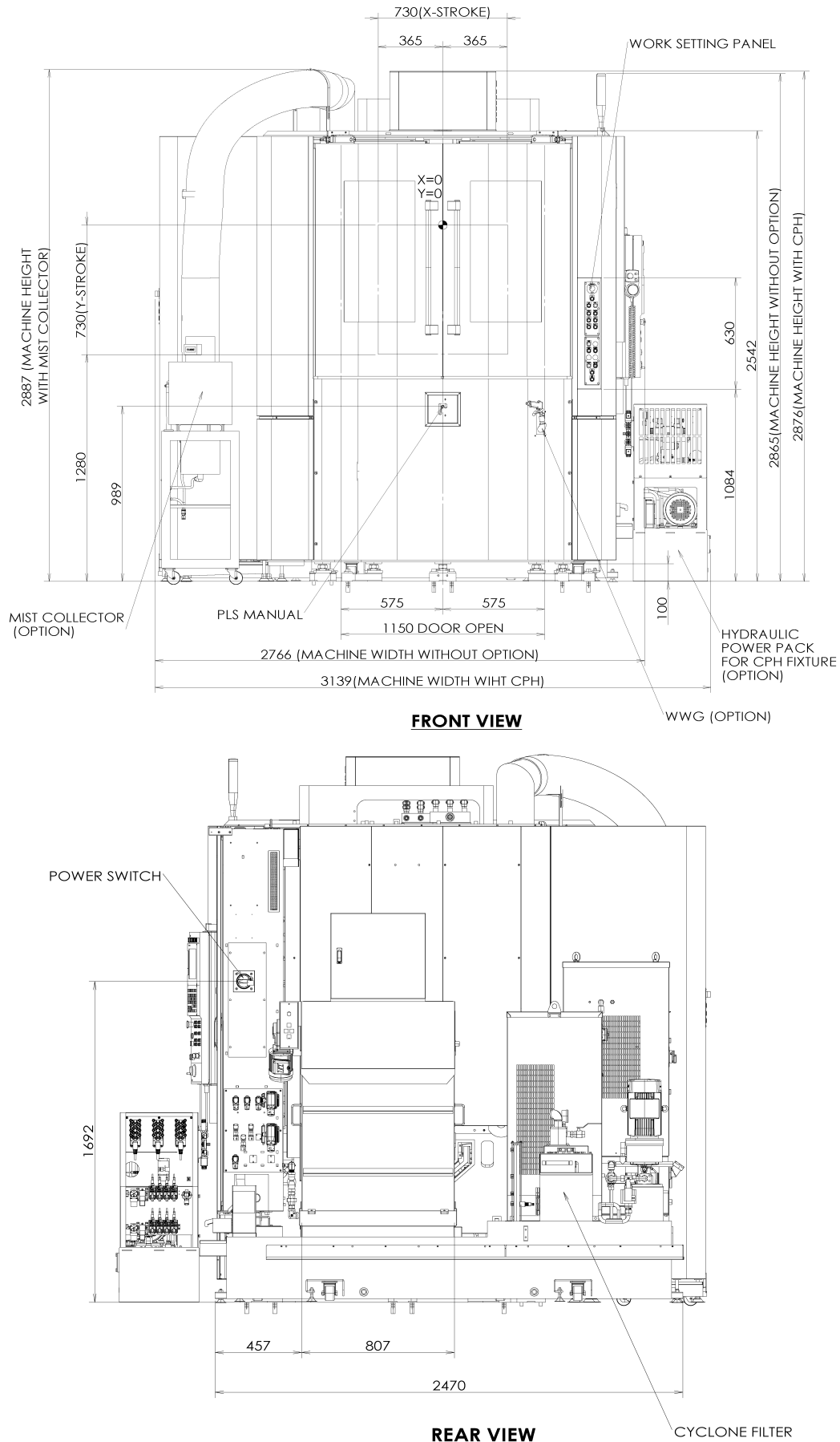


Figure 1.3 Options Machine General Views & Dimensions 1

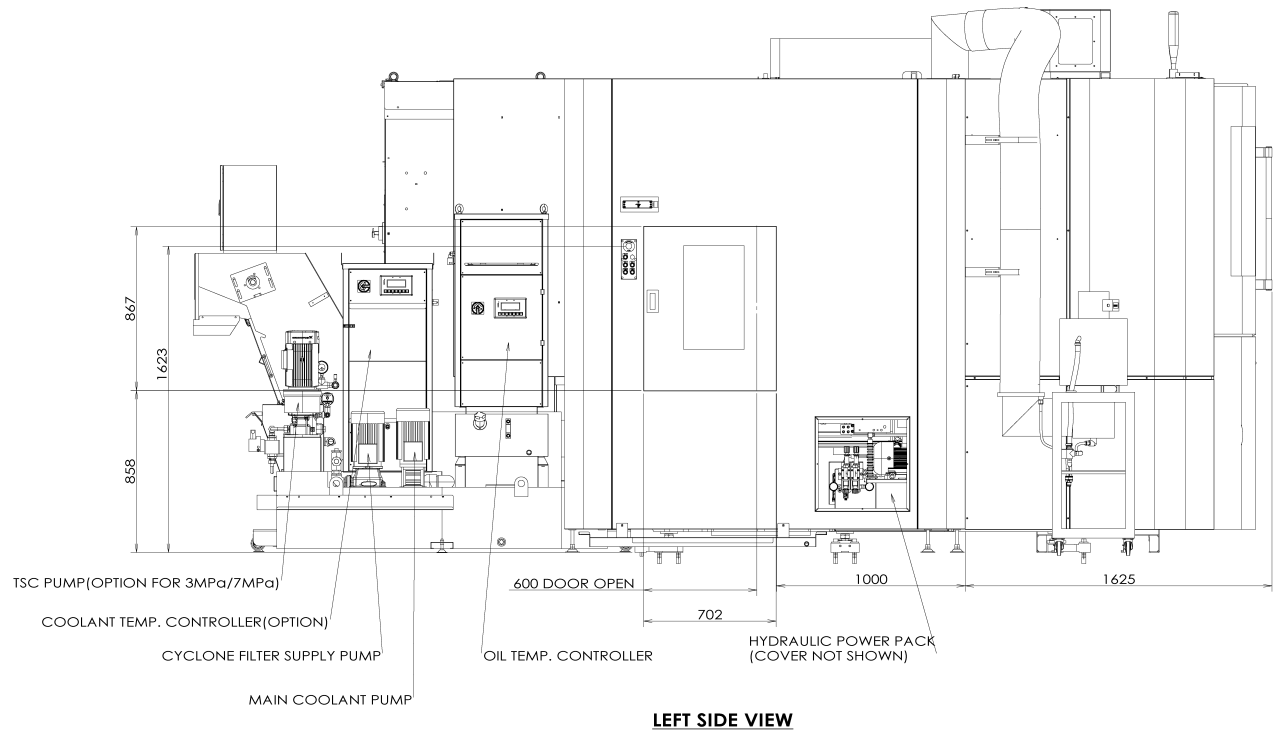
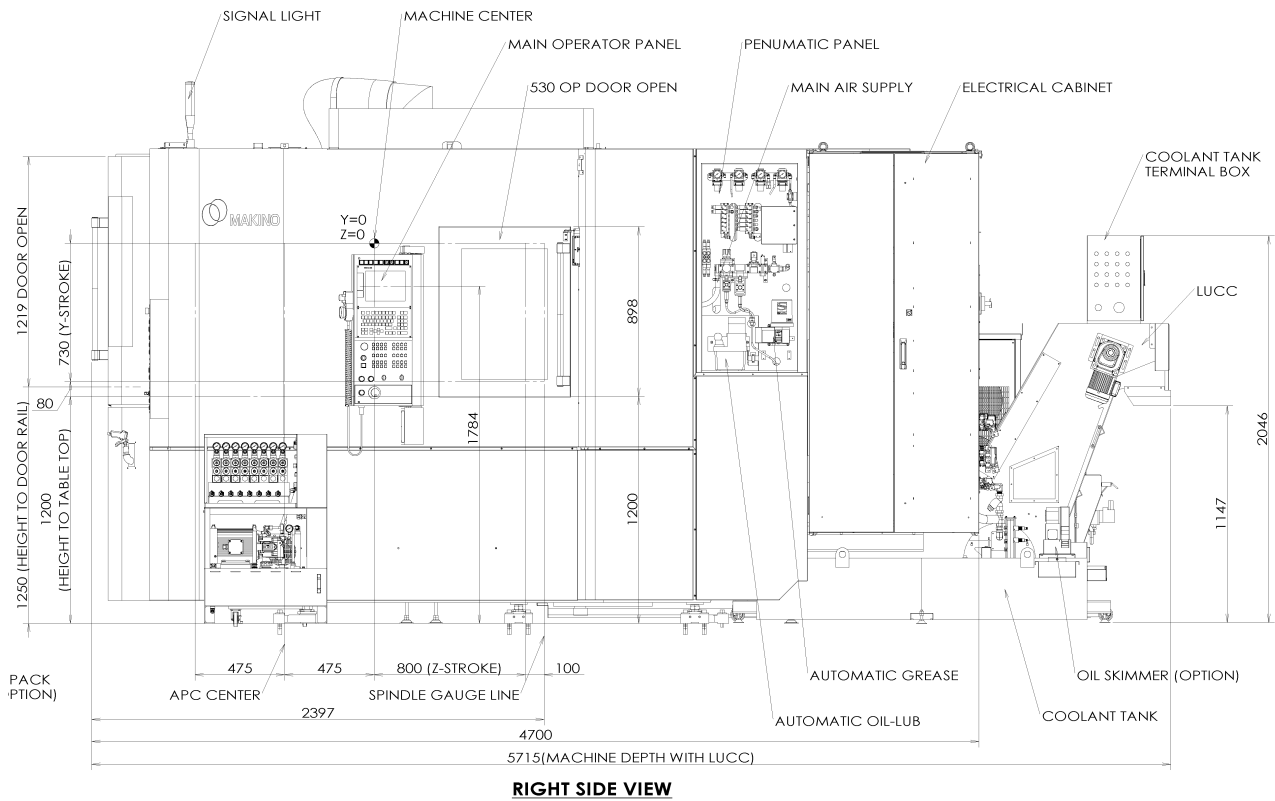
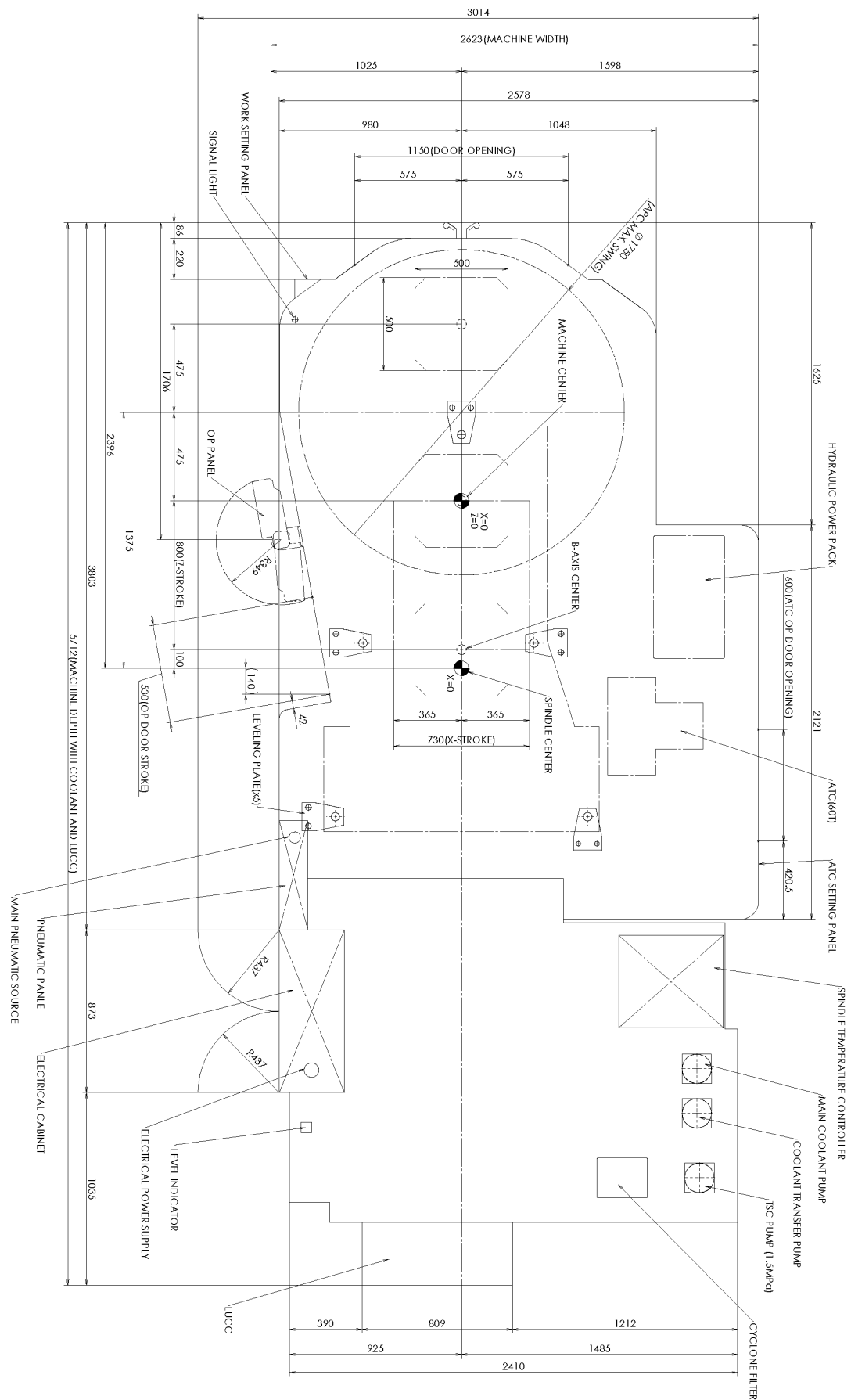


Figure 1.4 Options Machine General Views & Dimensions 2



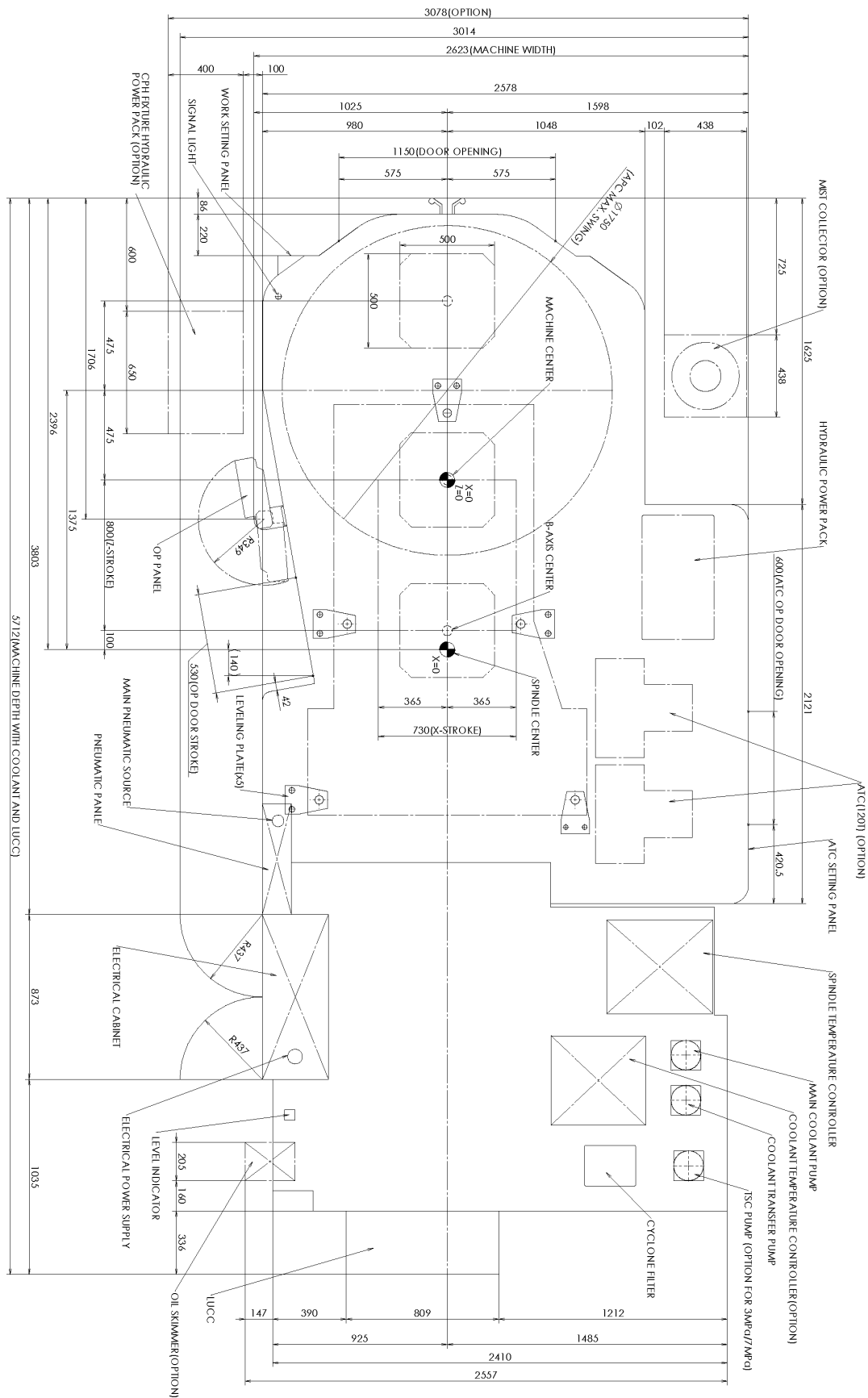


Figure 1.6 Option Machine Floor Plan

1.2 Preparation of Transport Route

Prepare the machine transport route by referring to the following tables and figures in this section.



Installing the machine

- The weight of the standard machine model is about 12,000 kg. To hoist the machine, use a crane that can fully support this weight.

Safety information for transporting

- Be careful not to subject the machine to sudden impacts when hoisting and transporting it. Never go under the machine while it is being hoisted.

By Roller Transport

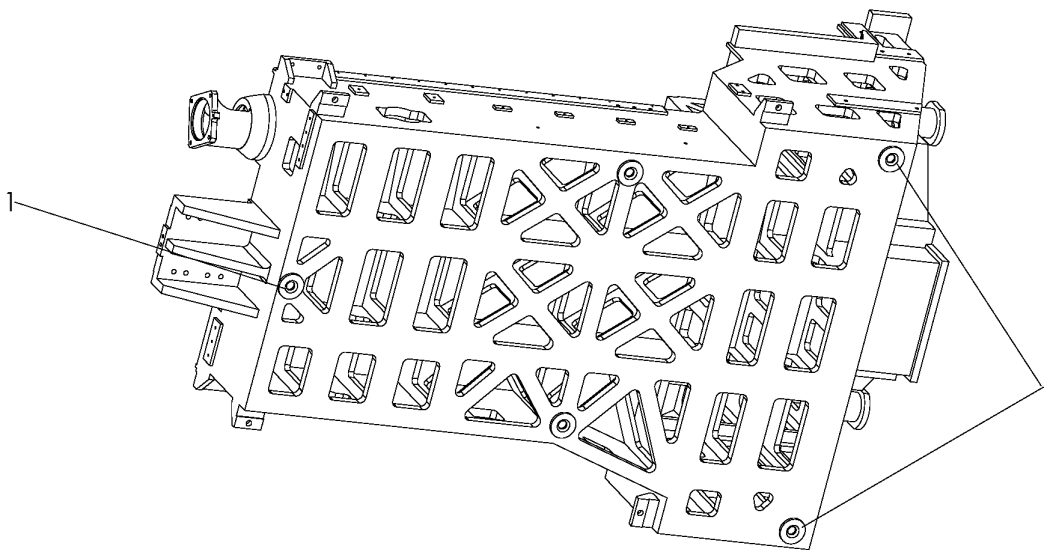


Figure 1.7 Roller Locations

No.	Name
1	Roller positions hole: $\varnothing$ 30mm

By Forklift Transport

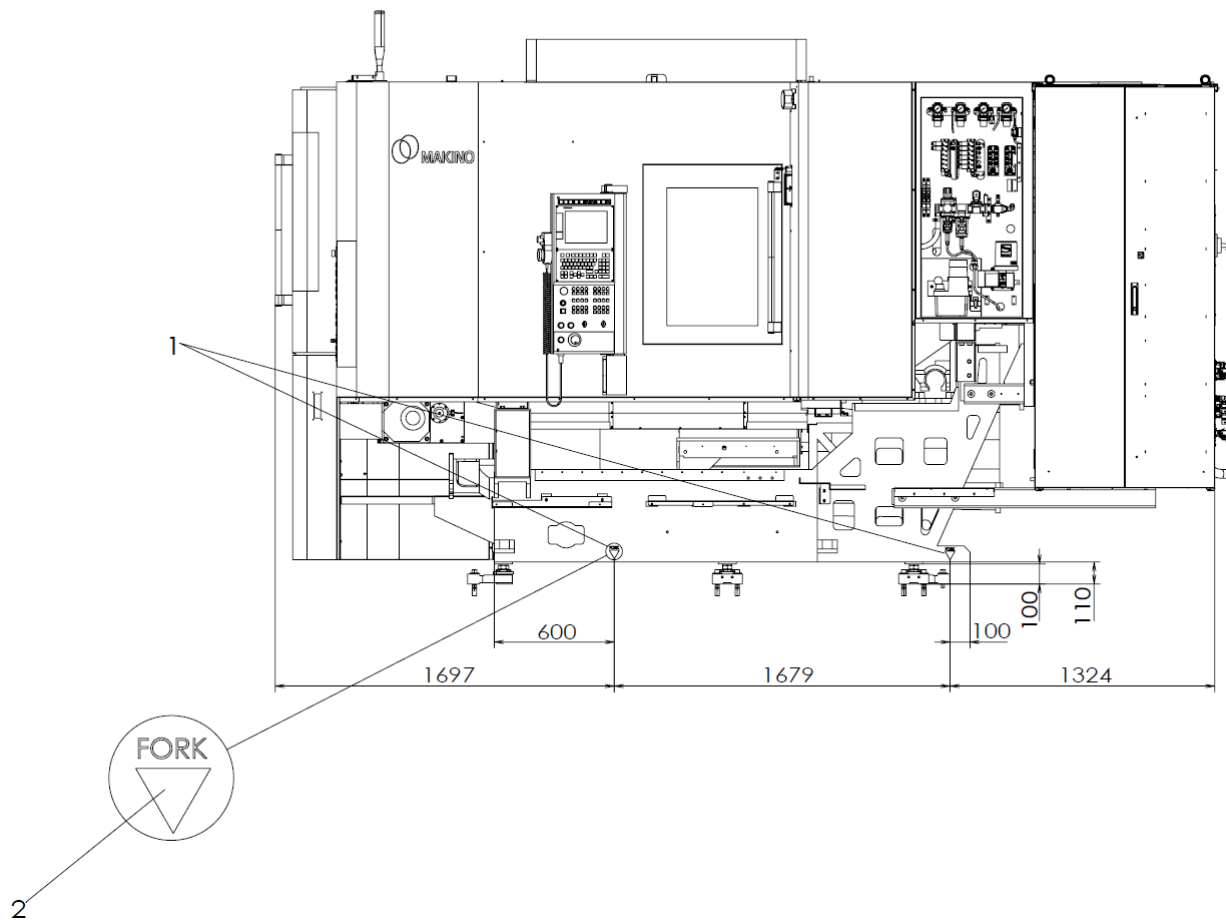


Figure 1.8 Forklift Insertion Slots

No.	Name	No.	Name
1	Forklift Slot Symbol Location	2	Forklift Slot Symbol

By Crane Transport

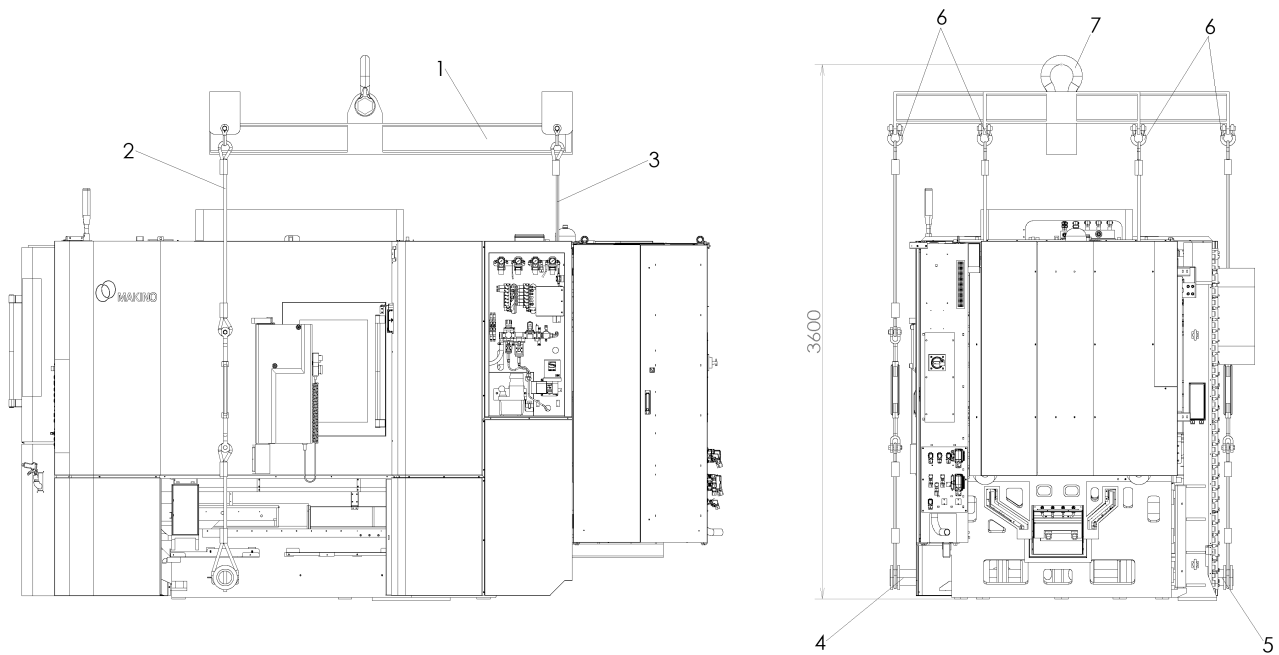


Figure 1.9 Overview of Hoisting Attachments

No.	Name	No.	Name
1	Beam, Lifting Fixture	5	Hook Rod (Right)
2	Rope (Front)	6	Shackle anchor
3	Rope (Back)	7	Hook
4	Hook Rod (Left)	-	-

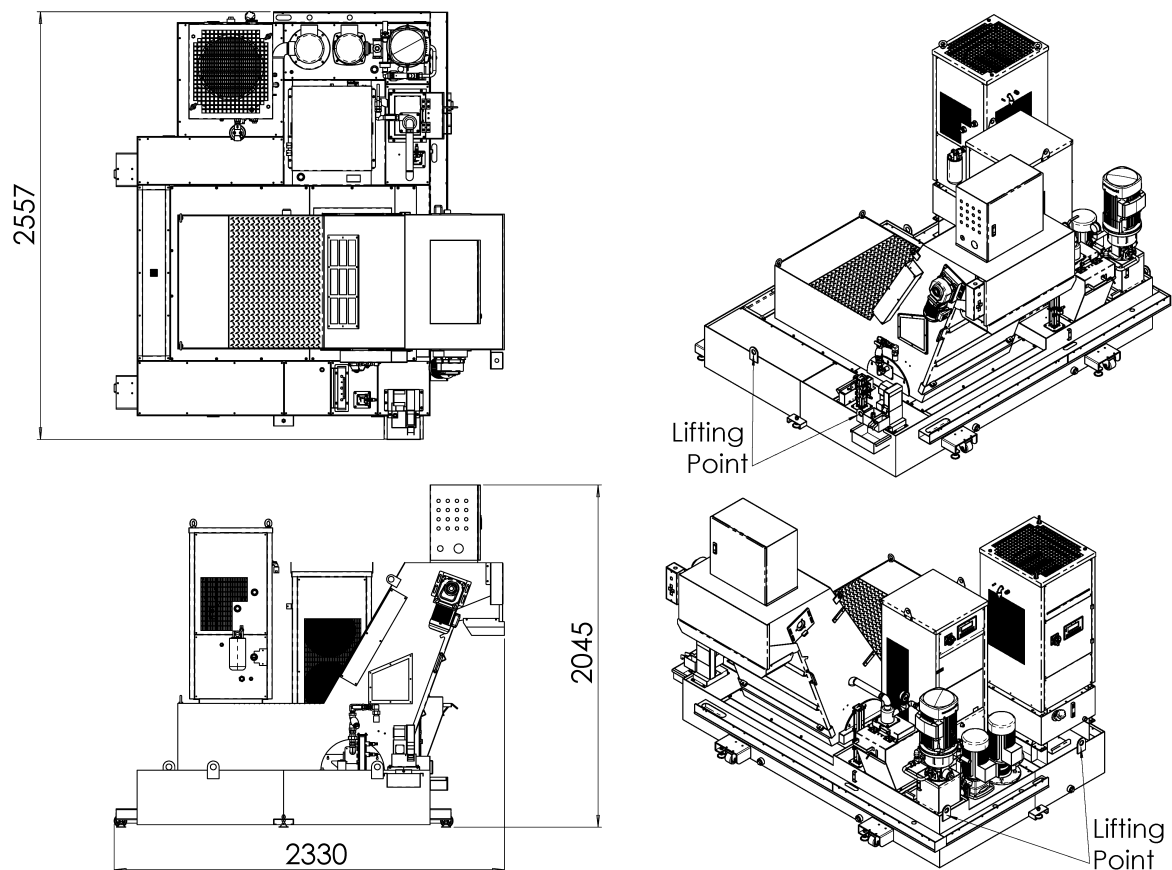


Figure 1.10 Standard Coolant Tank with Options Dimensions for Lifting

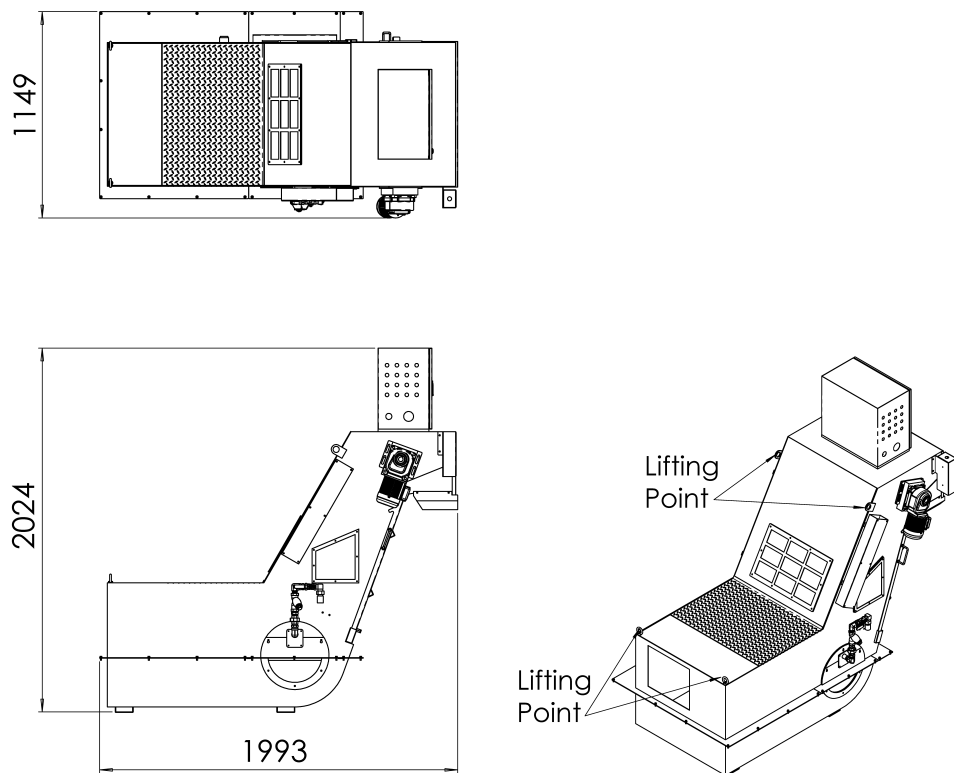


Figure 1.11 Drum Conveyor Dimensions for Lifting

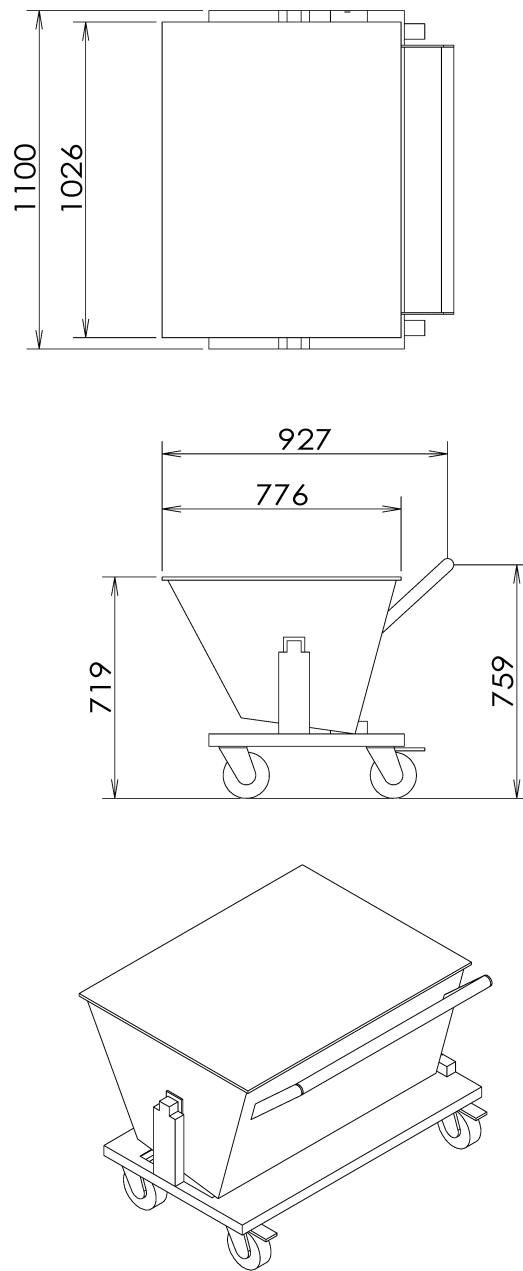


Figure 1.12 Chip Trolley Dimensions

1.3 Preparation of Installation & Transportation Equipment

Prepare equipment such as crane, forklift, skates, capable of supporting the size and weight of the machine prior to transportation.

When transport equipment is required, contact your Makino service representative for further assistance.

Refer to the table below for the approximate weights of each item during transportation.

Table 1.2 Machine Weight During Transportation

Item		Approximate Weight
Main body	Standard Machine (SPD-15K)	12,000 kg
Loose items	Coolant Tank & LUCC	1030 kg
	LUCC only	570 kg
	Chip Trolley (Option)	100 kg
	Mist Collector (Option)	45 kg
	Coolant Temperature Controller (Option)	146 kg
	Air Dryer (Option)	30 kg

1.4 Preparation of Electric & Air Supply

Table 1.3 Electrical Power Source Requirements

Item	Description
Power Supply (3 phase)	400 VAC $\pm$ 10%, 50Hz/60Hz
Maximum Total Power Consumption	90 kVA
Main Power Transformer Capacity (If supply voltage is different than 400 VAC $\pm$ 10%)	2.29 kVA
Circuit Breaker	100 A
Power cable size	35 mm ² with 400V input
Protective ground	D-class grounding work (grounding resistance:100 Ω)
Grounding cable	30mm ² or more (600V insulated cables specified by JIS C3307)

Table 1.4 Air Source Requirements

Item	Description						
Pressure	0.5 $\pm$ 0.1 MPa						
Flow rate (Maximum/Average)	700 L/min for Std 900 L/min for Std + Fixture air blow						
Air Quality	Dew point temperature: -20°C or less. Compliant with class 2.5.2 specified by JIS B 8392-1 (equivalent to ISO 8573-1) Maximum particles number/1m² for particle diameter x: <table border="1"> <tr> <td>0.001 < x $\leq$ 0.005 mm</td><td>100 or less</td></tr> <tr> <td>0.0005 < x $\leq$ 0.001 mm</td><td>6,000 or less</td></tr> <tr> <td>0.0001 < x $\leq$ 0.0005 mm</td><td>400,000 or less</td></tr> </table> Dew point at maximum pressure: +7°C or less (Absolute Pressure: 0.8 MPa) Max. oil concentration: 0.1 mg/m³ or less	0.001 < x $\leq$ 0.005 mm	100 or less	0.0005 < x $\leq$ 0.001 mm	6,000 or less	0.0001 < x $\leq$ 0.0005 mm	400,000 or less
0.001 < x $\leq$ 0.005 mm	100 or less						
0.0005 < x $\leq$ 0.001 mm	6,000 or less						
0.0001 < x $\leq$ 0.0005 mm	400,000 or less						
Air Source Connection	PT 3/8"						
Minimum Air Hose Inner $\varnothing$	19 mm						
Air Filter	5 μ m $\pm$ 0.01 μ m						

-NOTES-

- 1 The air quality varies according to the factory circumstances. Use a particle counter to confirm that the air quality values satisfy the required values.
- 2 An air dryer is mandatory to be installed.
- 3 Periodic maintenance must be performed to maintain an optimum air supply. When maintenance of the filters is neglected, filter pollution and damage to the filter may occur in a short period of time.
- 4 Besides potential damage to machine components, issues with air quality may affect the performance of tool measurement systems and linear scale.

1.5 Preparation of Foundation

The foundation must be constructed to fully support the machine weight.

Refer to the information below for the minimum recommended specifications. Where possible, construct the foundation exceeding these conditions.

- The quality of the foundation is greatly influenced by the geological structure of the installation location. To construct the best possible foundation, consult the designated builder referring to the foundation conditions.
- When the axis is moved while the maximum load is placed on the table, change in the foundation surface level must be within 0.005 mm/m. If this condition is satisfied, the machine accuracy can be maintained for a considerable period.
- Finish the foundation surface as level as possible for machine installation. This facilitates leveling of the machine and the installation and movement of attached units.
- Issues with the foundation may cause vibrations on the machine as well as affect the surrounding area.

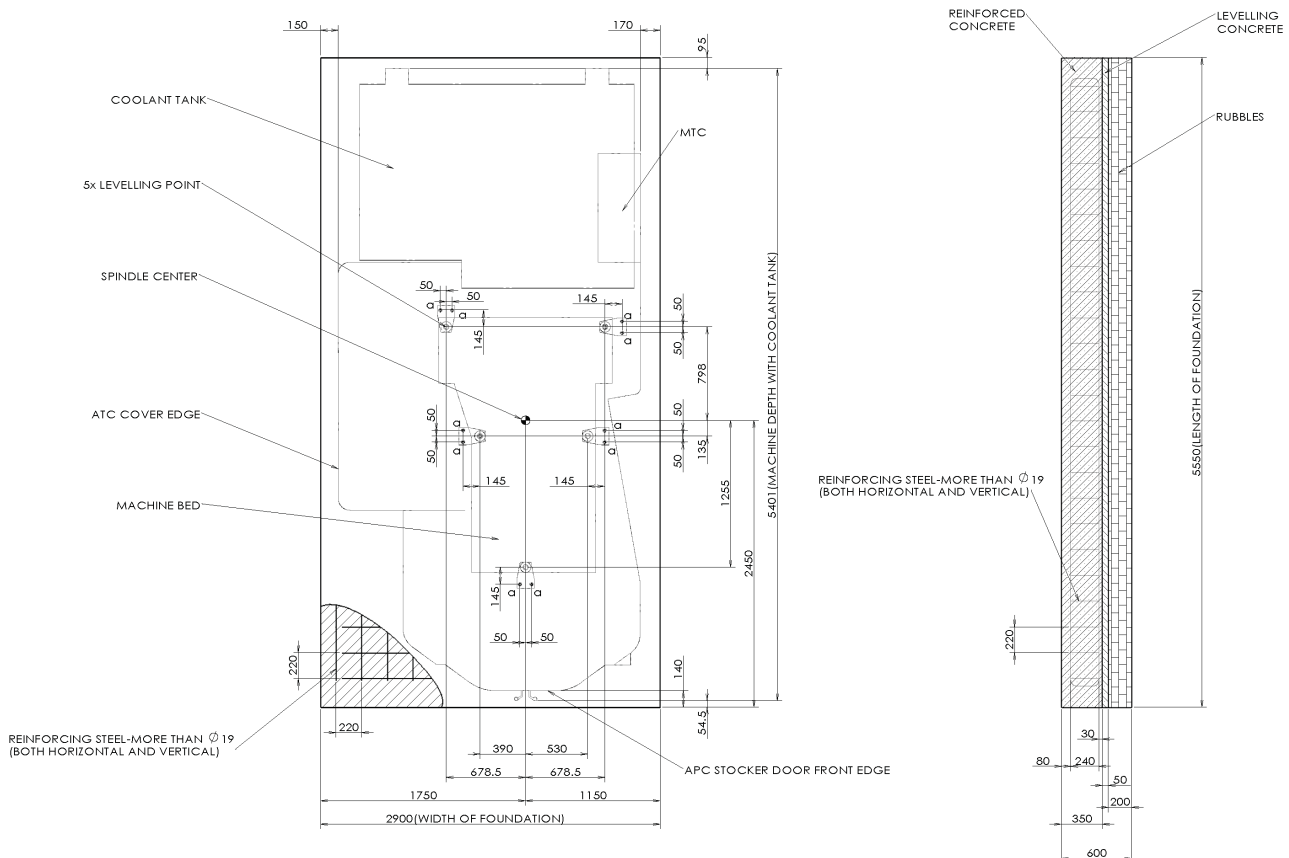


Figure 1.13 Recommended Machine Foundation

Table 1.5 Recommended Foundation Conditions

Items	Description
Foundation Thickness	350 mm
Allowable Soil Bearing Capacity	5 tonne/m ²
Concrete	FC250 or more
Machine Support Point	5 points
Vertical & Horizontal Reinforcing Steel	More than Ø 19 mm
Rubbles	Medium or large size crushed stones
Leveling Concrete Thickness	50 mm
Concrete Weight	15 tonne (Specific Gravity: 2.3 g/cm ³)
Approximate Machine Weight on the Foundation	12 tonne

1.6 Preparation of Tools

Prepare the tools needed for the various work procedures.

Table 1.6 Tools Required for Installation

Name	Part No.	Notes	Qty
Hammer drill	-	-	1
Anchor drill	-	22 mm diameter	1
Pile hammer	-	-	1

Table 1.7 Tools Required for Tool Magazine Centering Adjustment

Name	Part No.	Notes	Qty
Indicator	-	Min display increment: 2 μ m or less	1
Magnet stand	-	-	1
Feeler gauge	-	Thickness 0.1 mm	1
Tool change lever	1HS310B03A	(HSK-A63 standard accessory)	1
Centering holder	J7M705A100501	For HSK-A63	1
Ring gauge	J7M705A100601	For HSK-A63	2
Centering shaft	A2T98-A2033E	For HSK-A63	1
ATC alignment tool	-	For HSK-A63	1
ATC-spindle gap tool	-	GO-NOGO gauge	1
ATC push-pull tool	-	-	2

1.7 Preparation of Fixtures and Hoisting Attachments

Confirm the following points prior to installation work:

- Lifting fixture specific designed to lift JH5 machine.
- Inspect for any machine damage.
- Check all attachments and accessory units with the shipping and packing check list.

Table 1.8 Parts to Be Prepared (Lifting Fixture)

Name	Part No.	Qty.
Lifting Parts	2HS320A01A	1

Table 1.9 Hoisting Attachment List

Name	Part No.	Qty.
Beam, Lifting Fixture	2HS320A2005=X	1
Shackl, Anchor SWL 6.5 TON	7VS150A703-XX	4
Rope Back 2	2HS320A2004=X	1
Rope Back	2HS320A2003=X	2
Rope Front	2HS320A2006=X	2
Lifting Bar Left	2HS320A2002=X	1
Lifting Bar Right	2HS320A2001=X	1
Eye Bolt M10	Z273K1110000	2
Hook	Z259B2550000	1

The Lifting fixture can be provided by Makino Regional Office on request or can be purchased by the customer as machine optional equipment.

1.7.1 Hoisting Attachments



- Special hoisting attachments are available specifically for loading, unloading, and relocating the machine. (Special accessory)
- The total weight of the hoisting attachments is 600 kg, and so be extremely careful when handling them.
- Only qualified operators are allowed to handle the hoisting attachments.
- The axes must be locked when hoisting the machine.

Overview of Hoisting Attachments

The hoisting attachments of the machine consist of lifting beam, hook rods, shackles, wire ropes and hooks connecting these parts. The height from the floor including the hoisting attachments is about 3600 mm.

The weight of the machine including both the machine body and hoisting attachments is about 12,000 kg.

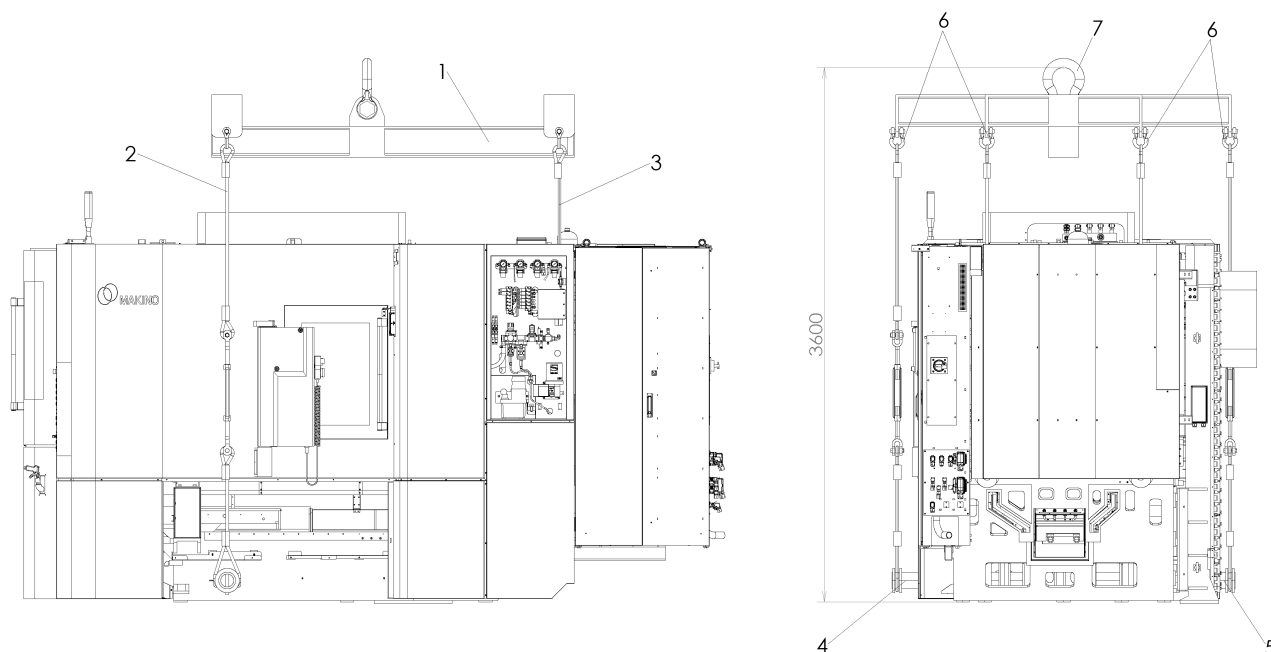


Figure 1.14 Overview of Hoisting Attachments

No.	Name	No.	Name
1	Beam, Lifting Fixture	5	Hook Rod (Right)
2	Rope (Front)	6	Shackle anchor
3	Rope (Back)	7	Hook
4	Hook Rod (Left)	-	-

Mounting the Hoisting Attachments

A Mounting the Hook Rod

- 1) Turn the protrusions [1] on the hook rod [2] so that they are on the side, and insert the rod into the cast hole [3] on the bed side. (Insert by aligning with the shape of the cast hole [3].) Remove the eyebolt [4].

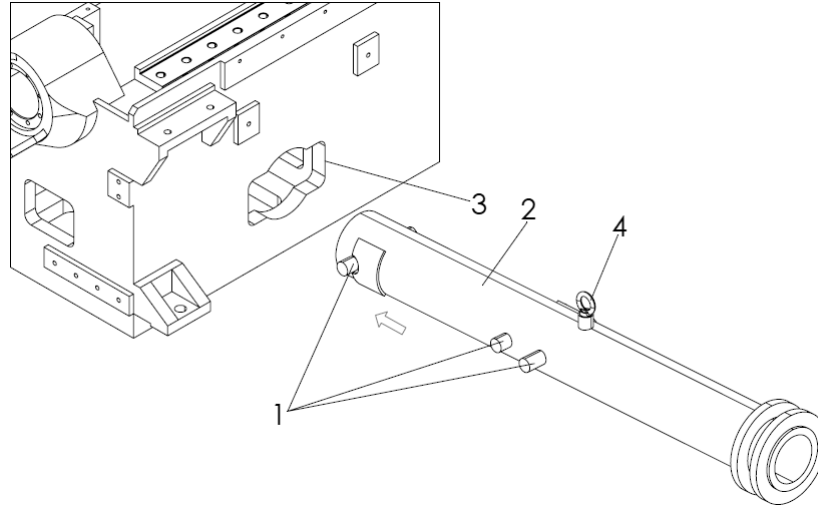


Figure 1.15 Mounting the Hook Rod 1

No.	Name	No.	Name
1	Protrusions	3	Cast hole
2	Hook rod	4	Eyebolt

- 2) Push in the hook rod until the outer wall [1] is positioned between the two protrusions of the hook rod and fitted inside.

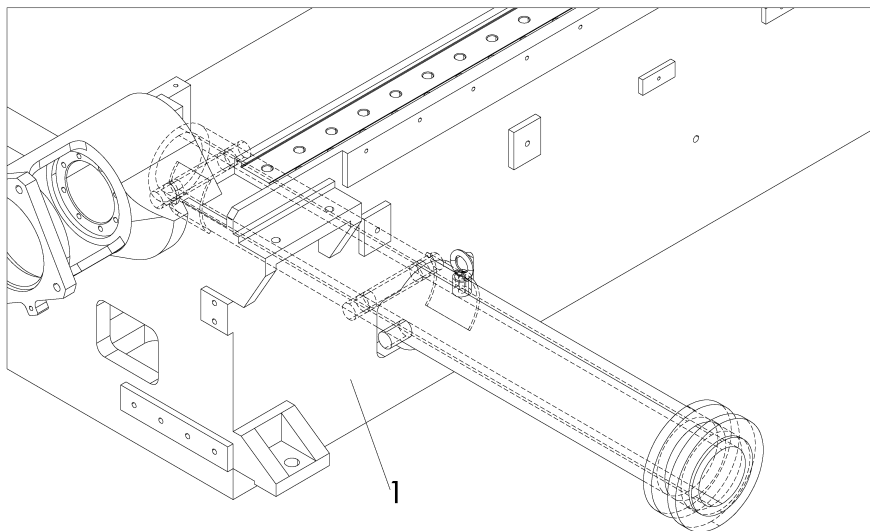


Figure 1.16 Mounting the Hook Rod 2

No.	Name
1	Outer wall

- 3) Turn the hook rod by 90° so that the protrusions are facing upward.

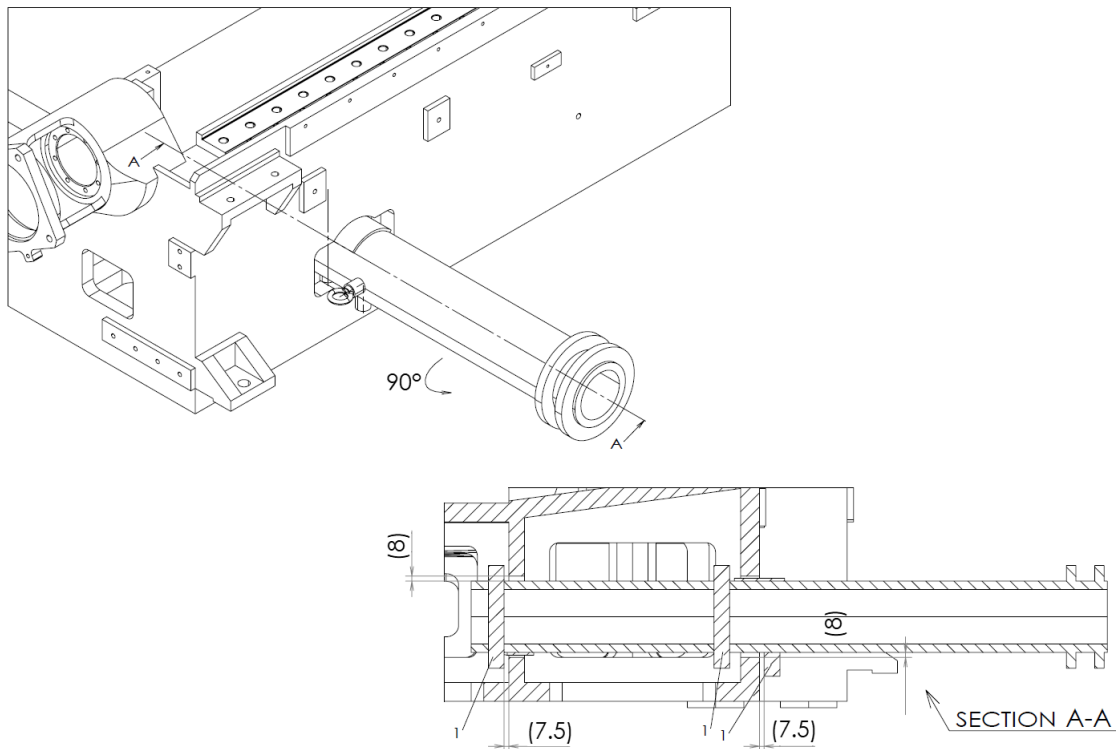


Figure 1.17 Mounting the Hook Rod 3

No.	Name
1	Protrusions

- 4) Mount the rope [2] to the groove at the end of the hook rod [1] . (See cross-sectional view along B-B) Once the hook rod is successfully secured in place, perform the hoisting operation.

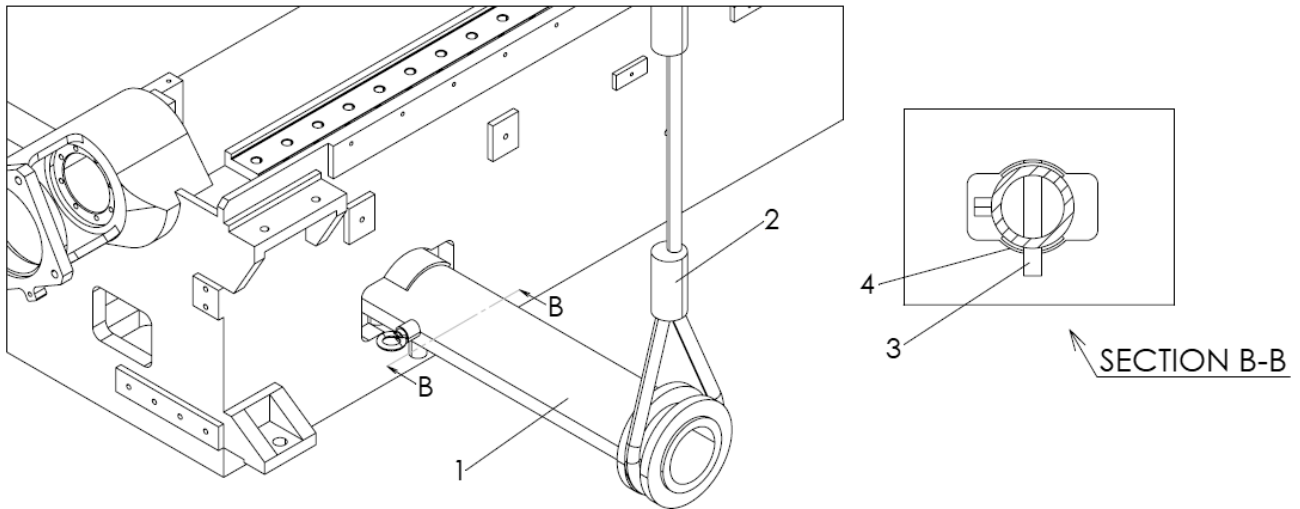


Figure 1.18 Mounting the Hook Rod 4

No.	Name	No.	Name
1	Hook rod	3	Protrusions
2	Rope (Front)	4	Cast hole bottom section

B Adjusting the Wires

- 1) Mount the hoisting attachments to the crane. When mounting to the crane, adjust the length of the adjustment ropes [1] at top of the machine so that the lifting beam is horizontal.

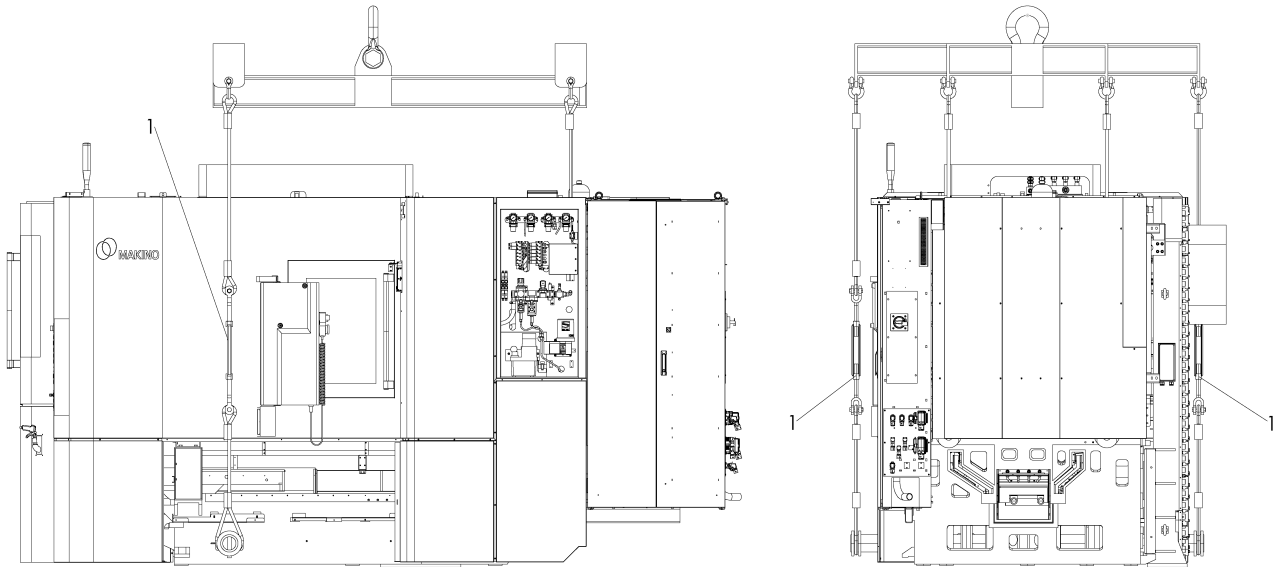


Figure 1.19 Adjusting the Wires

No.	Name
1	Adjustment ropes

C Hoisting the Machine Body

Before hoisting the machine body, check that the bed is not anchored to the floor.

1.8 Hoisting Peripheral Equipment

This section explains how to hoist the peripheral equipment.

1.8.1 Hoisting the Coolant Tank & LUCC



- Assign only qualified workers to slinging work and crane operation. Before the machine body can be hoisted by a crane, it must be confirmed that all the ropes are strong enough for the weight and that the machine body is handled in a well-balanced position. If a heavy load falls due to breaking of a rope or loss of balance during crane operation, the load may be damaged and workers may be seriously injured.

Hoist the equipment with a crane.

- Weight of coolant tank with Drum conveyor: 1,030 kg.

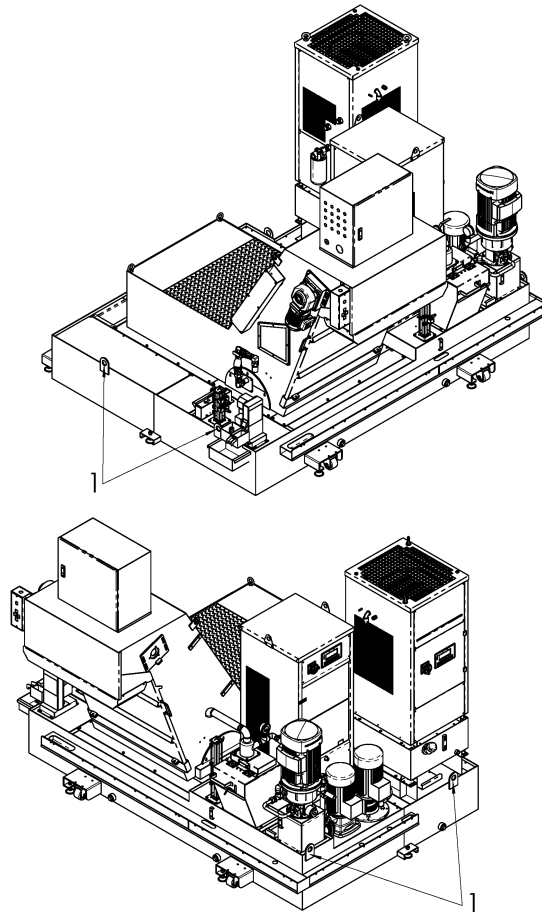


Figure 1.20 Coolant Tank with Drum Conveyor (Option) / Provisional Placement

No.	Name
1	Lifting Point

- Weight of LUCC: 570 kg.

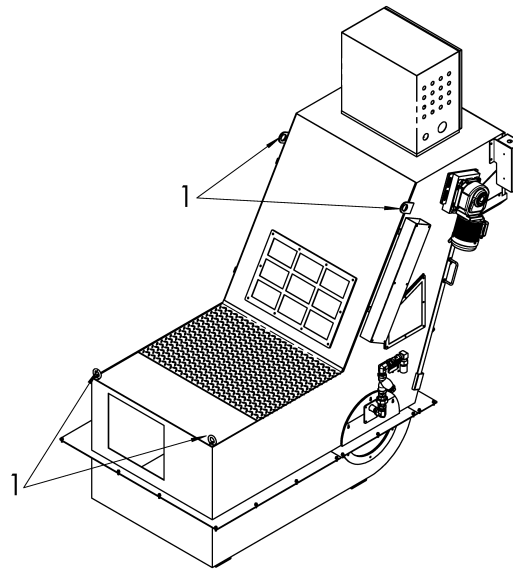


Figure 1.21 Lift Up Chip Conveyor

No.	Name
1	Lifting Point

1.8.2 Hoisting the Coolant Temperature Controller



- Assign only qualified workers to slinging work and crane operation. Before the machine body can be hoisted by a crane, it must be confirmed that all the ropes are strong enough for the weight and that the machine body is handled in a well-balanced position. If a heavy load falls due to breaking of a rope or loss of balance during crane operation, the load may be damaged and workers may be seriously injured.

Hoist the equipment with a crane.

- Weight of Coolant lubricating oil temperature controller: 146 kg.

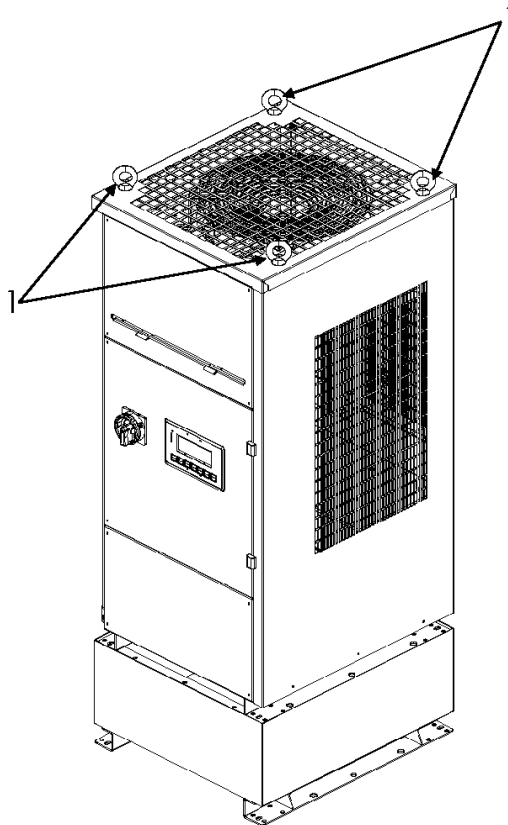


Figure 1.22 Hoisting the Coolant Lubricating Oil Temperature Controller

No.	Name
1	Lifting Point

1.8.3 Hoisting the Air Dryer



- **Assign only qualified workers to slinging work and crane operation. Before the machine body can be hoisted by a crane, it must be confirmed that all the ropes are strong enough for the weight and that the machine body is handled in a well-balanced position. If a heavy load falls due to breaking of a rope or loss of balance during crane operation, the load may be damaged and workers may be seriously injured.**

Hoist the equipment with a crane.

- Weight of air dryer: 30 kg.

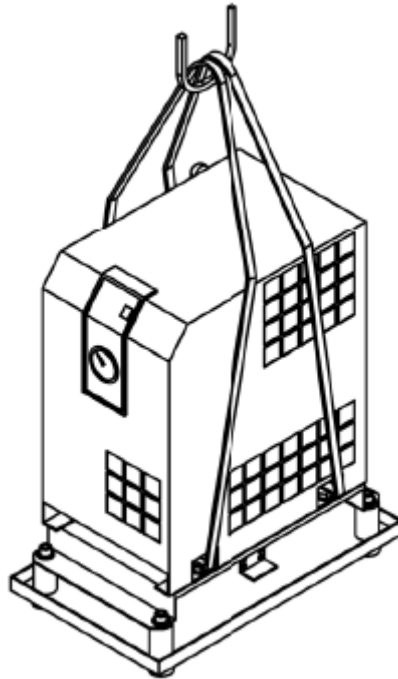


Figure 1.23 Hoisting the Air Dryer

1 Installation Work

1.1 Machine Grouting Method

- 1) Mark the machine positioning location on the floor considering machine front edge/splash guard reference dimensions as shown in [Chapter 2 Figure 1.13](#) .
- 2) Mark the position of holes "a" with reference to the machine location as shown in [Chapter 2 Figure 1.13](#) .
- 3) Install the anchor as shown in installation procedure. (Refer to [Figure 1.1](#) )

Anchor Details

Refer to [Figure 1.1](#)  for anchor locations.

- Name: JA anchor
- Model: JA-M16
- Manufacturer: Japan Drive-It
- Quantity: 10
- Prepared hole: Diameter 22 mm, Depth 65 mm

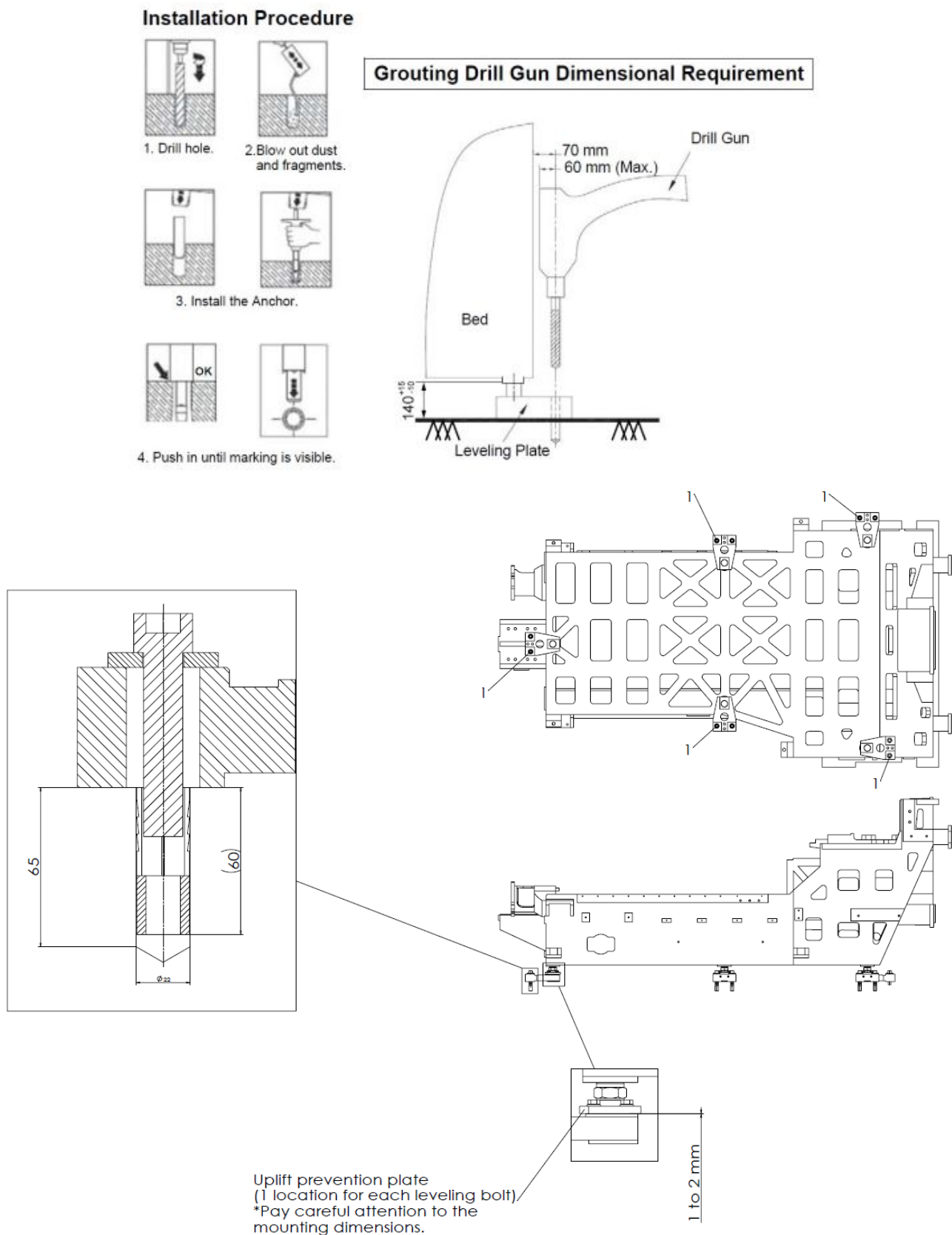


Figure 1.1 Anchor Locations

No.	Name
1	Anchors (2 locations each)

1.2 Installation of Machine Accessory Units

Installing the Accessory Units

If any peripheral devices are also provided as separate units at factory shipping, assemble these devices.

- Coolant tank
- Spindle oil temperature controller
- Coolant temperature controller
- Mist collector

Spindle Oil Temperature Controller Piping

Connect the piping by aligning the alignment marks on the piping.

- Connection piping
 - 1) Cooling oil IN
 - 2) Cooling oil OUT

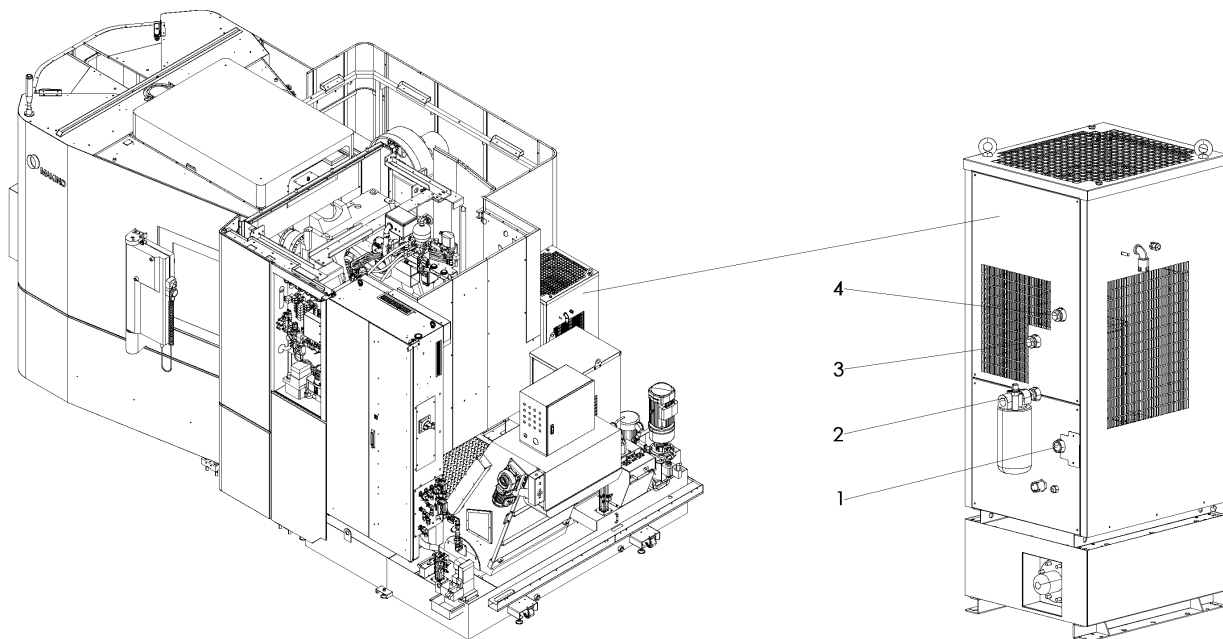


Figure 1.2 Spindle Oil Temperature Controller Piping

No.	Name	No.	Name
1	Main spindle oil supply port Rc 3/4"	3	Main spindle oil return port Rc 3/4"
2	Screw oil supply port Rc 3/4"	4	Screw oil return port Rc 3/4"

Connecting the Coolant Tank

Insert the trough [2] into the trough connection port [1] of the coolant tank, and connect the coolant tank.

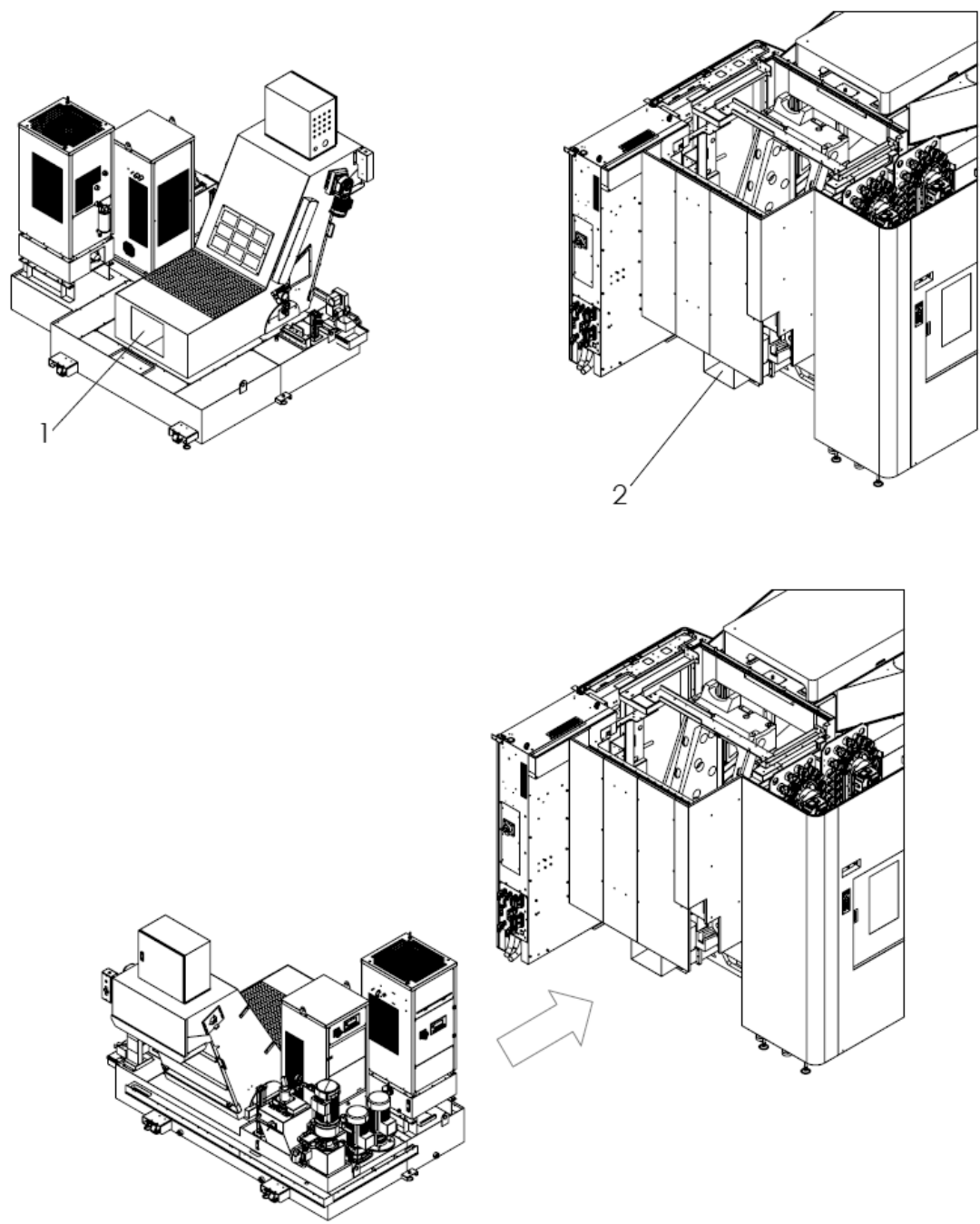


Figure 1.3 Coolant Tank Connections

No.	Name	No.	Name
1	Trough connection port	2	Trough

Coolant Piping

Connect the piping by aligning the alignment marks on the piping.

- Connection piping
 - 1) Piping from main pump: 2-inch union connection
 - 2) Through-spindle coolant hose: 3/8-inch hose connection
 - 3) Spiral cleaning coolant hose: 3/4-inch hose connection
 - 4) Spiral cleaning valve pneumatic tube: $\varnothing$ 6 mm

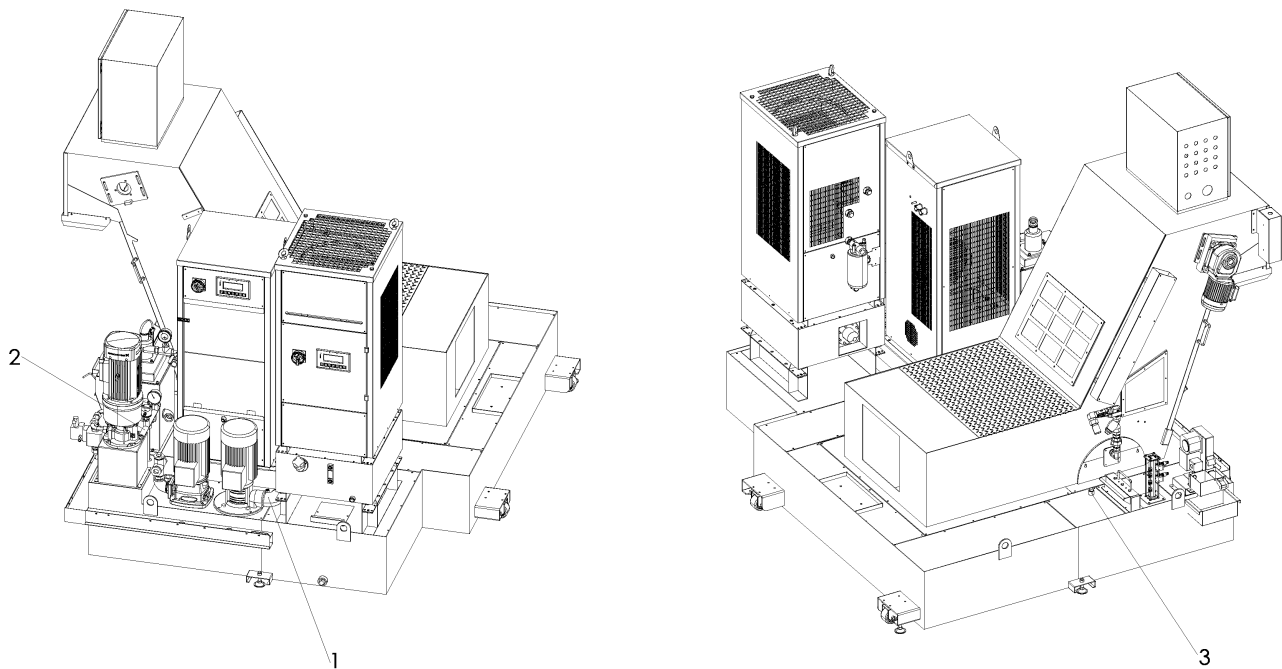


Figure 1.4 Coolant Piping Locations (Coolant Tank Side)

No.	Name	No.	Name
1	Main pump	3	Through-spindle coolant drain
2	Through-spindle coolant	-	-

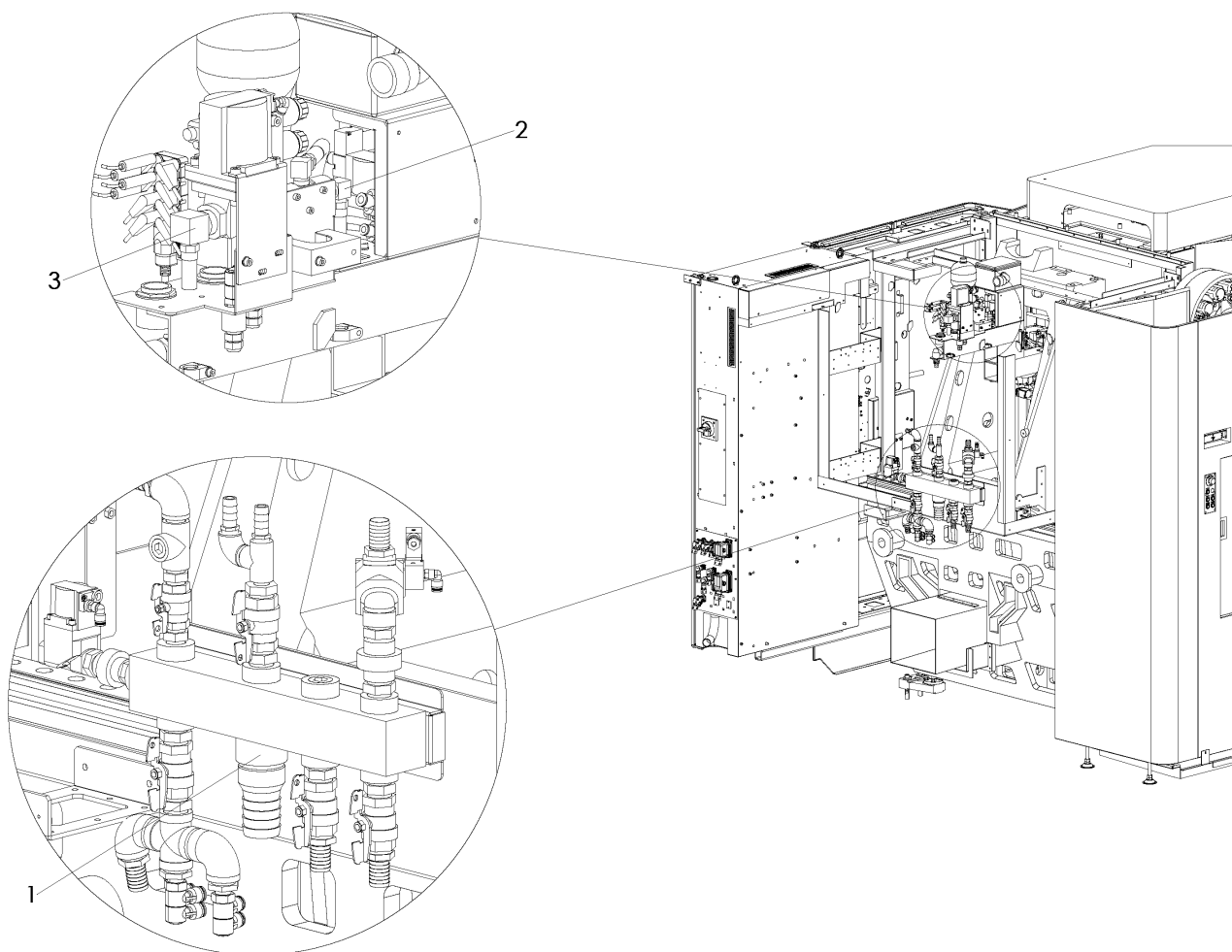


Figure 1.5 Coolant Piping Locations (Machine Side)

No.	Name	No.	Name
1	Main pump	3	Through-spindle coolant drain
2	Through-spindle coolant	-	-

Pneumatic Piping

Connect the air piping from the plant air source to the air supply port (Rc3/8) [1].

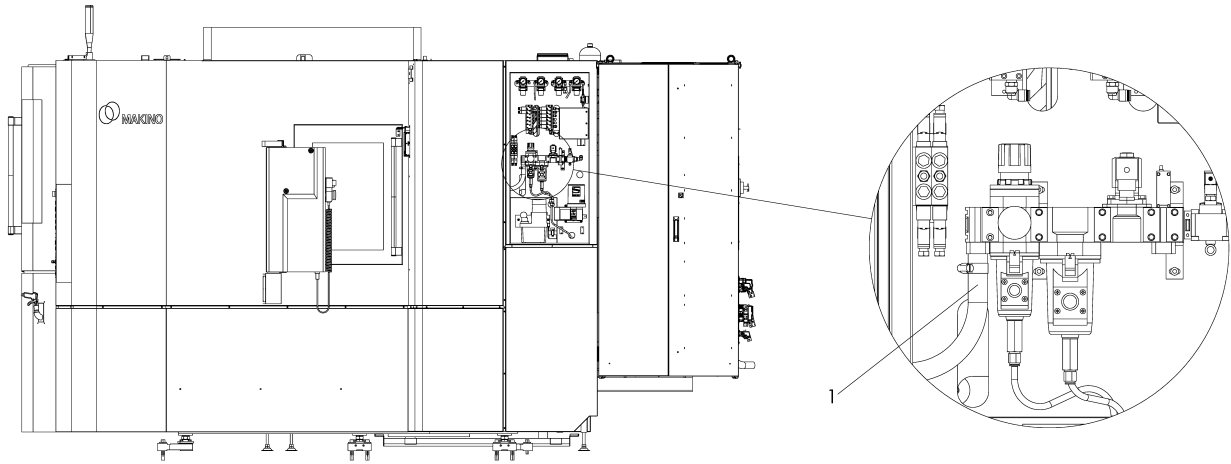


Figure 1.6 Pneumatic Piping Locations

No.	Name
1	Air supply port (Rc3/8)

Supplying Oil After Installation

If this machine will be shipped outside Singapore, be sure to drain all the hydraulic oil and cooling oil before shipment. Supply oil up to the upper limit level on the level gauge [2] before turning ON the power.

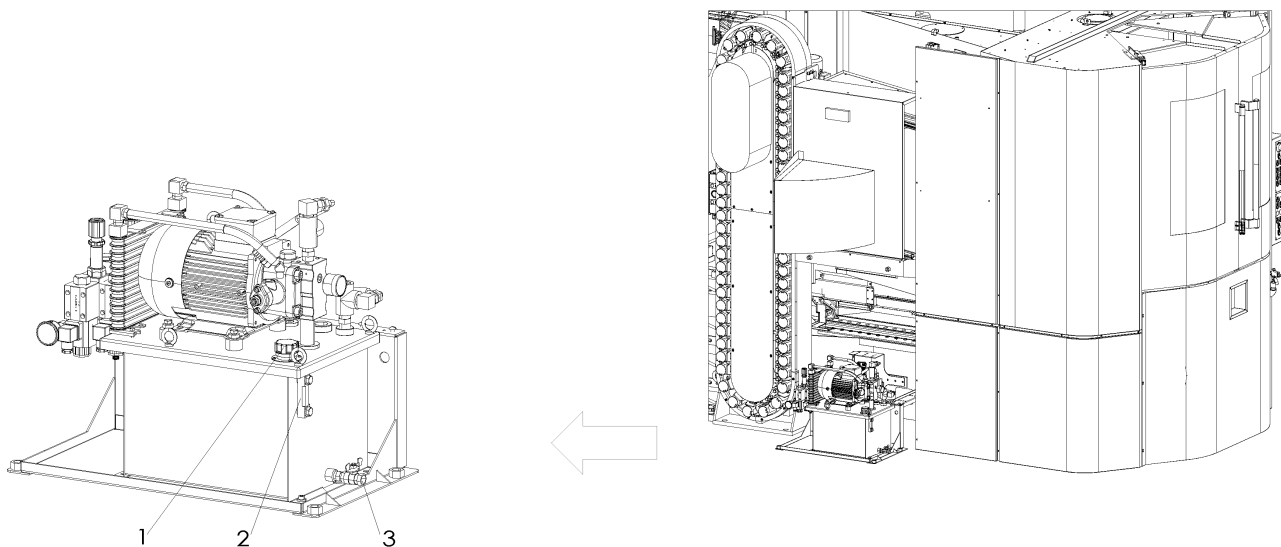


Figure 1.7 Supplying with Hydraulic Oil

No.	Name	No.	Name
1	Fill port	3	Drain port (Rc 3/8")
2	Level gauge		

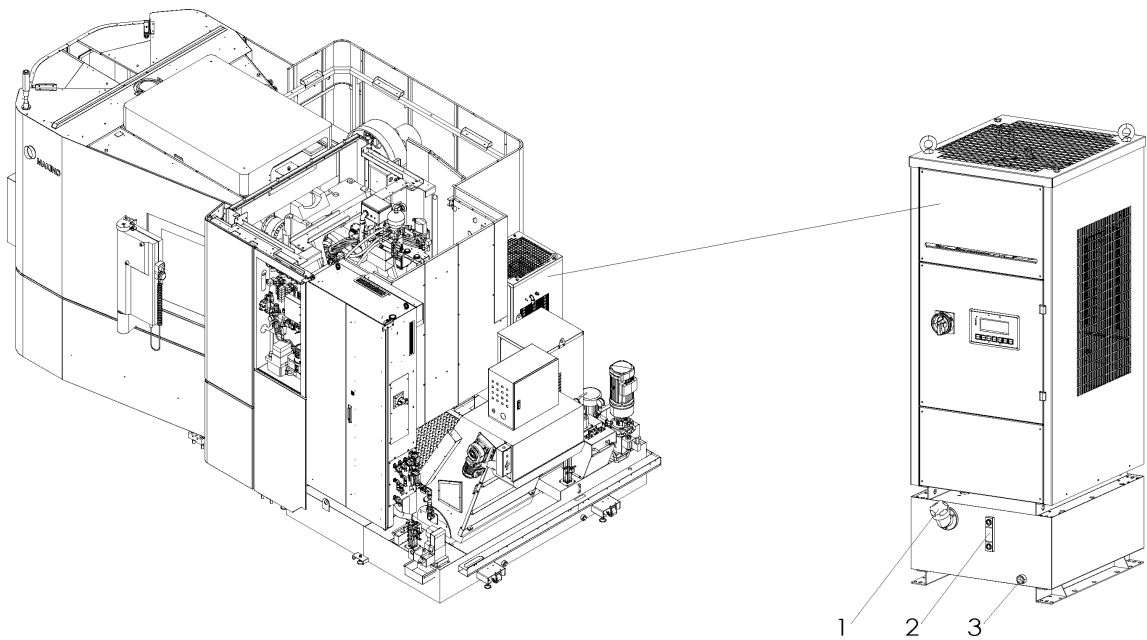


Figure 1.8 Supplying with Spindle and Screw Cooling Oil

No.	Name	No.	Name
1	Fill port	3	Drain port (G1/2")
2	Level gauge	-	-

1.3 Connecting Wiring to Machine

This section explains how to connect the main power supply from the factory and connect the ground to the machine body. (Refer to [Figure 1.9](#)^[70] and [1.10](#)^[71])

- 1) Confirm that the "Main Power" switch is turned off.
- 2) Feed the wiring for the power cables through the top or bottom of the Machine Tool Cabinet and connect it to the main power circuit breaker.
 - (a) Remove the mounting screws [2] and the "Main Power" switch cover [1].
 - (b) Connect the ground line to the ground connection ports [5].
 - (c) Connect the power cables [3] to the terminals of the power connection ports [4].
- 3) Check the Power source phase rotation with a phase rotation indicator. Confirm the phase connection by checking the Coolant pump fan (M2) [6] rotation. It must be rotating in counterclockwise direction.
- 4) Check the voltage is within the specification: 400 VAC $\pm$ 10%, 50Hz/60Hz.
- 5) Secure the power cable along the routing route with cable ties to prevent movement.
- 6) Mount the "Main Power" switch cover [1] and secure it with the mounting screws [2].

This completes connection of the wiring.

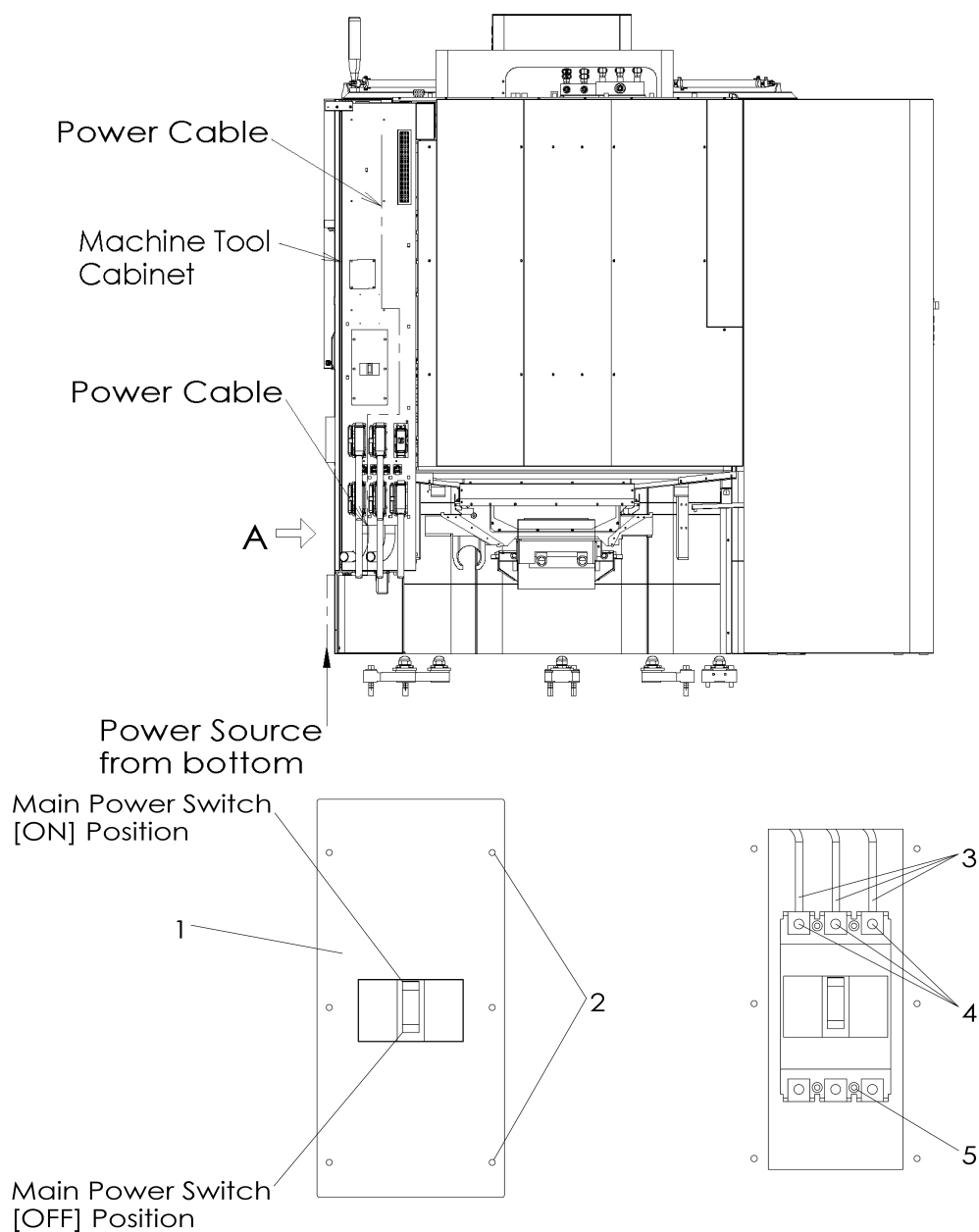


Figure 1.9 Connecting the Wiring

No.	Name	No.	Name
1	Cover	4	Power Connection Ports
2	Mounting Screws	5	Power Connection Ports
3	Power Cable	-	-

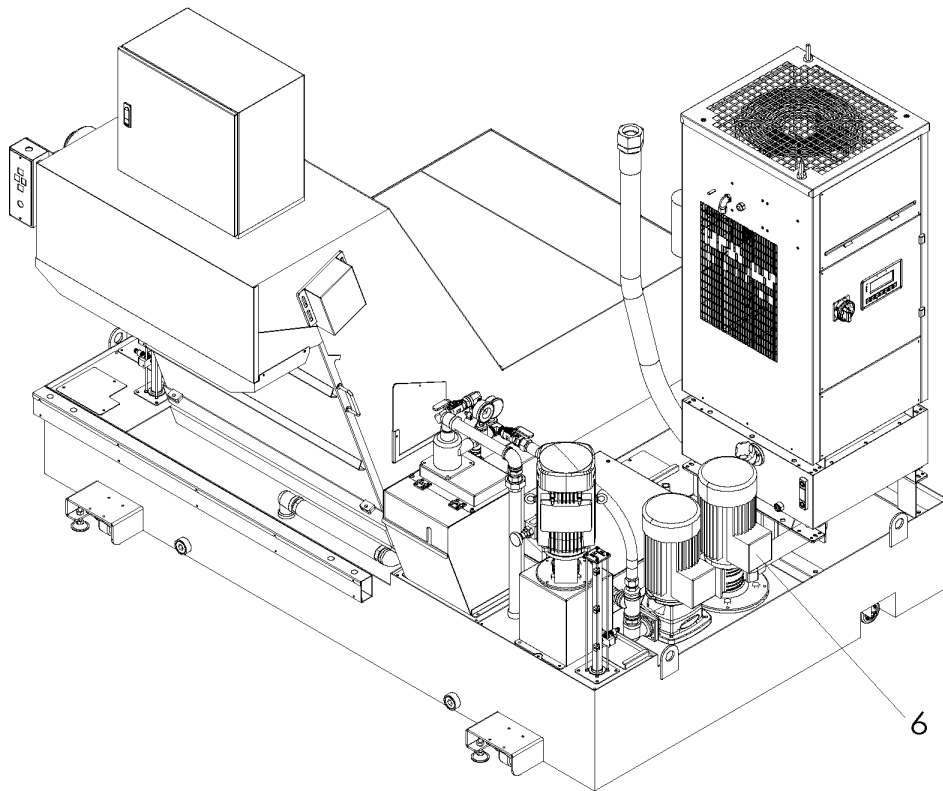


Figure 1.10 Check Phase Rotation

No.	Name
6	Coolant Pump Fan (M2)

1.4 Wiring Connection of Peripheral Devices

Electrical Wiring

Connect the power cables of the peripheral devices to the connectors on the side of the machine controller.

- Coolant tank power supply
- Coolant tank signal
- Spindle oil temperature controller

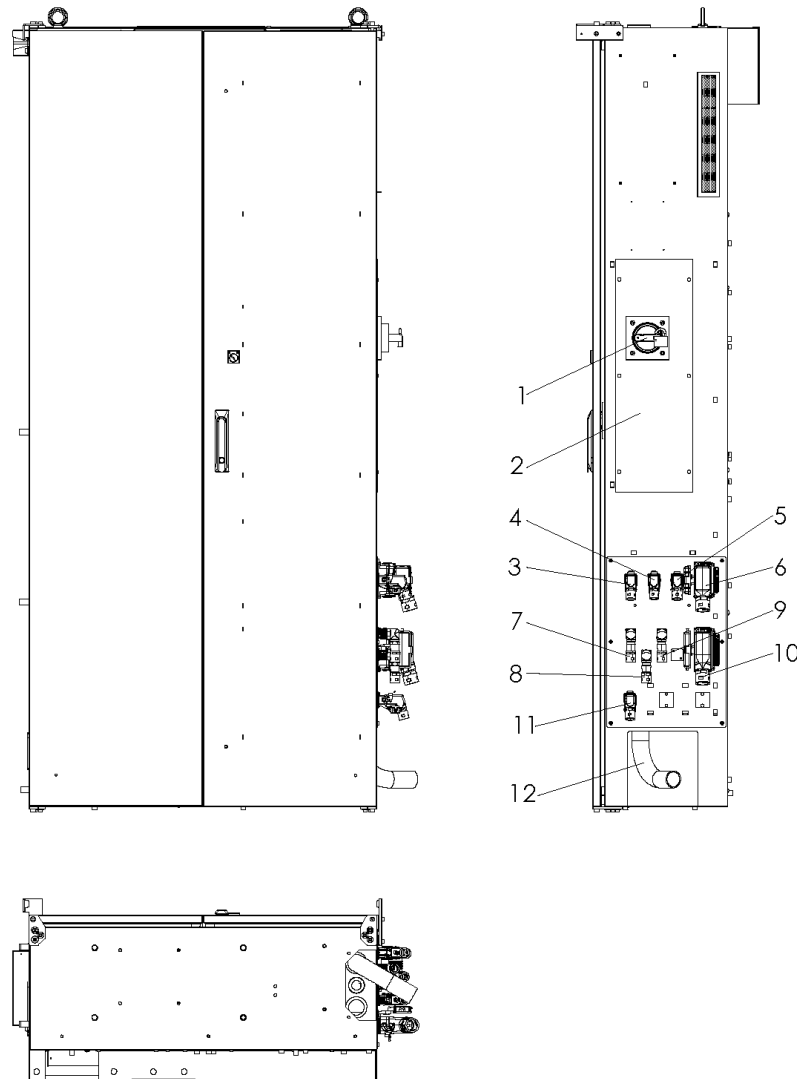


Figure 1.11 Machine Controller

No.	Name	No.	Name
1	Main circuit breaker switch	7	Fixture 1 SOL/PS
2	Main circuit breaker cover	8	Air check
3	Spindle oil temperature controller connector	9	Fixture 2 SOL/PS
4	Air Dryer	10	CPH Power
5	Coolant tank power connector	11	Robot Interface
6	Coolant tank signal controller	12	Main Power Cable

1.5 Piping and Wiring Connection of Mist Collector

Mist Collector are options. If available, connect the piping and wiring from the machine to the Mist Collector as shown in [Figure 1.12](#).

Piping: Duct and Drain (Mist Collector)

Wiring: Power Cable (Mist Collector)

- When the piping and wiring have been completed, place the Mist Collector following the layout and then lock the casters beneath the stand.
- Be careful that the piping, hoses and cables on the machine side are not subjected to pressure.

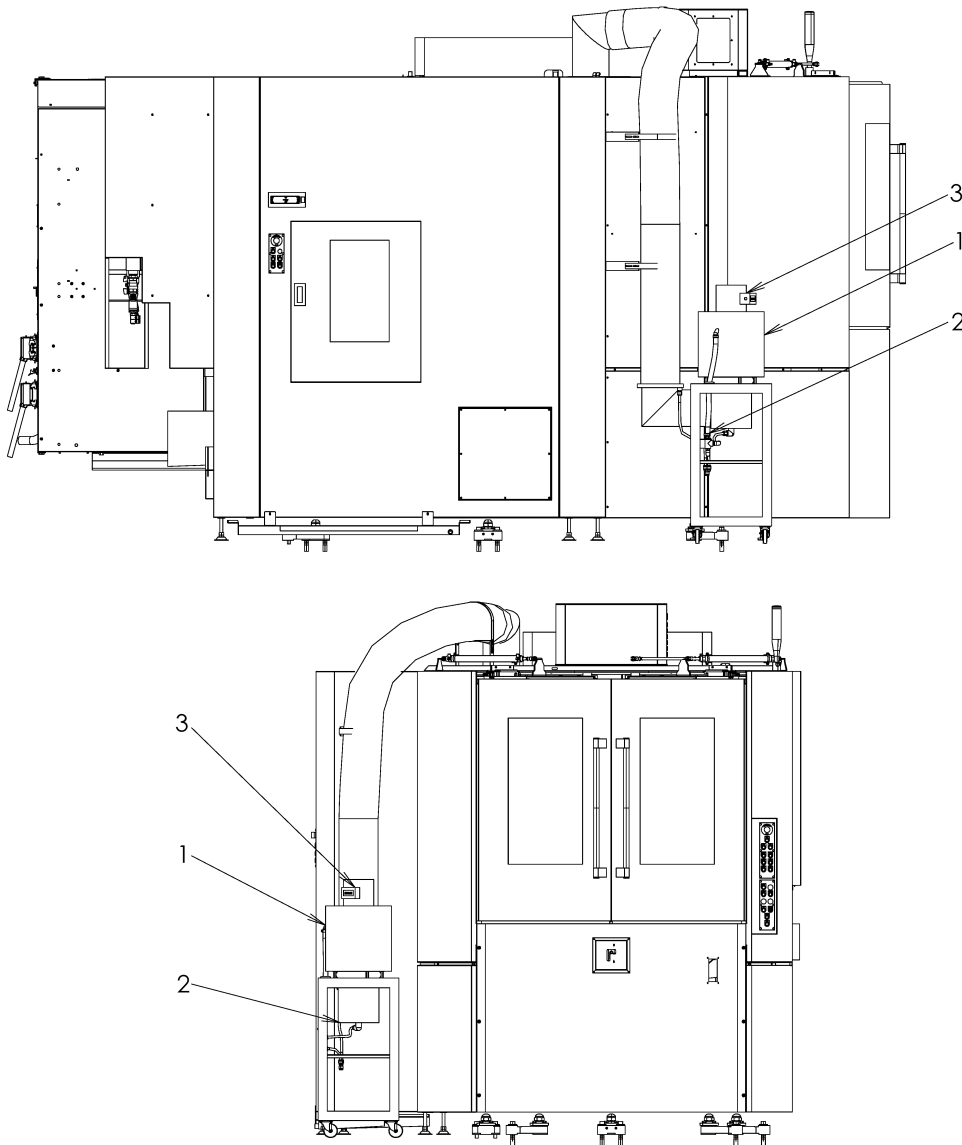


Figure 1.12 Mist Collector (Option) on Standard Machine Piping/Wiring

No.	Name	No.	Name
1	Mist Collector	3	Mist Collector Cable
2	Mist Collector Drainage	-	-

1.6 Piping and Wiring Connection of Air Dryer

Air Dryer are option. If available, connect the Air tube to the air regulator and wiring from the Electrical cabinet (MTC) to the Air dryer as shown in [Figure 1.13](#)⁷⁴.

- 1) Connect the air dryer cable [2] to the MTC [3]. (Refer to [1.4 Wiring Connection of Peripheral Devices](#)⁷²)
- 2) Connect main air supply hose to main air inlet [4].

This completes the connection of piping and wiring for the air dryer.

No.	Air Dryer working condition
1	Working Fluid : Compressed Air
2	Inlet Air Temperature : 5 ~ 50°C
3	Inlet Air Pressure : 0.15 ~ 1.0Mpa
4	Ambient Temperature : 2 ~ 40°C (Relative Humidity of 85% or less)

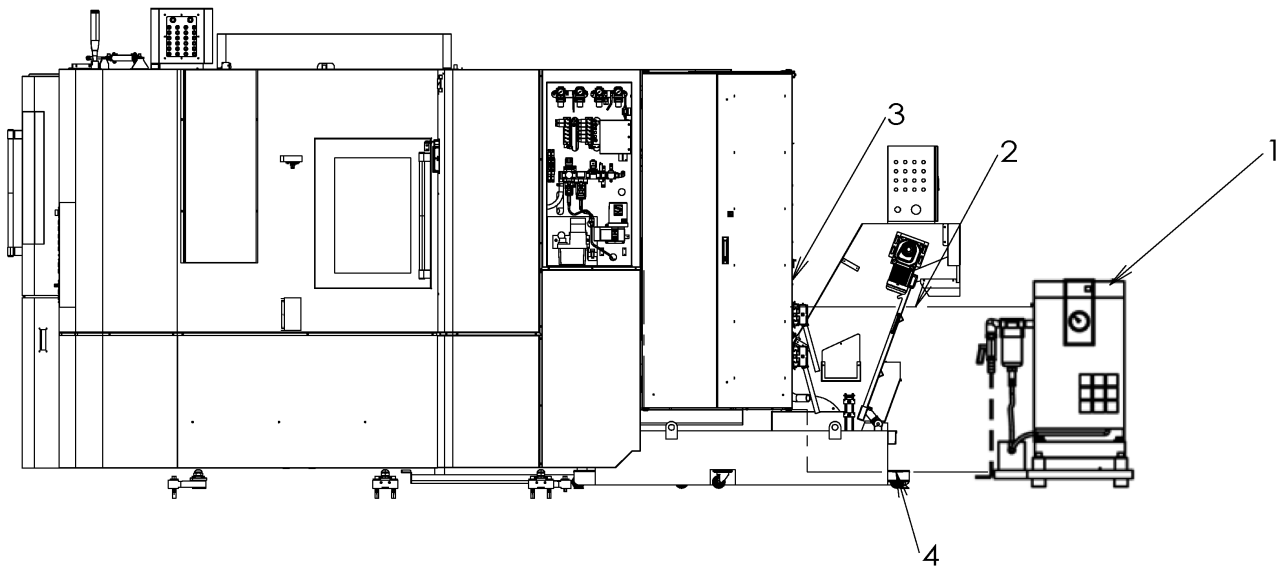


Figure 1.13 Installing the Air Dryer

No.	Name	No.	Name
1	Air Dryer	3	MTC
2	Air Dryer Cable	4	Main Air Inlet

1.7 Connecting Power Supply and Air Source to Machine



- Read the "SAFETY" chapter thoroughly and ensure adequate precautions are taken before performing any work operations.
- When performing any work, be sure to place a warning tag on each operation panel, main power switch, and other locations to ensure that other personnel do not inadvertently operate the machine (Refer to [Chap. 1 Safety 1.6 Implementing Lock-Out/Tag-Out](#).)

1.7.1 Connecting Air Source to Machine

This section explains how to connect the pneumatic piping from the factory to the machine body.

(Refer to [Figure 1.14](#) and [1.15](#))

- 1) Connect the pneumatic piping to the air source connection port [1].

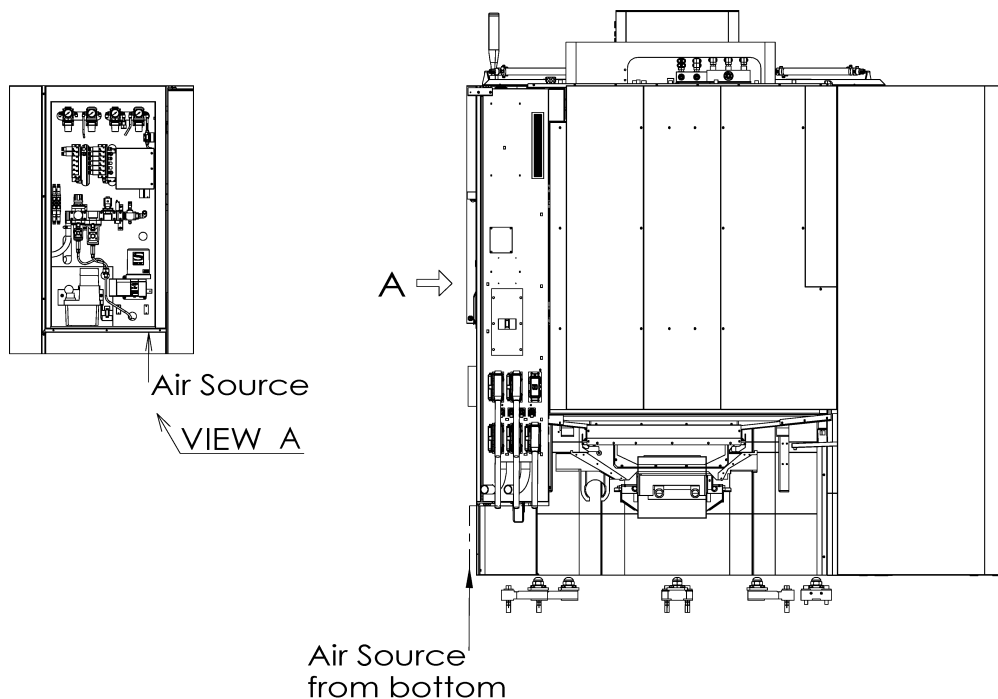


Figure 1.14 Connecting the Pneumatic Piping

- 2) With Air Dryer: Connect the air source hose to the Air Dryer inlet and connect another hose from the Air Dyer outlet to the air source inlet.

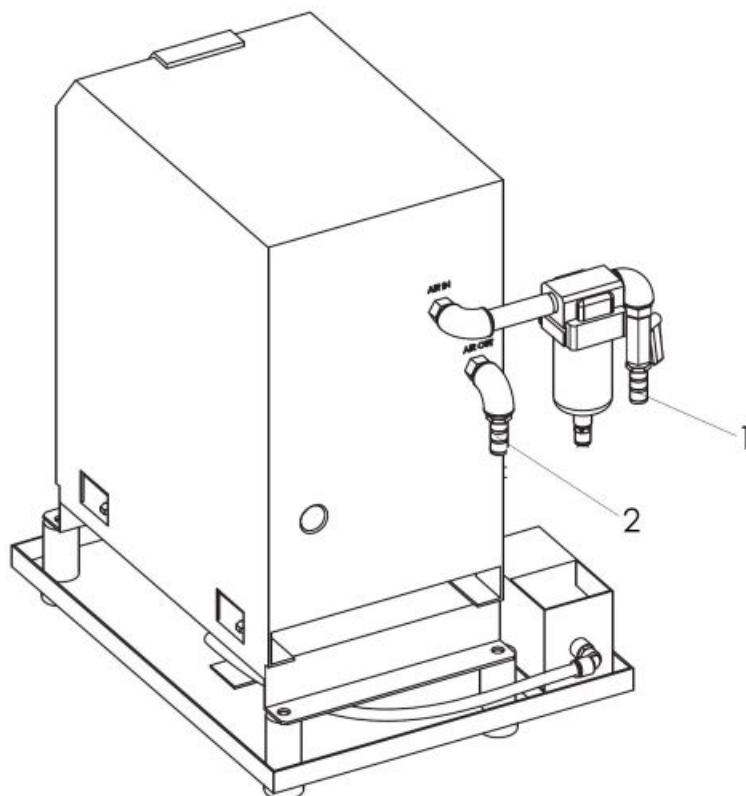


Figure 1.15 Connecting the Air dryer Pneumatic Piping

No.	Name	No.	Name
1	Inlet	2	Outlet

This completes the connection of the pneumatic piping.

1.8 Lubrication

This section explains how to supply lubricant and cutting fluid and confirm the oil levels.

1.8.1 Lubricant and Hydraulic Oil List

Table 1.1  lists the recommended lubricants and hydraulic oils.



- Obtain an SDS (Safety Data Sheet) from the supplier of the lubricant or hydraulic oil, and handle the lubricant or hydraulic oil according to the contents of the material safety data sheet. The SDS contains information about health and safety, and emergency responses.
- When handling oils, cutting fluids, and other chemical substances, wear protective gloves and safety goggles according to the instructions in the SDS.
- Lubricants and hydraulic oils affect the precision and performance of the machine. Therefore, be sure to use the oils recommended by Makino. If using lubricant or hydraulic oil other than those recommended by Makino, contact Makino service representatives.
- Among the recommended oils in "[Table 1.1 Recommended Lubricants and Hydraulic Oils](#) ". If using lubricating oil or hydraulic oil other than those recommended by Makino, the original lubricant oil must be drained completely from the tank before refilling.

Table 1.1 Recommended Lubricants and Hydraulic Oils

Peripheral Device	Recommended Oil/Grease/Coolant
Automatic Grease Supply Unit	Lube Corporation LHL-X100 cartridge
Spindle Temperature Controller	Mobil Velocite Oil No. 3
Spindle Hydraulic Power Pack	ISO3448 VG32
Fixture Hydraulic Power Pack (Option)	(Castrol Hyspin AWS32, Mobil DTE24)

1.8.2 Automatic Grease Supply Unit

Automatic Grease Supply unit supplies grease to all axis Linear motion guide and Ballscrew to keep them lubricant at all time. Replace the Grease cartridge before is empty.

Lubricating Grease: LHL-X100 cartridge

Manufacturer: Lube Corporation

Makino part number: Z192A1600010 (700mL)

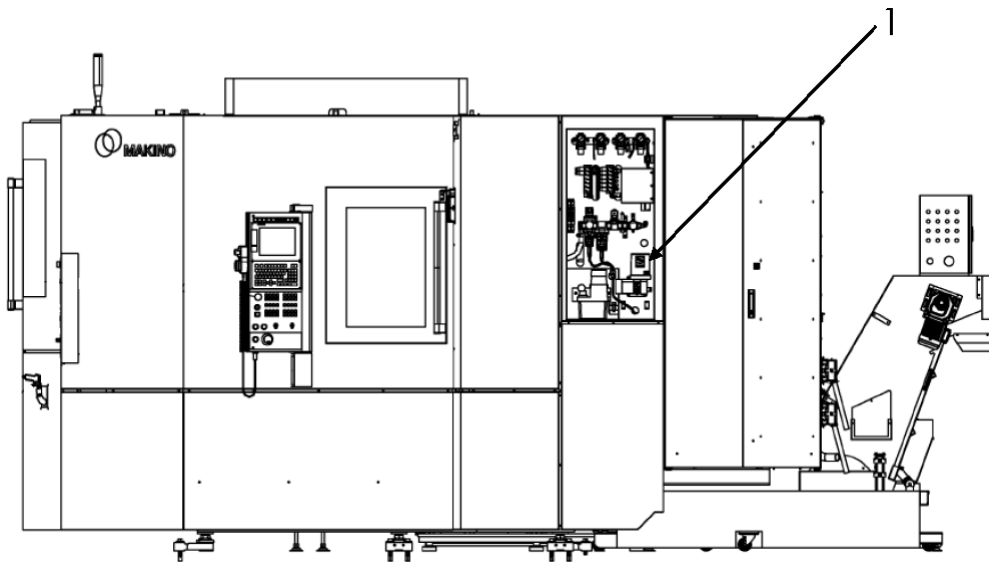


Figure 1.16 Automatic Grease Unit

No.	Name
1	Cartridge

1.8.3 Spindle Oil Temperature Controller

Coolant circulate inside the spindle head to cool down the spindle during operation so that it does not get too hot. Check the oil level with the level gauge [2] during installation.

Lubricating oil: Mobil Velocite Oil No. 3

Manufacturer: Mobil

Makino part number: Z479Z-0003-00 (20 liters)

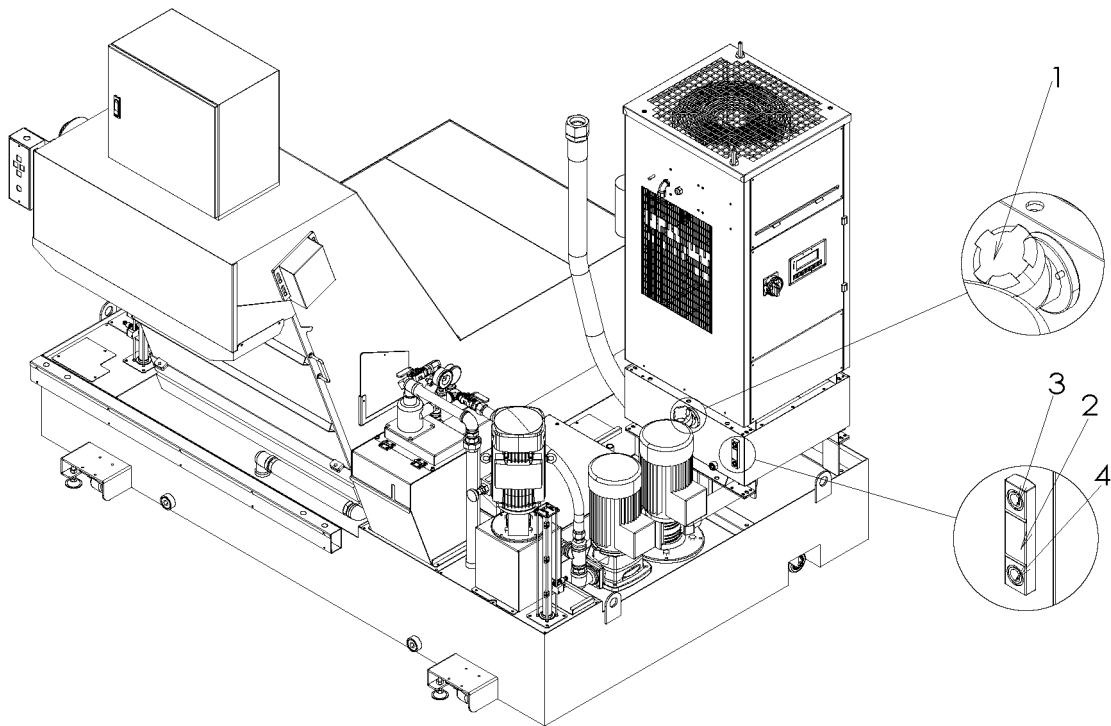


Figure 1.17 Spindle Cooler

No.	Name	No.	Name
1	Oil Fill Port	3	Upper Limit
2	Coolant Level Gauge	4	Lower Limit

1.8.4 Hydraulic Unit

Tool clamping/unclamping are performed by hydraulic pressure supplied by the hydraulic unit. During installation, inspect that the oil level is between the level gauge [1] marks.

Hydraulic Oil: Castrol Hyspin AWS 32, Mobil DTE 24 (VG 32)

Manufacturer: Castrol/Mobil

Procurement number: Z479A0101012 (20 liters)

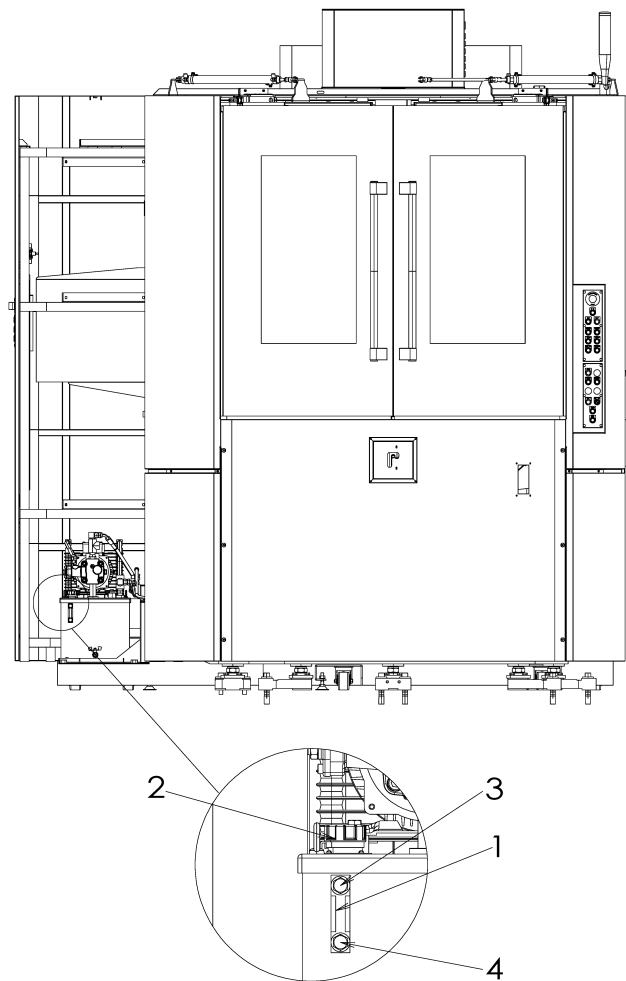


Figure 1.18 Spindle Hydraulic Power Pack Unit

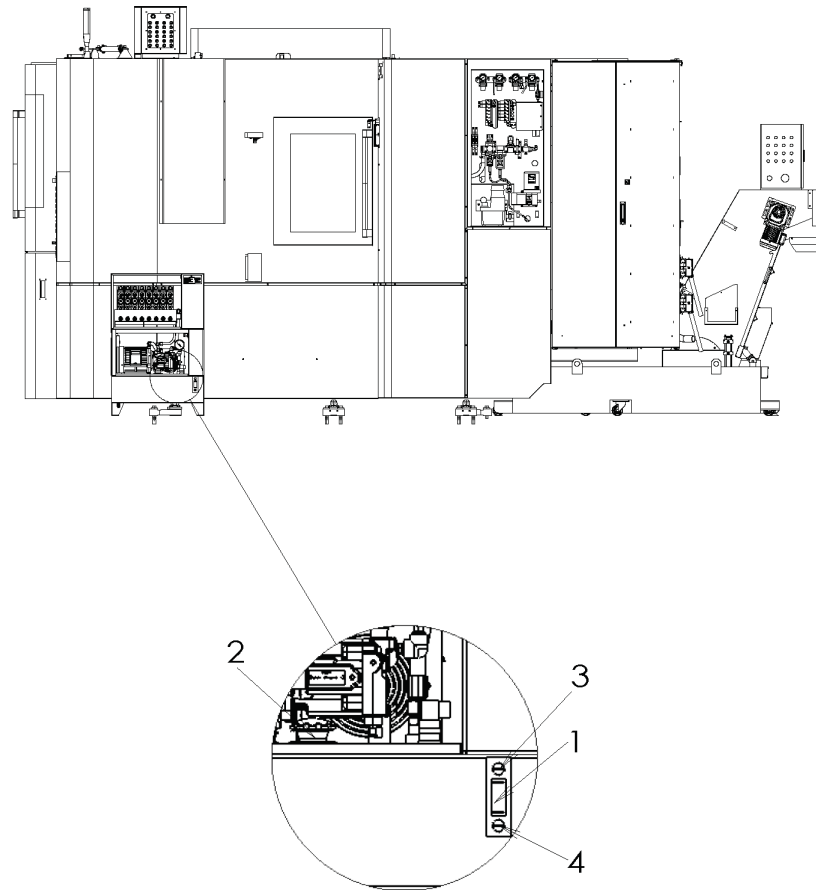


Figure 1.19 Hydraulic Fixture Power Pack Unit

No.	Name	No.	Name
1	Coolant Level Gauge	3	Upper Limit
2	Oil Fill Port	4	Lower Limit

2 Work after Power Connection

2.1 Removal of Fixtures

Removing the Fixing Brackets (Axis Locks)



WARNING

- Before starting this work, confirm that the machine power is turned ON and that all machine operations are stopped.
- This procedure requires the operator to enter inside the machine, and so careful attention is needed during the work.
- Be sure to place a warning tag on each operation panel, main power switch, and other locations when performing work to ensure that other personnel do not turn ON the power inadvertently which could result in the machine being operated.
- When operating the machine, maintain clear communication with everyone in the area to ensure operator safety.



WARNING

- If you attempt to move a feed axis without removing the fixing bracket, the machine could be damaged.

The fixing brackets are required when relocating the machine. After removing them, store them for future use.

X-axis

- 1) Remove the washer [4] and socket head cap screw [5] and remove the fixing bracket [3] (yellow painted part).

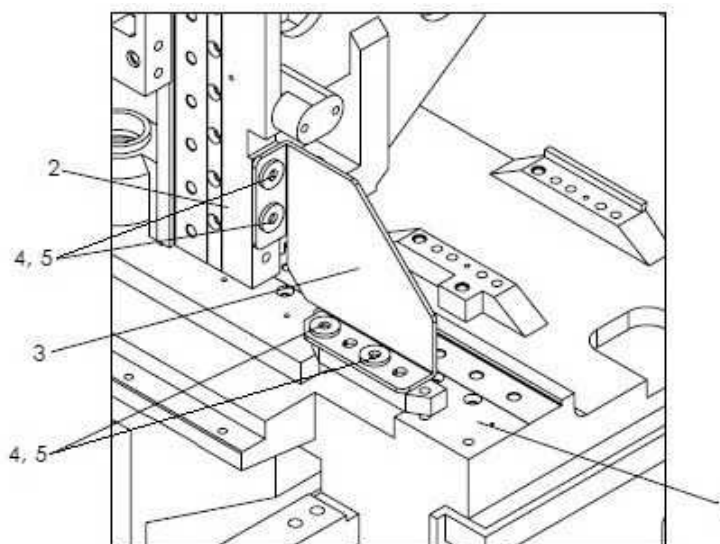


Figure 2.1 Fixing Bracket (X-axis)

No.	Name	No.	Name
1	Bed	4	4x Washer DIN6340-13
2	Column	5	Socket head cap screw (M12x35)
3	Fixing bracket	-	-

Y-axis

- 1) Turn ON the power.
- 2) Remove the washers [5] and socket head screws [6] to remove the fixing bracket [3] [4].

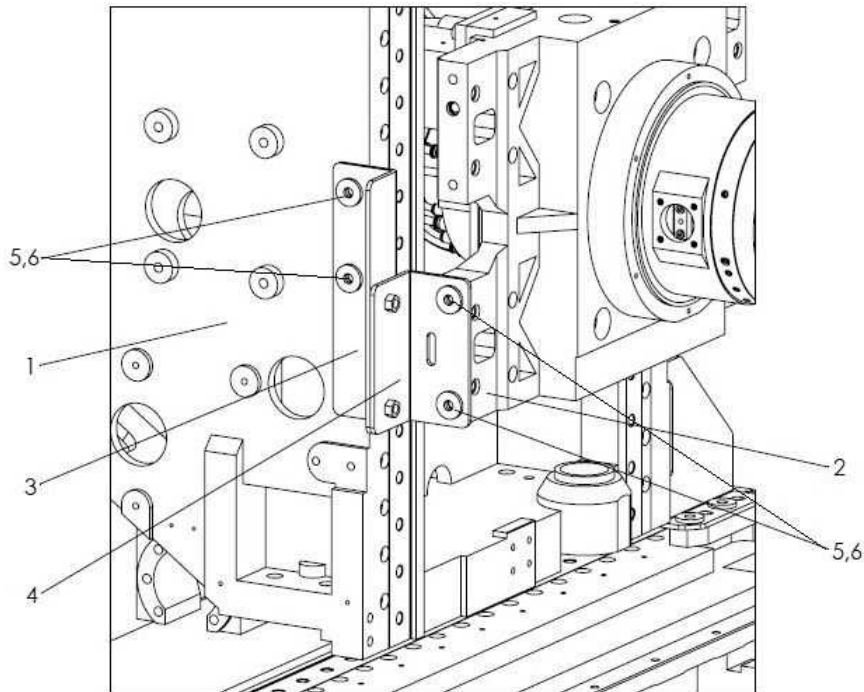


Figure 2.2 Fixing Bracket (Y-axis)

No.	Name	No.	Name
1	Column	4	Fixing Bracket, Housing
2	Housing, Spindle	5	4x WASHER DIN6340-13
3	Fixing Bracket, Column	6	4x Socket Head Cap Screw (M12x35)

Z-axis

- 1) Remove the washer [4] and socket head cap screw [5] mounted to the fixing bracket [3] (yellow painted part).

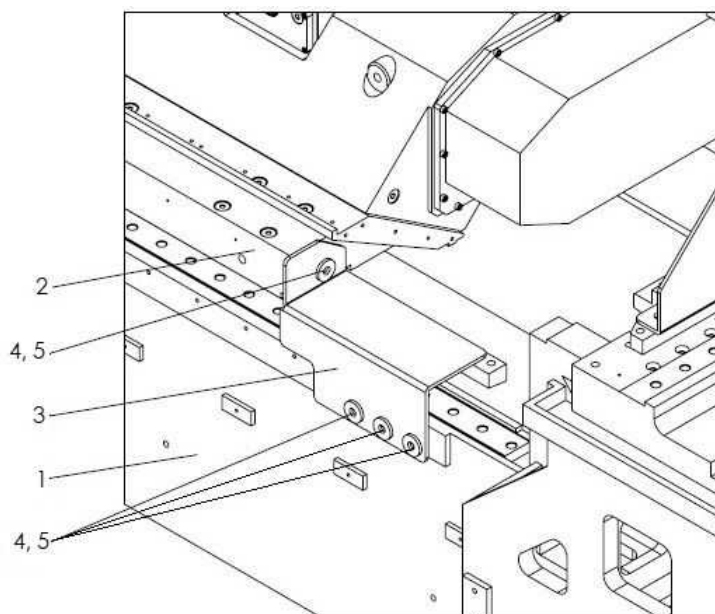


Figure 2.3 Fixing Bracket (Z-axis)

No.	Name	No.	Name
1	Bed	4	4x Washer DIN6340-13
2	Slide	5	4x Socket head cap screw (M12x35)
3	Fixing bracket	-	-

2.2 Level Adjustment

Adjust the machine installation level.

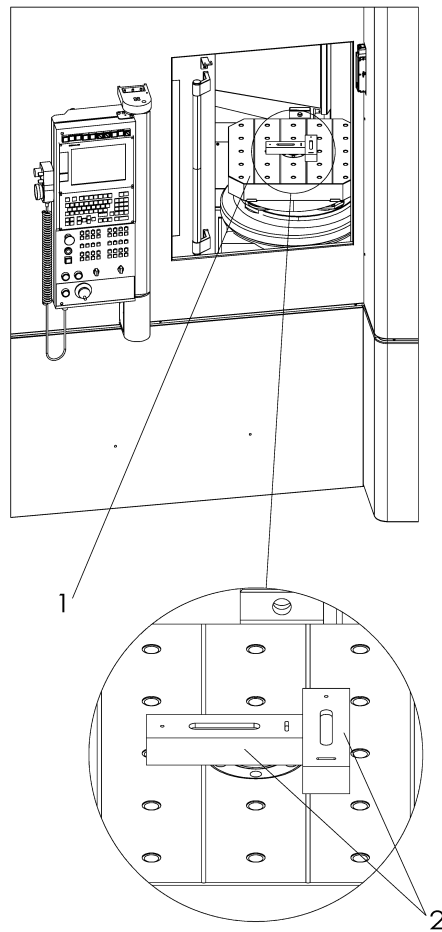


Figure 2.4 Machine Accuracy Checking

No.	Name	No.	Name
1	Table	2	Level Gauge

- 1) Move the X-axis, Y-axis and Z-axis to the center of stroke.
- 2) Place two level gauges [2] on the table. One level gauge on the X-axis direction another on Y-axis direction.
- 3) If the leveling bubble is not centered in each level gauge, perform the adjustment procedure below.

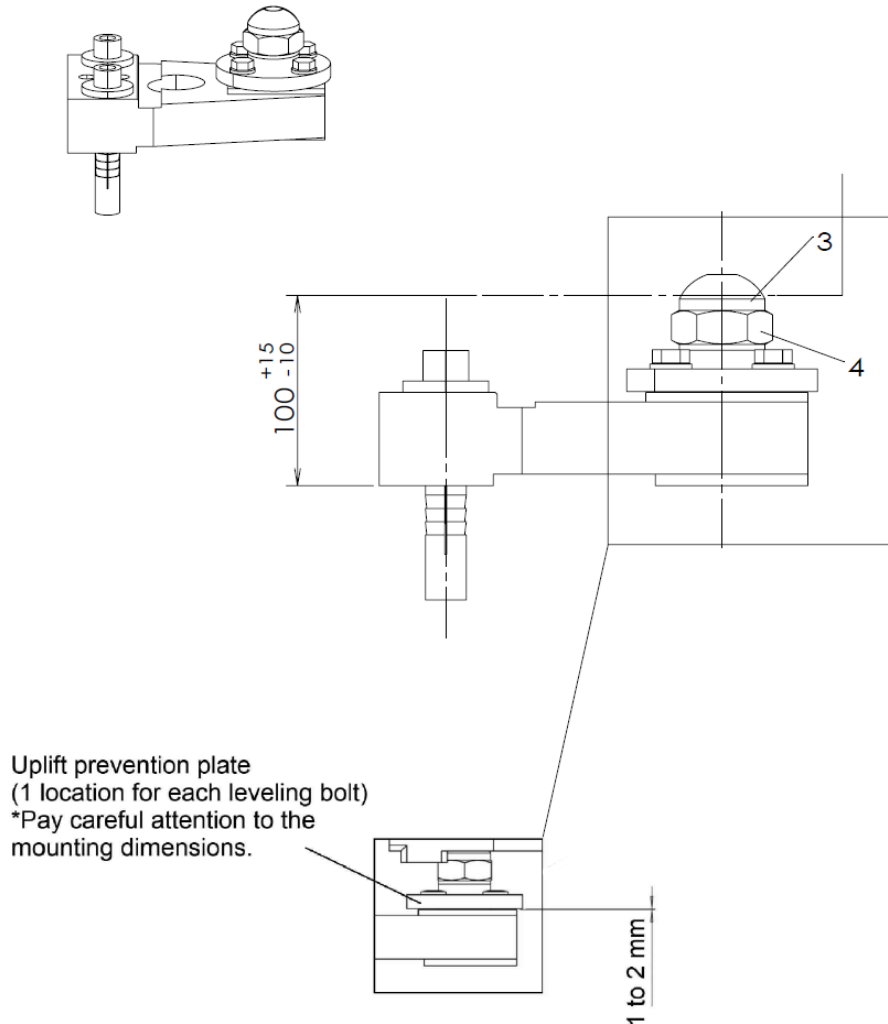


Figure 2.5 Components for Level Adjustment

No.	Name	No.	Name
3	Leveling Bolt	4	Lock Nut

- a) Loosen the set screw [4] of the leveling plate to be adjusted.
- b) Adjust the leveling bolt [3] so that the leveling bubble is centered in each level.



- **Loads must be applied equally to each leveling bolt. Uneven loads can cause distortion of the bed.**

- c) Tighten the set screw [4].
- 4) Use the portable manual pulse generator to move the X-axis and Y-axis position to both ends of the stroke and check that the level gauge bubble position is still centered.
- 5) Ensure all bolts and nuts fully tightened and marked.



- **Operation with the operator doors open is very dangerous. Use extreme care when operating the machine.**

After leveling adjustment is completed, then proceed the static accuracy check, following "MAKINO customer service's installation check sheet ".

2.3 Tool Magazine Adjustment

- 1) Use the ATC-spindle gap tool to verify the clearance between the ATC arm and spindle.
- 2) Adjust the clearance by repositioning the ATC unit using the ATC push-pull block.
- 3) Verify the ATC tool change position.
- 4) Install the tool magazine level pad; adjust the bolt to achieve contact.
- 5) Tighten the nut below ATC with Hand tight is sufficient. No specific torque required.

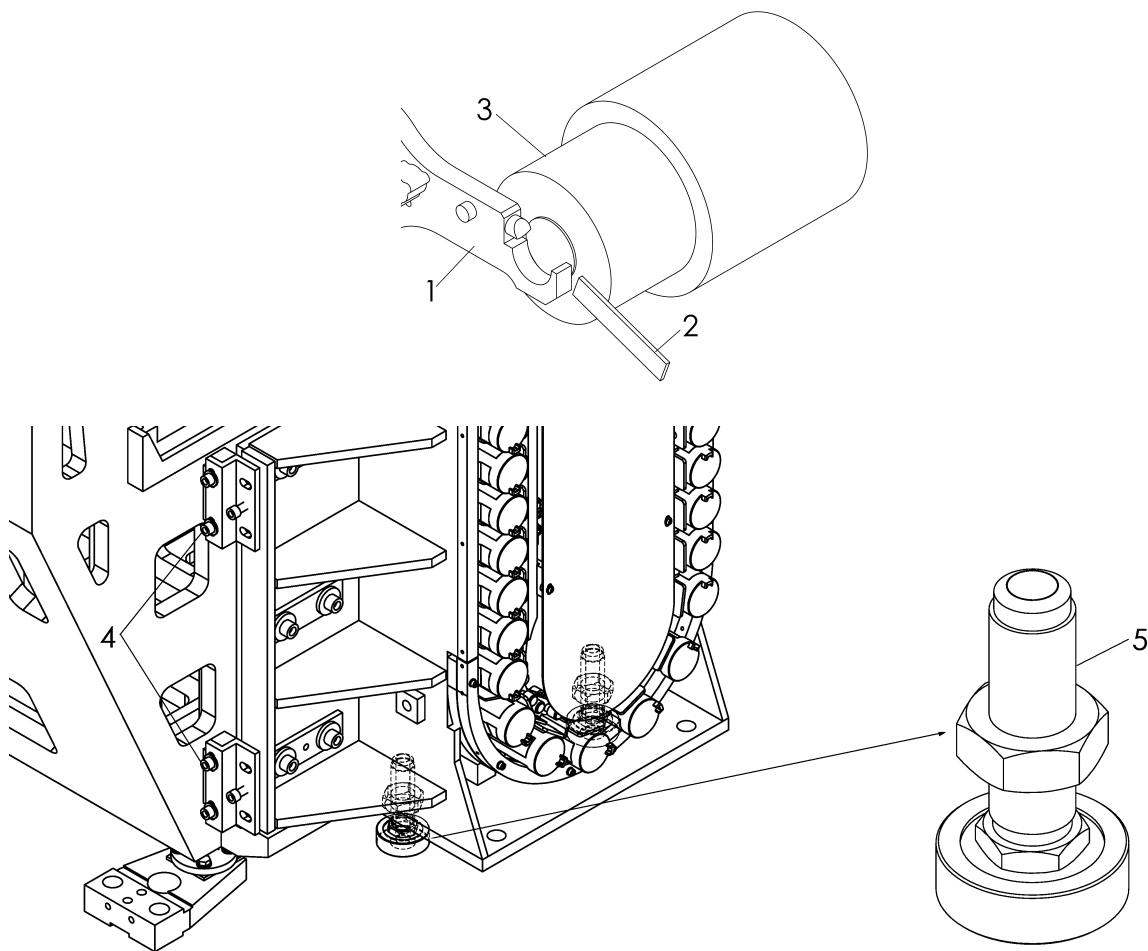


Figure 2.6 Adjusting the tool magazine

No.	Name	No.	Name
1	ATC arm	4	ATC push-pull tool
2	ATC-spindle gap tool	5	Tool magazine level pad
3	Spindle	-	-

2.4 Mounting Covers

1) Mount the covers of the machine.

This completes the mounting of the covers.

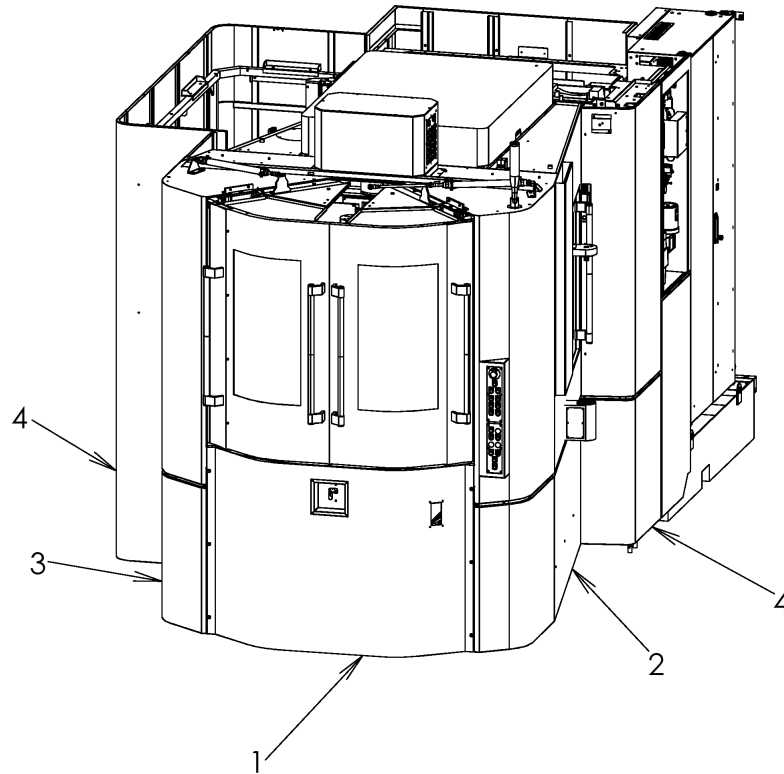


Figure 2.7 Mounting the Machine Covers (Standard Machine)

No.	Name	No.	Name
1	Front Skirt	3	Left Skirt
2	Right Skirt	4	Side Skirt

2.5 Installing the Tool Magazine



- Before starting this work, confirm that the machine power is turned OFF and that all machine operations are stopped.
- This procedure requires the operator to enter inside the machine or work within a small area, and so careful attention is needed during the work.
- Be sure to place a warning tag on each operation panel, main power switch, and other locations when performing work to ensure that other personnel do not turn on the power inadvertently which could result in the machine being operated.
- When operating the machine, maintain clear communication with everyone in the area to ensure operator safety.

ATC covers Installation Procedure

- 1) Install 1x M8 Eyebolt onto ATC arm cover [1] hosting hole [2] to lift and position at the installation location. Secure the ATC arm cover by using 9x M5x10 Truss head screws.

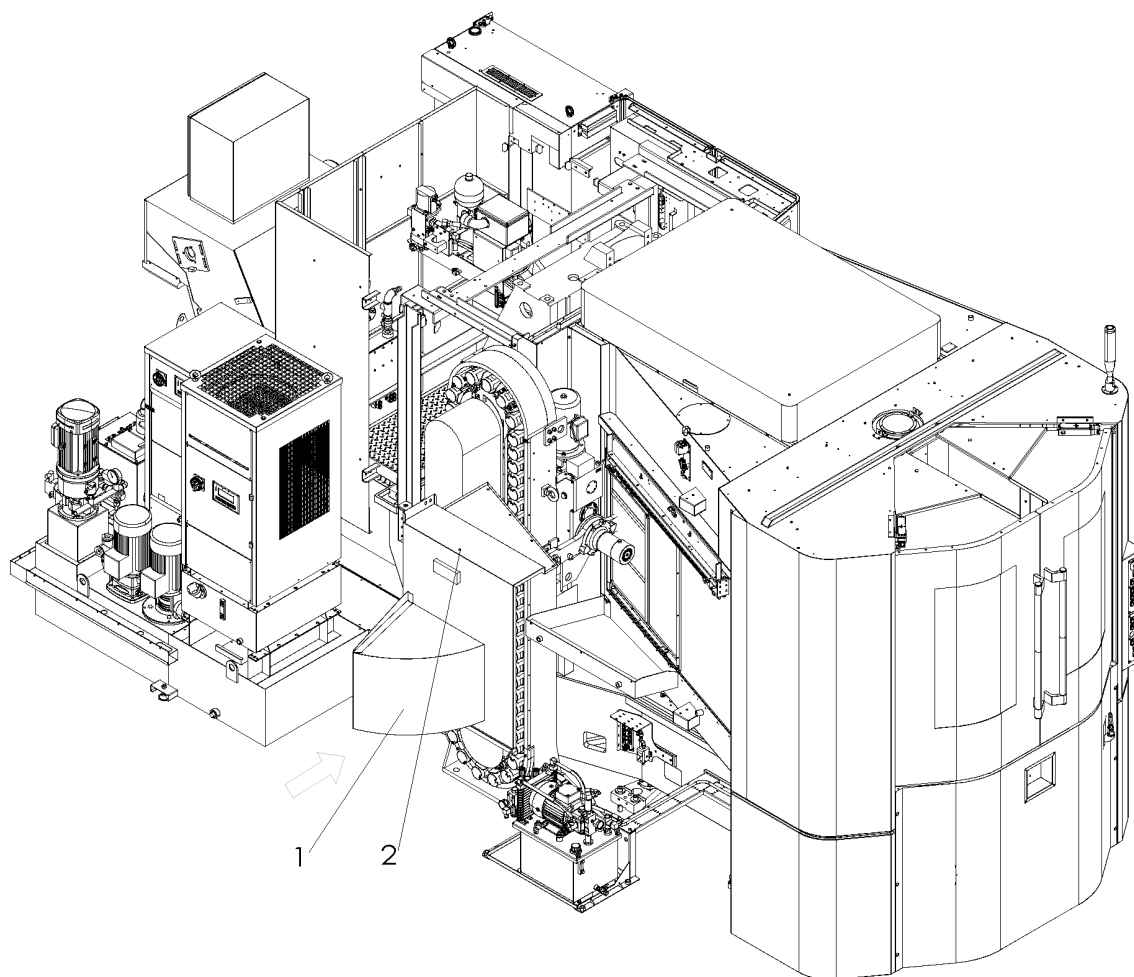


Figure 2.8 ATC covers Installation Procedure 1

No.	Name	No.	Name
1	ATC arm cover	2	Hosting hole

- 2) Install 3x M12 Eyebolt onto ATC cover [1] hosting hole [2] to lift and position at the installation location. Secure the ATC cover by using 8x M8 washers (OD18mm) and 8x M8x25 Socket head screws.

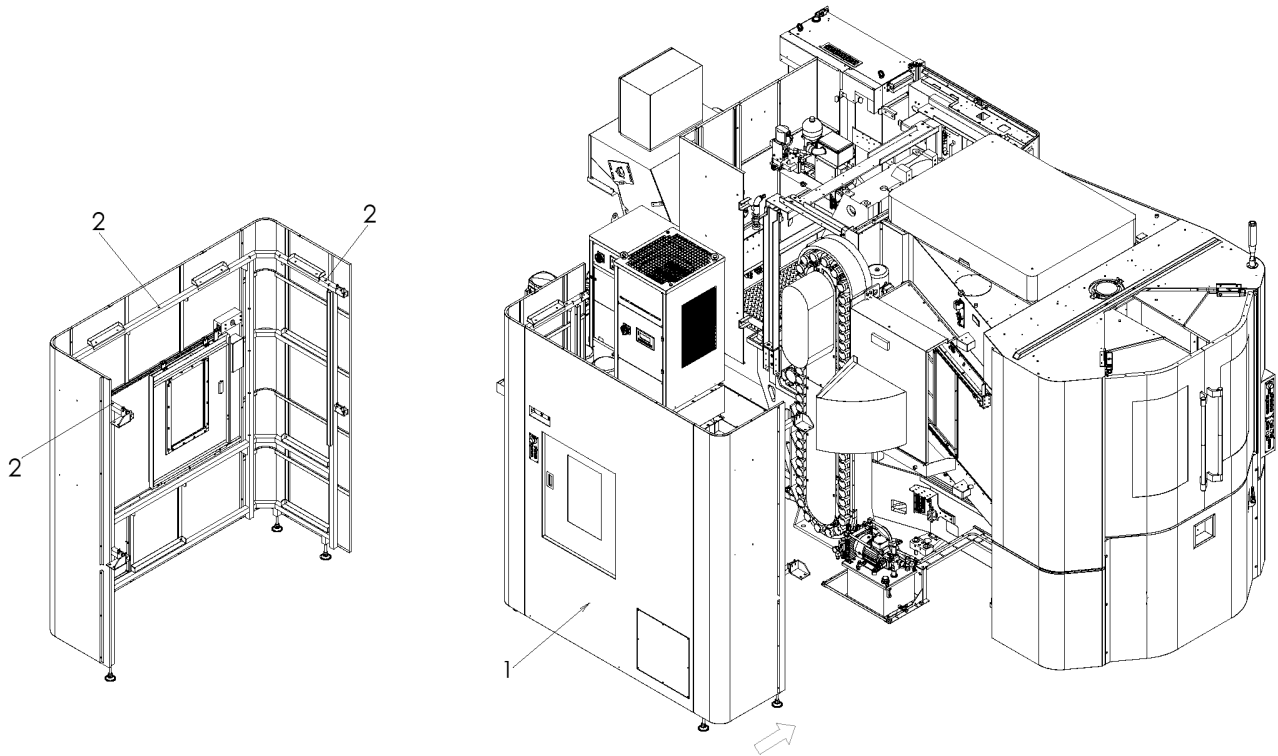


Figure 2.9 ATC covers Installation Procedure 2

No.	Name	No.	Name
1	ATC cover	2	Hosting hole

- 3) Install 1x M8 Eyebolt onto ATC inner cover [1] hosting hole [2] to lift and position at the installation location. Secure the ATC cover by using 4x M5x10 Truss head screws on ATC cover side and then 3x M6x10 Truss head screws on ATC side.

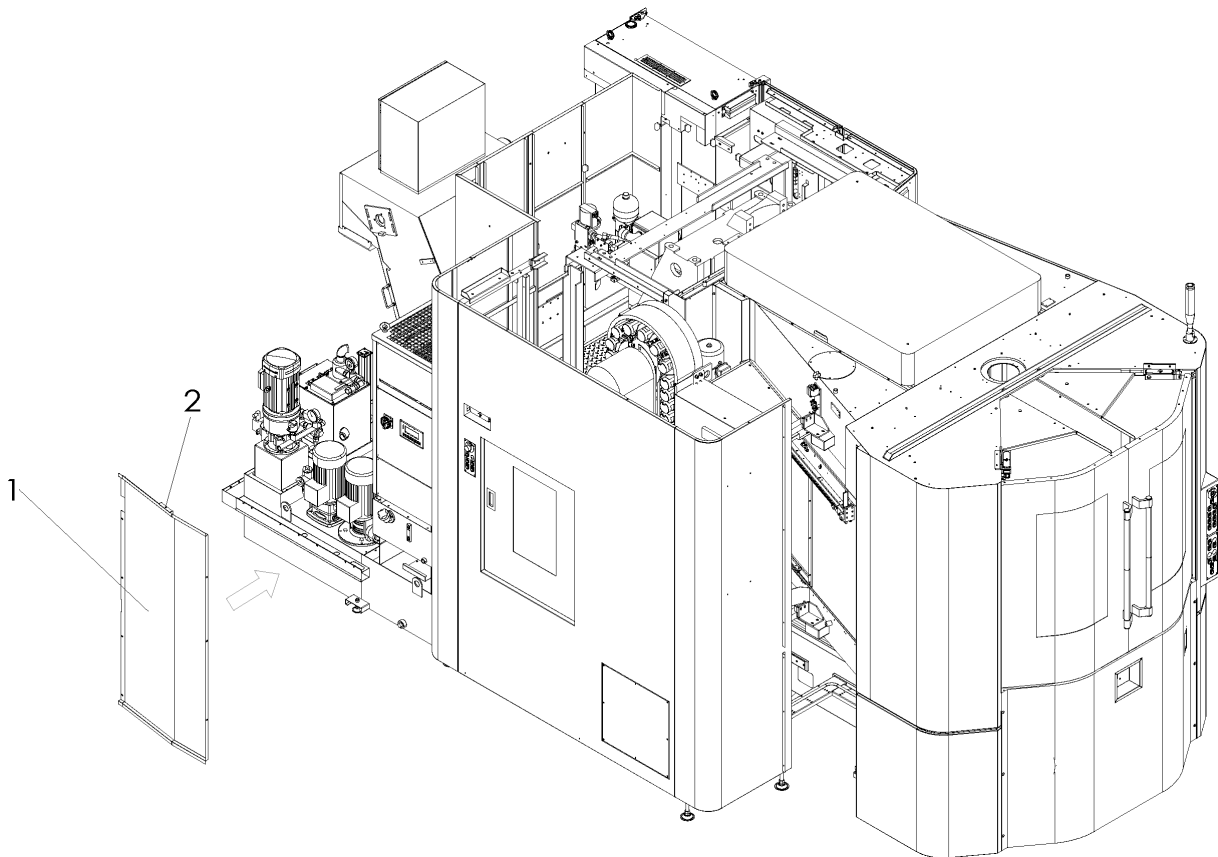


Figure 2.10 ATC covers Installation Procedure 3

No.	Name	No.	Name
1	ATC inner cover	2	Hosting hole

- 4) Slide the Chip Tray, ATC cover [1] under the ATC cover and position it under the ATC magazine. Secure the Chop tray by using 2x M5 washer and 2x M5x10 Truss head screws.

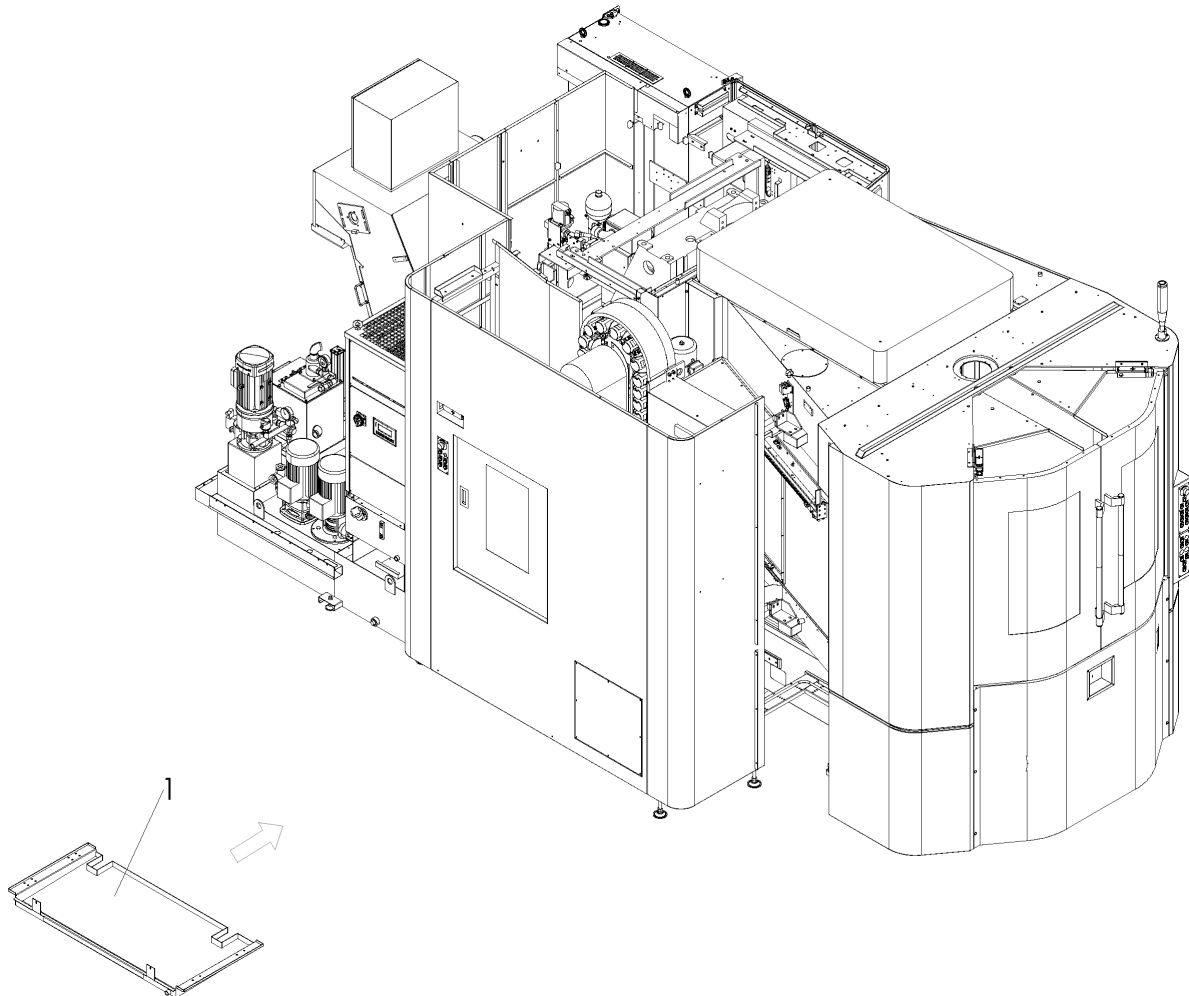


Figure 2.11 ATC covers Installation Procedure 4

No.	Name
1	Chip Tray, ATC

- 5) Install Cover Top LH [1] using 6x M5x10 Truss head screws and then Cover bottom middle [2] using 9x M5x10 Truss head screws.

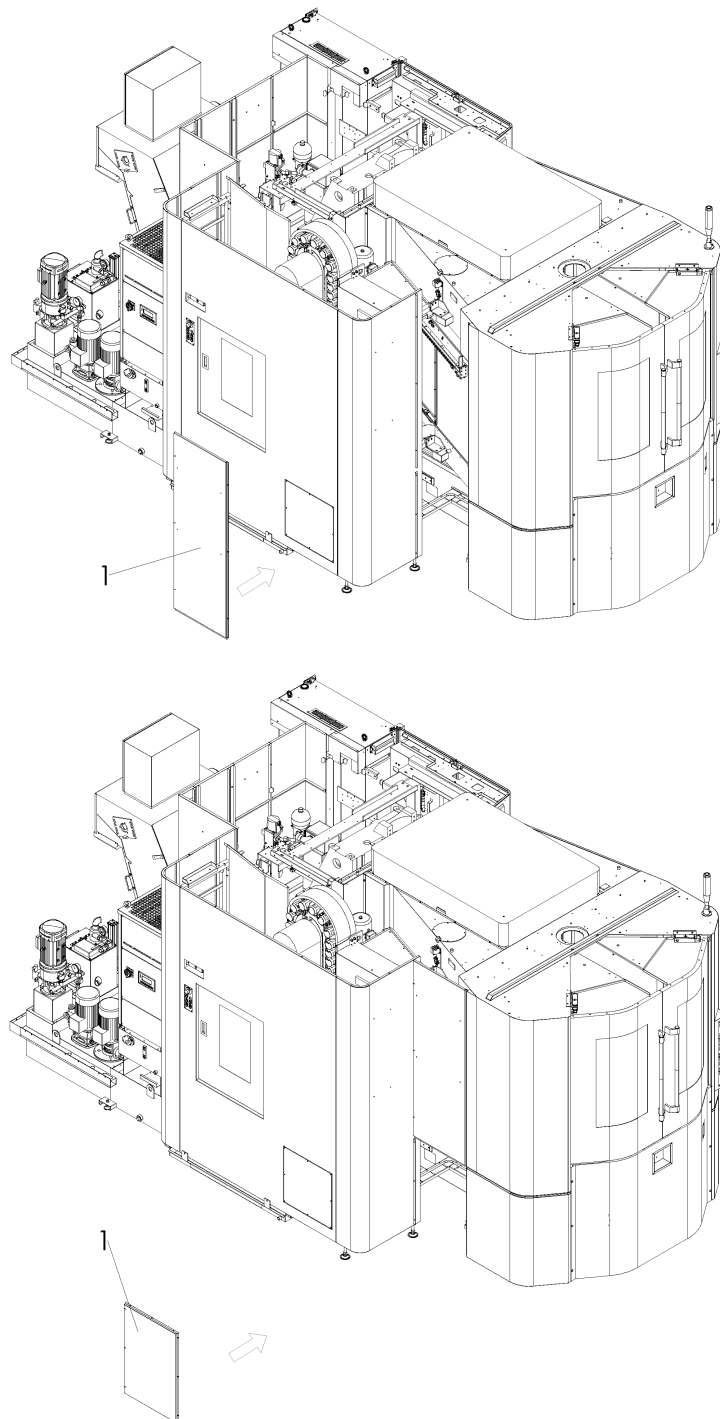


Figure 2.12 ATC covers Installation Procedure 5

No.	Name	No.	Name
1	Cover Top LH	2	Cover bottom middle

6) ATC covers installation complete.

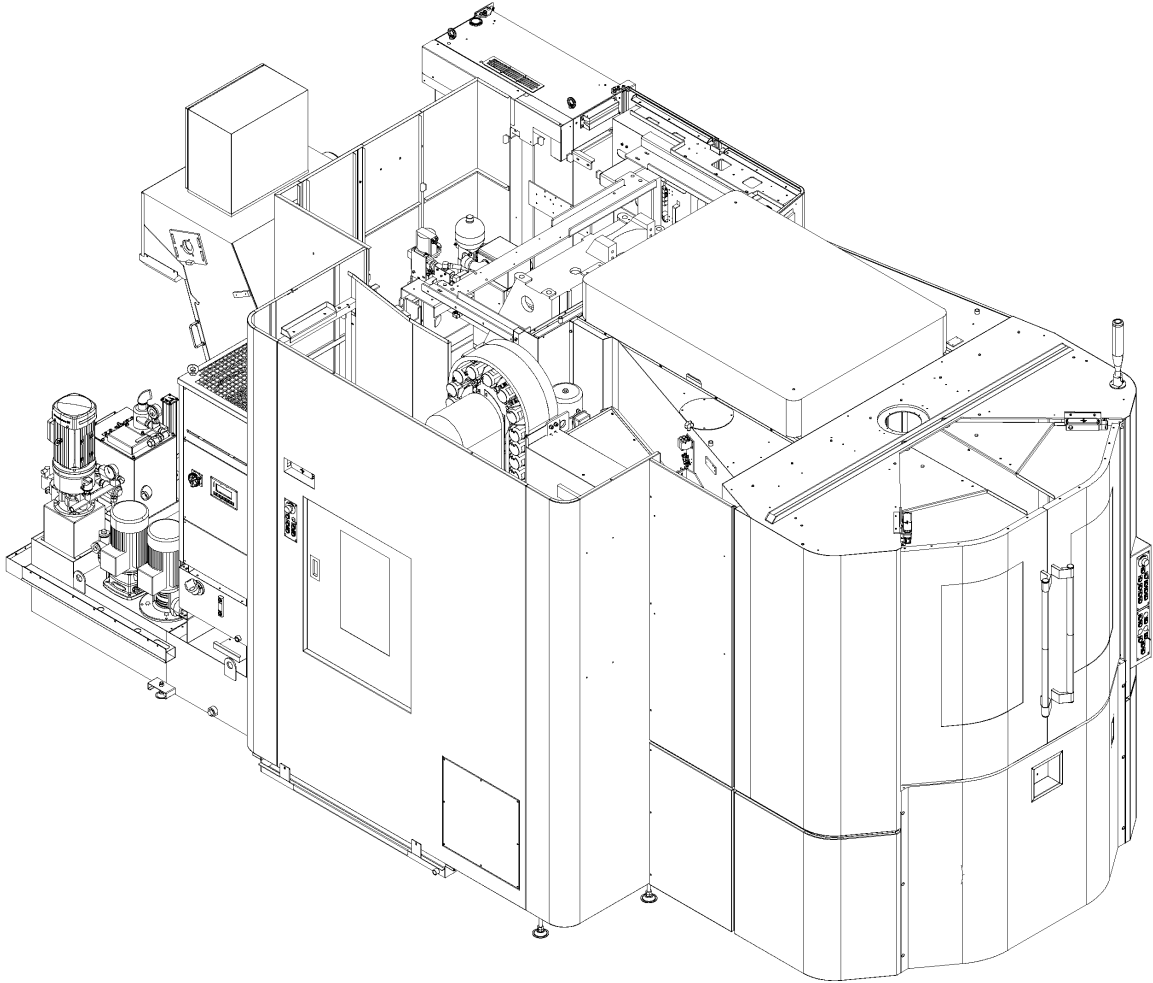


Figure 2.13 ATC cover Installation Procedure 6

2.6 Checking Static Accuracy

After leveling adjustment is completed, then proceed the static accuracy check, following "MAKINO customer service's installation check sheet ":

- Straightness of X-axis motion in the horizontal XY plane.
- Straightness of X-axis motion in the horizontal XZ plane.
- Straightness of Y-axis motion in the horizontal YX plane.
- Straightness of Y-axis motion in the horizontal YZ plane.
- Straightness of Z-axis motion in the horizontal ZX plane.
- Straightness of Z-axis motion in the horizontal ZY plane.
- Squareness between coordinates axes.
- Spindle movement in Y-axis direction.
- Round out of Spindle taper.
- Parallelism between the Spindle center line and Y-axis motion in the vertical YX plane.
- Parallelism between the Spindle center line and Y-axis motion in the vertical YZ plane.

2.7 Supply of Coolant



- The use of non-forming and water-soluble coolant (A1-class emulsion type bases on JIS) is strongly recommended. The use of A2-class soluble type, A3-class chemical solution type or synthetic type based coolant can damage the machine. (Examples: Peeling of paint, deterioration of rubber parts.) Consult with the manufacturer before use.
- Coolant used must be non-corrosive (both to rubber seals and metals).
- Do not use anti-foaming additives unless absolutely necessary to prevent clogging of the filter for the Drum unit. If an anti-foaming additive must be used, the concentration must be properly controlled.
- If use of an oil-based coolant cannot be avoided, a fire extinguisher system must be provided. (Refer to [Chapter 1, 1.3 Precautions for Potential Fire Hazard](#) )
- Dilute coolant in accordance with the manufacturer's instructions.
- Dilute coolant outside the Coolant Tank.
- Follow the manufacturer's instructions regarding the concentration of the coolant. (If the instructions are not followed, it may result in the formation of mold, foaming, corrosion and / or other problems.)
- Supply an adequate amount of coolant. If there is not an adequate volume of coolant, machining may be interrupted due to abnormal coolant circulation.
- Exercise extreme caution to prevent contamination while coolant is being supplied since the fill port is on the clean tank.
- Exercise extreme caution to avoid spilling of coolant.

Procedure to supply of coolant to coolant tank. (Refer to [Figure 2.14](#) )

- 1) Remove the Maintenance cover [2] to access the coolant fill port.
- 2) Supply coolant from the fill port with the Coolant Tank pumps OFF.
- 3) Return back Maintenance cover.
- 4) Confirm that the coolant level is in between the upper limit [3] and the lower limit [4] of the coolant level gauge with the Coolant Tank pumps [1] ON.

This completes the supply of coolant to the Coolant Tank.

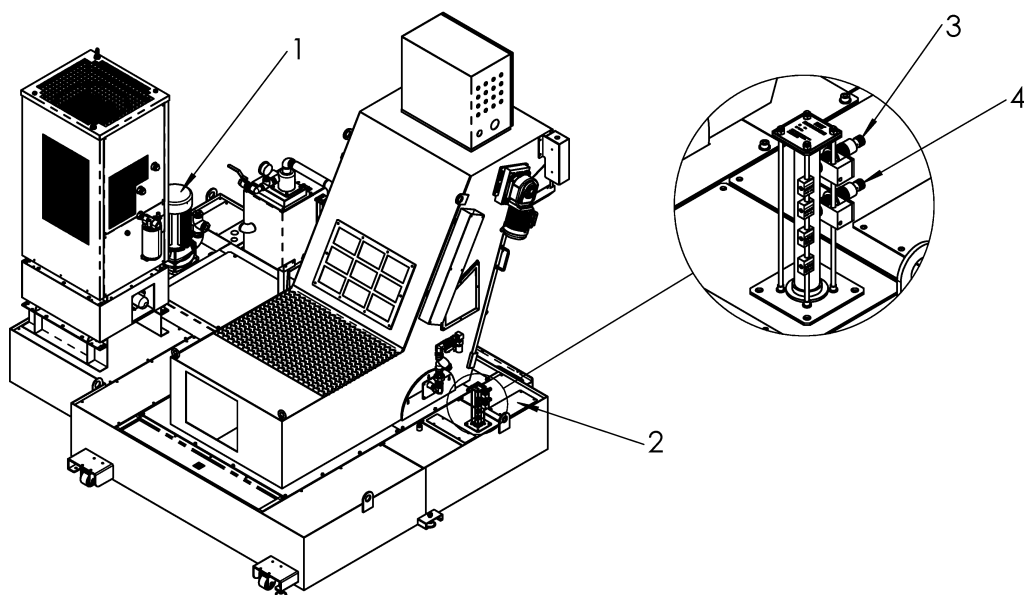


Figure 2.14 Coolant Supply

No.	Name	No.	Name
1	Coolant Tank Pump	3	Upper Limit
2	Maintenance Cover	4	Lower Limit

2.8 Before Starting Operation

Once machine installation and startup are completed, you are ready to start machine operation. Generally, Makino installation staff are present at the installation site until the customer can start actual machining, but the safety information and work items for starting operation are presented below, and so please use them as a checklist.

- Install and assemble machine
- Remove the fixing brackets (axis locks)
- Connect the main power supply and conduct power test
- Connect the pneumatic source and check the pressure
- Check and fill with hydraulic oil
- Check and fill with cooling oil
- Check and fill with cutting fluid
- Check the feed axis lubricating grease
- Check the alarm display after the power is turned on, and check the status (unusual noises or fluid leaks) of each machine component
- Check the parameters
- Check operation of the safety devices (emergency stop buttons, door interlock, etc.)
- Check the basic operation (spindle rotation, axis feeding, and various auxiliary functions besides the automatic tool changer (ATC))
- Check the static accuracy
- Check the reference position of the tool magazine
- Check the alignment of the tool magazine (Confirm Section 8 “Centering Alignment” in Chapter 6 of Maintenance Manual (H020I-M1E-XA) to make adjustment)

After the above checks are completed, register the tool data to the machine based on the customer's management method. A program tool number can be registered to the tools that are used for enabling use of the automatic tool change function (M6 and T functions).

A program number with the same number as each pod (tool storage location) in the tool magazine is already registered at shipping, and so the tool change operation can still be performed automatically even if no tool numbers are registered. To use your own unique tool number, register, and then perform the next tool related operations.

- Load tool to tool magazine
- Check the automatic tool changer (ATC) related auxiliary functions (M6 operations, T functions)
- Check the coolant nozzle discharge direction

2.9 Test Run

The main operations for running the machine are performed from the main operation panel. Almost all operations for the NC unit and the machine are possible from the main operation panel.

After the main power switch (main circuit breaker) of the machine controller is turned ON, pressing the READY button prepares the machine for all operations. For details, see the Instruction Manual, "Basic Operation".

If any problem occurs in the machine, an alarm number and message appear on the display of the main operation panel. For details on alarms, see the Maintenance Manual, "Supplement to Control Software".

When shutting OFF the main power of the machine, be sure to follow the specified power-off procedure. For details, see the Instruction Manual, "Basic Operation".

3 Checking Axis Reference Positions

3.1 Axis Reference Position Return

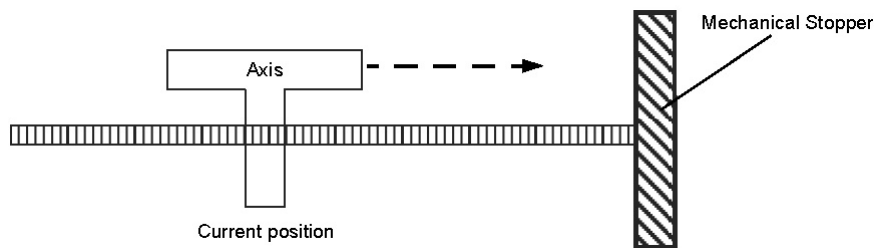
Perform reference position return following the procedures below. This procedure is to be performed after machine installation is completed.

3.1.1 Butt Type Reference Position Return

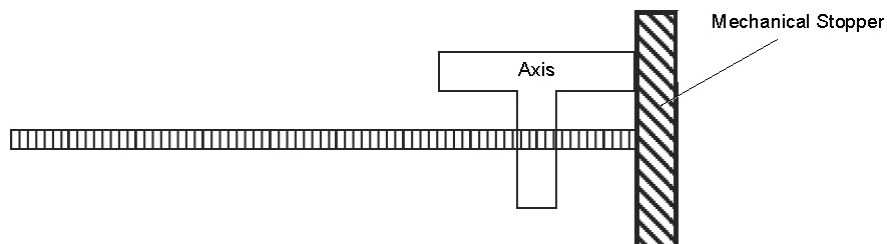
The Butt Type Reference method takes the axis reference with respect to the mechanical stopper. To perform reference position return follow the procedures below.

Operation Procedure

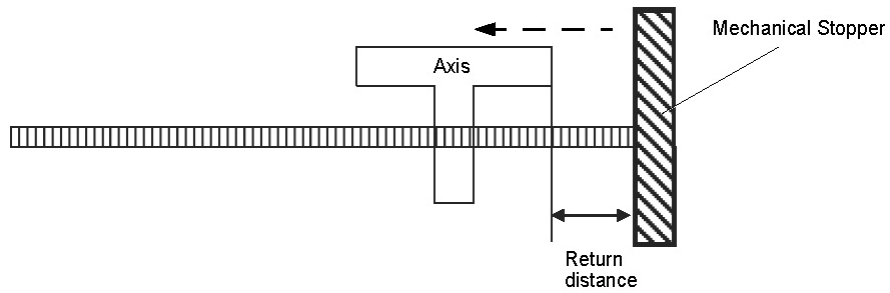
- 1) Select **[REFERENCE]** mode on the main operation panel.
- 2) Select the axis to be referenced by pressing the Axis Selection switch. Z-axis should be selected first.
- 3) Press **[CYCLE START]**.
- 4) The selected axis move towards the mechanical stopper in the direction that have been set in parameter at a constant feedrate.



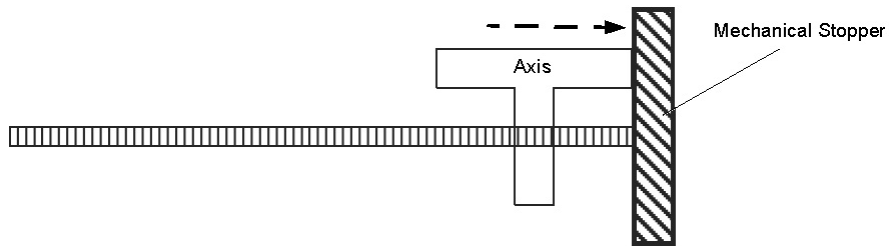
- 5) The axis butts against the mechanical stopper until the torque limit is reached.



- 6) After the axis butts the mechanical stopper the axis is retracted in opposite direction at the feed rate specified in parameter to the return distance specified in parameter.



- 7) From the return position the axis butts again(second time) the mechanical stopper in the direction set in parameter at constant feed rate.



- 8) After the axis butts the mechanical stopper the axis is retracted in opposite direction at the feed rate and distance specified in parameter.

Table 3.1 Setting for Butt Type Reference Position Return

	X-Axis	Y-Axis	Z-Axis
Operating Direction	(+)	(-)	(+)
Axis moving speed towards mechanical stopper (1st attempt)	1000mm/min		
Axis moving speed towards mechanical stopper (2nd attempt)	500mm/min		
Axis returning speed from mechanical stopper	2000mm/min		
Axis returning distance	7.5mm	7mm	10mm

Establishing Reference Point with Respect to Table

The reference point is established with respect to table center in full stroke as per machine specification. During machine building stage, follow the method below to achieve reference point from table center.

- 1) Carry out the procedure as explained in section 3.1.1 ¹⁰¹ Butt Type Reference Position Return.
- 2) Check the machine axis stroke. Note down the shift value required to achieve the stroke with respect to table center.
- 3) Input the required shift value for each respective axis in the Butt Type Reference Maintenance Screen. (Refer to Figure 3.1 ¹⁰³)

To display Butt Type Reference Screen:

Press iHMI **[Menu]** hard button → select **[Makino]** Screen → select **[BUTT REF]** → Butt Type Reference Screen is displayed.

On the Butt Type Reference Screen, turn ON the Maintenance Mode, select axis and input the required shift value.

- 4) Repeat step 1 to 3 till required stroke is achieved with respect to table center.

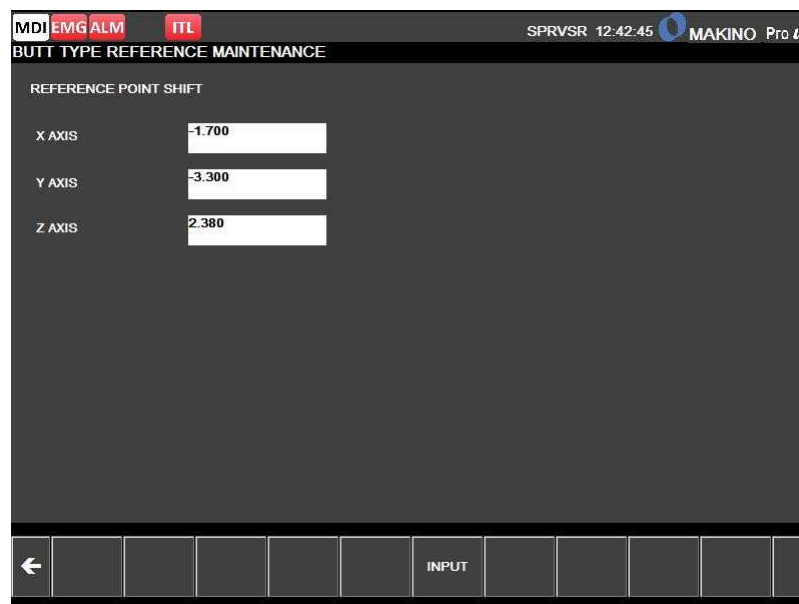


Figure 3.1 Butt Type Reference Maintenance Screen

Revision Record

Manual Title	JH5 INSTALLATION MANUAL Professional <i>u</i>
Manual No.	H020I-I2E-XB

No.	Manual No. (Month/Year Issued)	Contents	Remarks
1	H020I-I2E-XA (06/2025)	Newly issued	
2	H020I-I2E-XB (07/2026)	Updated: Chapter 2 - 1.6 Preparation of Tools Chapter 3 - 2.2 Level Adjustment -2.3 Tool Magazine Adjustment	Newly added



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